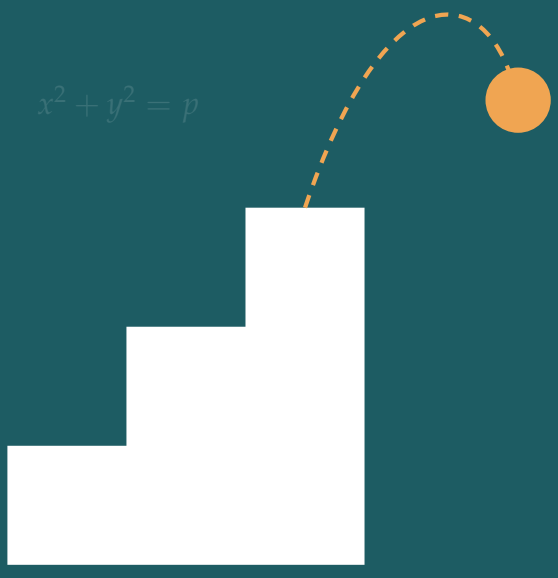
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

1996 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

1996 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

1996 Worked Solutions

Worked solutions to all 42 questions of the 1996 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 1996, Question 1

Pure • Optimisation of Cylinder

> **STEP I 1996, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/96-s1-q1

The key idea

Two quantities describe the tin, its radius r and its height h. The volume constraint lets us remove one of them, so S becomes a function of r alone, and we can minimise it by ordinary differentiation. The optimal tin turns out to have a very tidy shape. Once you see that shape, both sphere questions come down to looking at a single cross-section through the axis.

Solution

Minimising the surface area

Let the tin have radius $r > 0$ and height $h > 0$. Then

$$V = \pi r^2 h, \qquad S = 2\pi r^2 + 2\pi r h,$$

where $2\pi r^2$ comes from the two ends and $2\pi rh$ from the curved side. For fixed V we have $h = V/(\pi r^2)$, so $2\pi rh = 2V/r$ and

$$S(r) = 2\pi r^2 + \frac{2V}{r}, \qquad r > 0.$$

Differentiating,

$$\frac{\mathrm{d}S}{\mathrm{d}r} = 4\pi r - \frac{2V}{r^2} = \frac{2(2\pi r^3 - V)}{r^2}.$$

This is negative for $r^3 < V/(2\pi)$ and positive for $r^3 > V/(2\pi)$. So S decreases and then increases, and its least value on $r > 0$ is at

$$r^3 = \frac{V}{2\pi}.$$

This is the global minimum, not just a local one, because it is the only stationary point and the derivative changes sign from $-$ to $+$ there. (You can also see it from the formula: $S \to \infty$ both as $r \to 0^+$ and as $r \to \infty$.)

At this radius $V = 2\pi r^3$, so

$$h = \frac{V}{\pi r^2} = 2r.$$

The best tin is exactly as tall as it is wide. Then $2V/r = 4\pi r^2$, so

$$\begin{aligned} S &= 2\pi r^2 + 4\pi r^2 = 6\pi r^2 = 6\pi \left(\frac{V}{2\pi}\right)^{2/3} \\ &= 3 \cdot 2\pi \cdot (2\pi)^{-2/3} V^{2/3} = 3\,(2\pi)^{1/3} V^{2/3} = 3\,(2\pi V^2)^{1/3}, \end{aligned}$$

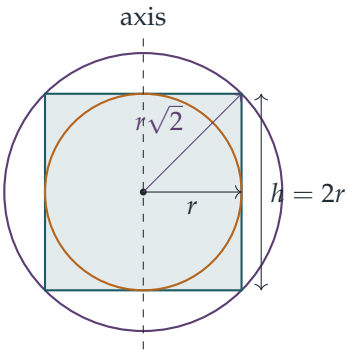
which is the required minimum.

Ask yourself Why work with r rather than h?

Either variable works. Eliminating h is easier, though, because h appears only linearly in both formulae, so the substitution $h = V/(\pi r^2)$ gives a simple function of r. Eliminating r would bring in $\sqrt{V/(\pi h)}$, and the algebra gets messier.

The largest sphere inside the tin

From now on the tin has $h = 2r$. Cut it by a plane through its axis: the cross-section is a $2r \times 2r$ square.



Any sphere inside the tin has radius $\rho \le r$. To see this, project it onto the plane of the base: it gives a disc of radius ρ lying inside the base disc of radius r. Also $2\rho \le h = 2r$, which gives the same bound. The sphere of radius r centred at the centre of the tin (the amber circle in the cross-section) touches both ends and the curved side, so it fits. Using $r^3 = V/(2\pi)$, the largest sphere therefore has volume

$$\tfrac{4}{3}\pi r^3 = \tfrac{4}{3}\pi \cdot \frac{V}{2\pi} = \tfrac{2}{3}V,$$

as required.

The smallest sphere containing the tin

Take a point on the rim of the top end and the diametrically opposite point on the rim of the bottom end. These are opposite corners of the square cross-section, so their

distance apart is

$$\sqrt{(2r)^2 + h^2} = \sqrt{8r^2} = 2\sqrt{2}\,r.$$

Any sphere containing the tin contains both points, so its diameter is at least $2\sqrt{2}\,r$ and its radius at least $\sqrt{2}\,r$.

Ask yourself How do we know a sphere of radius $\sqrt{2}\,r$ is actually big enough?

Centre it at the centre O of the tin. Any point of the tin is at distance $\rho \le r$ from the axis and at height z above or below O with $|z| \le r$. Its distance from O is therefore $\sqrt{\rho^2 + z^2} \le \sqrt{r^2 + r^2} = \sqrt{2}\,r$. So the whole tin lies inside this sphere (the plum circle passes through all four corners of the square).

The smallest sphere into which the tin fits therefore has radius $\sqrt{2}\,r$ and volume

$$\tfrac{4}{3}\pi\left(\sqrt{2}\,r\right)^3 = \tfrac{4}{3}\pi \cdot 2\sqrt{2}\,r^3 = \tfrac{8\sqrt{2}}{3}\pi \cdot \frac{V}{2\pi} = \frac{4\sqrt{2}}{3}\,V.$$

Checks

- With $V = 1$, minimising $2\pi r^2 + 2/r$ by brute force over a fine grid of r gives $S \approx 5.5358$, which agrees with $3(2\pi)^{1/3} \approx 5.5358$.
- The ratio of the two sphere volumes should be $(\sqrt{2})^3 = 2\sqrt{2}$, the cube of the ratio of the radii. Indeed $\frac{4\sqrt{2}}{3}V \big/ \frac{2}{3}V = 2\sqrt{2}$.
- The order makes sense: $\frac{2}{3}V < V < \frac{4\sqrt{2}}{3}V \approx 1.886\,V$, so the inner sphere is smaller than the tin and the outer one larger.

Common slips

- Forgetting one or both ends, writing $S = \pi r^2 + 2\pi rh$ or $S = 2\pi rh$. That gives a different optimal shape.
- Showing $\mathrm{d}S/\mathrm{d}r = 0$ without saying why the stationary point is a minimum.
- Taking the radius of the outer sphere to be r or h, rather than half the space diagonal of the tin.

Beyond STEP

This is constrained optimisation: minimise $S(r,h)$ subject to $V(r,h)$ being fixed. At university you will meet Lagrange multipliers, which handle this without eliminating a variable. The condition $\nabla S = \lambda \nabla V$ gives $h = 2r$ almost at once. The two spheres also illustrate the "inscribed" and "circumscribed" balls of a convex body, which are central in convex geometry. For more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 1996, Question 2

Pure • Beta Integrals

STEP I 1996, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q2

The key idea

The integrands in (i) and (ii) are the same function written two ways: $\alpha x + (1-x) = 1 + (\alpha - 1)x$. So one integral, worked out in two ways, gives two descriptions of the same polynomial in α. Part (i) gives a closed form, part (ii) gives the coefficients as integrals, and comparing coefficients does the rest. No integration by parts is needed.

Solution

Part (i)

Substitute $u = 1 + (\alpha - 1)x$, so $\mathrm{d}u = (\alpha - 1)\,\mathrm{d}x$; when $x = 0$, $u = 1$ and when $x = 1$, $u = \alpha$. Since $\alpha \neq 1$ we may divide by $\alpha - 1$:

$$\int_0^1 \big(1 + (\alpha - 1)x\big)^n \,\mathrm{d}x = \frac{1}{\alpha - 1}\int_1^\alpha u^n \,\mathrm{d}u = \frac{1}{\alpha - 1}\left[\frac{u^{n+1}}{n+1}\right]_1^\alpha = \frac{\alpha^{n+1} - 1}{(n+1)(\alpha - 1)},$$

as required. (The substitution is fine even if $\alpha < 1$ or $\alpha < 0$: it is linear, and the formula $\int_1^\alpha u^n \,\mathrm{d}u$ holds with the limits in either order. Equivalently, just differentiate $\big(1 + (\alpha - 1)x\big)^{n+1} / \big((n+1)(\alpha - 1)\big)$.)

Part (ii)

Expand the integrand by the binomial theorem, treating αx and $1 - x$ as the two terms:

$$\big(\alpha x + (1 - x)\big)^n = \sum_{k=0}^{n} \binom{n}{k} (\alpha x)^k (1 - x)^{n-k} = \sum_{k=0}^{n} \alpha^k \binom{n}{k} x^k (1 - x)^{n-k}.$$

This is a finite sum, so we may integrate term by term; α is a constant as far as the integration over x is concerned:

$$\int_0^1 \big(\alpha x + (1 - x)\big)^n \,\mathrm{d}x = \sum_{k=0}^{n} \alpha^k \left[\binom{n}{k} \int_0^1 x^k (1 - x)^{n-k} \,\mathrm{d}x\right].$$

Each bracket is a number not depending on α, so this is a polynomial in α of degree at most n, and the coefficient of α^k for $0 \le k \le n$ is $\binom{n}{k}\int_0^1 x^k(1-x)^{n-k}\,\mathrm{d}x$, as required.

Part (iii)

Write $P(\alpha)$ for the polynomial in (ii). Because $\alpha x + (1-x) = 1 + (\alpha-1)x$, part (i) says that for every $\alpha \neq 1$,

$$P(\alpha) = \frac{\alpha^{n+1}-1}{(n+1)(\alpha-1)} = \frac{1}{n+1}\left(1+\alpha+\alpha^2+\cdots+\alpha^n\right),$$

using the geometric series factorisation $\alpha^{n+1}-1 = (\alpha-1)(1+\alpha+\cdots+\alpha^n)$.

Ask yourself We only know these two polynomials agree for $\alpha \neq 1$. Why may we compare coefficients?

The difference of the two sides is a polynomial of degree at most n that vanishes at every $\alpha \neq 1$, so it has infinitely many roots. A nonzero polynomial of degree at most n has at most n roots, so the difference is the zero polynomial: every coefficient matches. (Alternatively, both sides are continuous, so they also agree at $\alpha = 1$.)

On the right, every coefficient of α^k, $0 \le k \le n$, equals $\frac{1}{n+1}$. Comparing with (ii),

$$\binom{n}{k}\int_0^1 x^k(1-x)^{n-k}\,\mathrm{d}x = \frac{1}{n+1},$$

and so

$$\int_0^1 x^k(1-x)^{n-k}\,\mathrm{d}x = \frac{1}{(n+1)\binom{n}{k}} = \frac{k!\,(n-k)!}{(n+1)\,n!} = \frac{k!\,(n-k)!}{(n+1)!},$$

which is the required result.

Checks

- $k=0$: $\int_0^1 (1-x)^n\,\mathrm{d}x = \frac{1}{n+1}$, and the formula gives $\frac{n!}{(n+1)!} = \frac{1}{n+1}$.
- $n=2$, $k=1$: $\int_0^1 (x-x^2)\,\mathrm{d}x = \frac{1}{2}-\frac{1}{3} = \frac{1}{6} = \frac{1!\,1!}{3!}$.
- Numerically, $\int_0^1 x^2(1-x)^3\,\mathrm{d}x \approx 0.016667 = \frac{1}{60} = \frac{2!\,3!}{6!}$; we checked (i) by Simpson's rule for random α and n, and (iii) exactly for all $0 \le k \le n \le 12$.
- A pleasing consequence: the coefficients of P are all equal, so $\int_0^1 \binom{n}{k}x^k(1-x)^{n-k}\,\mathrm{d}x$ does not depend on k.

Common slips

- Comparing coefficients without remarking that (i) was only proved for $\alpha \neq 1$; one sentence about polynomials (or continuity) closes the gap.

- Expanding $(1+(\alpha-1)x)^n$ in powers of $(\alpha-1)$ in (ii): that gives the right polynomial but not in powers of α. Expand $\big(\alpha x+(1-x)\big)^n$ instead.

Beyond STEP

The integral $\int_0^1 x^{a-1}(1-x)^{b-1}\,\mathrm{d}x$ is Euler's *beta function* $B(a,b)$, and the result here is the integer case of $B(a,b)=\Gamma(a)\Gamma(b)/\Gamma(a+b)$. In probability, it says that if p is chosen uniformly from $[0,1]$ and then n coins with head-probability p are tossed, every number of heads from 0 to n is equally likely, each with probability $\frac{1}{n+1}$ (this is exactly what "all coefficients equal" means), the starting point of Bayes's own famous essay. Grimmett and Welsh, *Probability: An Introduction*, develops the beta and gamma distributions behind this.

WORKED SOLUTION

STEP I 1996, Question 3

Pure • Divisibility of Integers

STEP I 1996, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q3

The key idea

Divisibility by a composite number such as 24 is easiest to handle one prime power at a time: $24 = 3 \times 8$ with 3 and 8 coprime, so we show divisibility by 3 and by 8 separately. Factorising gives us products of consecutive integers, and among any k consecutive integers one is a multiple of k. The other two parts yield to the same habit: factorise, or write n in a form that shows its remainder.

Solution

Part (i)

Taking out the common factor and using the difference of two squares,

$$n^5 - n^3 = n^3(n^2 - 1) = n^3(n-1)(n+1) = n^2\,(n-1)\,n\,(n+1).$$

Divisibility by 3. The numbers $n-1$, n, $n+1$ are three consecutive integers, so exactly one of them is a multiple of 3. Hence $3 \mid (n-1)n(n+1)$, and so $3 \mid n^5 - n^3$.

Divisibility by 8. We split into cases on the parity of n.

- If n is even, say $n = 2m$, then $n^3 = 8m^3$, which is a multiple of 8.
- If n is odd, then $n-1$ and $n+1$ are consecutive even numbers. One of any two consecutive even numbers is a multiple of 4, and the other is a multiple of 2, so $8 \mid (n-1)(n+1)$.

In either case $8 \mid n^3(n-1)(n+1) = n^5 - n^3$.

Ask yourself Why does knowing $3 \mid N$ and $8 \mid N$ give $24 \mid N$? Would $4 \mid N$ and $6 \mid N$ give the same?

Since 3 and 8 have no common factor, $8 \mid N$ means $N = 8a$, and then $3 \mid 8a$ forces $3 \mid a$ (as 3 is prime and doesn't divide 8). So $N = 24b$. The argument fails for 4 and 6, which

share the factor 2: the number 12 is divisible by both but not by 24.

Therefore $n^5 - n^3$ is divisible by 24 for every positive integer n, as required.

Part (ii)

Write $2^{2n} = 4^n$. Using the factorisation $x^n - 1 = (x-1)(x^{n-1} + x^{n-2} + \cdots + x + 1)$ with $x = 4$,

$$2^{2n} - 1 = 4^n - 1 = (4-1)\left(4^{n-1} + 4^{n-2} + \cdots + 4 + 1\right) = 3\left(4^{n-1} + \cdots + 1\right).$$

The bracket is an integer, so $2^{2n} - 1$ is divisible by 3.

Equivalently, in modular arithmetic: $4 \equiv 1 \pmod 3$, so $4^n \equiv 1^n = 1 \pmod 3$. A proof by induction also works: if $4^k - 1 = 3a$, then $4^{k+1} - 1 = 4(4^k - 1) + 3 = 3(4a + 1)$.

Part (iii)

Since $3 \mid n - 1$, write $n = 3k + 1$ for some integer $k \geq 0$. Factorise

$$n^3 - 1 = (n-1)(n^2 + n + 1).$$

The first factor is $n - 1 = 3k$. For the second,

$$\begin{aligned} n^2 + n + 1 &= (9k^2 + 6k + 1) + (3k + 1) + 1 \\ &= 9k^2 + 9k + 3 \\ &= 3(3k^2 + 3k + 1). \end{aligned}$$

So each factor contributes a 3:

$$n^3 - 1 = 3k \cdot 3(3k^2 + 3k + 1) = 9k(3k^2 + 3k + 1),$$

which is divisible by 9, as required. (When $n = 1$ we get $k = 0$ and $n^3 - 1 = 0$, which is divisible by 9 too.)

The point is that $n - 1$ supplies only one factor of 3; the second comes from $n^2 + n + 1$. A neat way to see that second factor is $n^2 + n + 1 = (n-1)^2 + 3n$, a sum of two multiples of 3 whenever $3 \mid n - 1$.

Checks

- Part (i): $n = 2$ gives $32 - 8 = 24$ and $n = 3$ gives $243 - 27 = 216 = 9 \times 24$. Since $n = 2$ gives exactly 24, no larger number divides every $n^5 - n^3$, so 24 is the best possible.
- Part (ii): $n = 1, 2, 3$ give 3, 15, 63.
- Part (iii): $n = 4$ gives $63 = 7 \times 9$ and $n = 7$ gives $342 = 38 \times 9$, matching $9k(3k^2 + 3k + 1)$ with $k = 1$ and $k = 2$.

We checked all three results by computer for every n below 5000 (and for $n < 2000$ in part (ii)), and confirmed that the highest common factor of the values of $n^5 - n^3$ is exactly 24.

Common slips

- Arguing "$(n-1)n(n+1)$ is divisible by 6, and n^2 is …" and losing track of where the factor 8 comes from. The even case needs the n^3, not just n.
- Concluding divisibility by 24 from divisibility by 4 and by 6; the factors must be coprime.
- In part (iii), stopping at "$n-1$ is divisible by 3, so $n^3 - 1 = (n-1)(\dots)$ is divisible by 3", which gives only one factor of 3.

Beyond STEP

All three parts are really statements in modular arithmetic: the integers modulo m form a ring, and part (iii) says that if $n \equiv 1 \pmod 3$ then $n^3 \equiv 1 \pmod 9$. This is the first step of a general phenomenon called *lifting the exponent*: if $p \mid n-1$ then $p^{j+1} \mid n^{p^j} - 1$. Fermat's little theorem ($n^p \equiv n \pmod p$) explains part (i) too, since it gives $3 \mid n^3 - n$ directly. Davenport's *The Higher Arithmetic* is a beautiful introduction to these ideas.

WORKED SOLUTION

STEP I 1996, Question 4

Pure • Integration by Completing Square

STEP I 1996, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q4

The key idea

Whenever a quadratic sits in a denominator, complete the square: $x^2 + 2ax + 1 = (x + a)^2 + (1 - a^2)$. The sign of $1 - a^2$ decides everything. If it is positive we have a "sum of squares" and an inverse tangent; if it is negative we have a difference of squares, which splits into partial fractions and gives logarithms. The real work is then tidying the answers into the given forms.

Solution

Throughout, write I for the integral over $[0, 1]$. Completing the square,

$$x^2 + 2ax + 1 = (x + a)^2 + 1 - a^2.$$

The case $|a| < 1$

Put $b = \sqrt{1 - a^2} > 0$. Using $\int \frac{\mathrm{d}u}{u^2 + b^2} = \frac{1}{b} \tan^{-1} \frac{u}{b}$ with $u = x + a$,

$$I = \frac{1}{b}\left[\tan^{-1} \frac{x + a}{b}\right]_0^1 = \frac{1}{b}(\alpha - \beta), \qquad \text{where } \alpha = \tan^{-1} \frac{1 + a}{b}, \quad \beta = \tan^{-1} \frac{a}{b}.$$

(The denominator is never zero on $[0, 1]$, since it is at least $b^2 > 0$, so there is no trouble with the integral.) We need a single inverse tangent, so we take the tangent of $\alpha - \beta$:

$$\tan(\alpha - \beta) = \frac{\dfrac{1 + a}{b} - \dfrac{a}{b}}{1 + \dfrac{(1 + a)a}{b^2}} = \frac{b}{b^2 + a + a^2} = \frac{b}{1 + a},$$

using $b^2 + a^2 = 1$. Finally

$$\frac{b}{1 + a} = \frac{\sqrt{(1 - a)(1 + a)}}{1 + a} = \sqrt{\frac{1 - a}{1 + a}},$$

which is valid because $1 + a > 0$.

Ask yourself Does $\tan(\alpha-\beta)=t$ really give $\alpha-\beta=\tan^{-1}t$?

Only if $\alpha-\beta$ lies in $(-\frac{\pi}{2},\frac{\pi}{2})$, the range of $\tan^{-1}$, so we must check this. Since $(1+a)/b>0$ we have $0<\alpha<\frac{\pi}{2}$, and $-\frac{\pi}{2}<\beta<\frac{\pi}{2}$. Also $\alpha>\beta$, because $\tan^{-1}$ is increasing and $(1+a)/b>a/b$. So $0<\alpha-\beta<\pi$. On that interval tan is positive only on $(0,\frac{\pi}{2})$, and we found $\tan(\alpha-\beta)>0$; hence $0<\alpha-\beta<\frac{\pi}{2}$ and

$$I=\frac{1}{\sqrt{1-a^2}}\tan^{-1}\sqrt{\frac{1-a}{1+a}}\qquad(|a|<1),$$

as required.

(A slicker view: with $a=\cos\varphi$, $0<\varphi<\pi$, we get $\alpha=\frac{\pi}{2}-\frac{\varphi}{2}$ and $\beta=\frac{\pi}{2}-\varphi$, so $\alpha-\beta=\frac{\varphi}{2}$, and indeed $\sqrt{(1-\cos\varphi)/(1+\cos\varphi)}=\tan\frac{\varphi}{2}$.)

The case $|a|>1$

Put $c=\sqrt{a^2-1}>0$. Now the denominator is a difference of two squares:

$$x^2+2ax+1=(x+a)^2-c^2=(x+a-c)(x+a+c),$$

and partial fractions give

$$\frac{1}{(x+a-c)(x+a+c)}=\frac{1}{2c}\left(\frac{1}{x+a-c}-\frac{1}{x+a+c}\right).$$

The roots of the quadratic are $-a\pm c$. Their product is 1 and their sum is $-2a$.

Suppose first that $a>1$. Then $a>c$ (as $a^2>a^2-1$), so both $x+a-c$ and $x+a+c$ are positive for $x\geq 0$. Hence

$$I=\frac{1}{2c}\left[\ln\frac{x+a-c}{x+a+c}\right]_0^1=\frac{1}{2c}\left(\ln\frac{1+a-c}{1+a+c}-\ln\frac{a-c}{a+c}\right).$$

Both fractions simplify, using $(a-c)(a+c)=a^2-c^2=1$:

- $(1+a-c)(a+c)=(a+c)+(a-c)(a+c)=1+a+c$, so $\dfrac{1+a-c}{1+a+c}=\dfrac{1}{a+c}$;
- $a-c=\dfrac{1}{a+c}$, so $\dfrac{a-c}{a+c}=\dfrac{1}{(a+c)^2}$.

Therefore

$$I=\frac{1}{2c}\left(-\ln(a+c)+2\ln(a+c)\right)=\frac{1}{2\sqrt{a^2-1}}\ln\left(a+\sqrt{a^2-1}\right),$$

which is the stated result, since $a+\sqrt{a^2-1}>0$ here and so equals its modulus.

What about $a<-1$? Then the roots $-a\pm c=|a|\pm c$ are both positive with product 1, so the smaller one, $|a|-\sqrt{a^2-1}$, lies strictly between 0 and 1. The integrand has a simple pole inside $[0,1]$, behaving like a constant times $1/(x-r)$ near the root r. So

the integral does not converge, and strictly the formula holds only for $a > 1$. (The same partial-fraction calculation with $\ln|\cdot|$ shows that the formula gives the *Cauchy principal value*, the limit of the integral with a small symmetric gap around r removed. That explains the modulus signs in the question.)

Checks

- $a = 0$: the integral is $\int_0^1 \frac{\mathrm{d}x}{1+x^2} = \frac{\pi}{4}$, and the formula gives $\tan^{-1} 1 = \frac{\pi}{4}$.
- $a = 2$: Simpson's rule gives 0.380173, matching $\frac{1}{2\sqrt{3}} \ln(2+\sqrt{3})$.
- As $a \to 1$ both branches tend to $\frac{1}{2} = \int_0^1 (1+x)^{-2}\,\mathrm{d}x$. For the second, $\ln(a+c) \approx c$ when c is small; for the first, $\tan^{-1} t \approx t$ and $\sqrt{(1-a)/(1+a)}/\sqrt{1-a^2} = 1/(1+a)$.

Common slips

- Writing $\tan^{-1} A - \tan^{-1} B = \tan^{-1} \frac{A-B}{1+AB}$ without checking that the difference lies in $(-\frac{\pi}{2}, \frac{\pi}{2})$.
- Forgetting that for $a < -1$ the denominator vanishes inside $[0,1]$.

Beyond STEP

The two answers are really one function of a. Writing $a = \cos\varphi$ in the first and $a = \cosh t$ in the second gives $\varphi/(2\sin\varphi)$ and $t/(2\sinh t)$, and with complex numbers ($\cosh t = \cos(\mathrm{i}t)$) one is the analytic continuation of the other. The divergence for $a < -1$ and the rescue by principal values are a first taste of how complex analysis handles poles on the path of integration. Needham's *Visual Complex Analysis* is a beautiful place to see this.

WORKED SOLUTION

STEP I 1996, Question 5

Pure • Complex Square Roots

> **STEP I 1996, Question 5**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/96-s1-q5

The key idea

Each part feeds the next. Part (i) is about matching rational and irrational parts, and it tells us that $4-2\sqrt{3}$ is a perfect square. Part (ii) asks for a complex square root, and after the usual "compare real and imaginary parts" step it collapses onto part (i). Part (iii) is a quadratic whose discriminant turns out to be four times the number in part (ii), so the square root we need is already sitting there.

Solution

Part (i)

Expanding,

$$(r+s\sqrt{3})^2 = (r^2+3s^2)+2rs\sqrt{3}.$$

We want this to equal $4-2\sqrt{3}$. Since r and s are rational, so are r^2+3s^2 and $2rs$. If $a+b\sqrt{3}=c+d\sqrt{3}$ with a,b,c,d rational and $b\neq d$, then $\sqrt{3}=(a-c)/(d-b)$ would be rational, which it isn't. So we may compare the rational parts and the coefficients of $\sqrt{3}$:

$$r^2+3s^2=4, \qquad rs=-1.$$

Then $r\neq 0$, so $s=-1/r$, and the first equation becomes

$$r^2+\frac{3}{r^2}=4 \iff r^4-4r^2+3=0 \iff (r^2-1)(r^2-3)=0.$$

The option $r^2=3$ gives $r=\pm\sqrt{3}$, which is not rational. So $r^2=1$, and the solutions are

$$r=1,\ s=-1 \qquad \text{and} \qquad r=-1,\ s=1,$$

that is, $r+s\sqrt{3}=\pm(1-\sqrt{3})$. Indeed $(1-\sqrt{3})^2=1-2\sqrt{3}+3=4-2\sqrt{3}$.

Part (ii)

Expanding $(p+q\mathrm{i})^2 = (p^2-q^2)+2pq\,\mathrm{i}$. Since p and q are real we can compare real and imaginary parts:

$$p^2-q^2 = 3-2\sqrt{3}, \qquad pq = 1-\sqrt{3}.$$

The second equation shows $p \neq 0$, so $q = (1-\sqrt{3})/p$ and $q^2 = (1-\sqrt{3})^2/p^2 = (4-2\sqrt{3})/p^2$. Substituting into the first equation and multiplying by p^2,

$$p^4 - (3-2\sqrt{3})\,p^2 - (4-2\sqrt{3}) = 0.$$

Ask yourself This is a quadratic in p^2 with nasty-looking coefficients. Is there a quick way to factorise it?

The two roots for p^2 must have sum $3-2\sqrt{3}$ and product $-(4-2\sqrt{3})$. Try -1: the other root would then be $4-2\sqrt{3}$, and indeed $-1+(4-2\sqrt{3}) = 3-2\sqrt{3}$. So

$$\left(p^2+1\right)\left(p^2-(4-2\sqrt{3})\right) = 0.$$

Since p is real, $p^2 \neq -1$, so $p^2 = 4-2\sqrt{3} = (\sqrt{3}-1)^2$ by part (i). Hence $p = \pm(\sqrt{3}-1)$, and $q = (1-\sqrt{3})/p = \mp 1$. The solutions are

$$p = \sqrt{3}-1,\ q=-1 \qquad \text{and} \qquad p = 1-\sqrt{3},\ q = 1.$$

In other words, the square roots of $(3-2\sqrt{3})+2(1-\sqrt{3})\mathrm{i}$ are $\pm\left((\sqrt{3}-1)-\mathrm{i}\right)$. As a check, $\left((\sqrt{3}-1)-\mathrm{i}\right)^2 = (4-2\sqrt{3})-1-2(\sqrt{3}-1)\mathrm{i}$, which is the required number.

Part (iii)

By the quadratic formula (which holds for complex coefficients, because completing the square works just as well over $\mathbb{C}$),

$$z = \frac{2 \pm \sqrt{4-4(1+\mathrm{i})(2\sqrt{3}-2)}}{2(1+\mathrm{i})} = \frac{1 \pm \sqrt{1-(1+\mathrm{i})(2\sqrt{3}-2)}}{1+\mathrm{i}},$$

where $\pm\sqrt{\ }$ stands for the two complex square roots. The quantity under the root is

$$1-(2\sqrt{3}-2)-(2\sqrt{3}-2)\mathrm{i} = (3-2\sqrt{3})+2(1-\sqrt{3})\mathrm{i},$$

which is exactly the number in part (ii). Its square roots are $\pm\left((\sqrt{3}-1)-\mathrm{i}\right)$, so

$$z = \frac{1 \pm \left((\sqrt{3}-1)-\mathrm{i}\right)}{1+\mathrm{i}}, \quad \text{giving} \quad z_1 = \frac{\sqrt{3}-\mathrm{i}}{1+\mathrm{i}}, \qquad z_2 = \frac{(2-\sqrt{3})+\mathrm{i}}{1+\mathrm{i}}.$$

Multiplying top and bottom by $1-\mathrm{i}$ (so the denominator becomes 2):

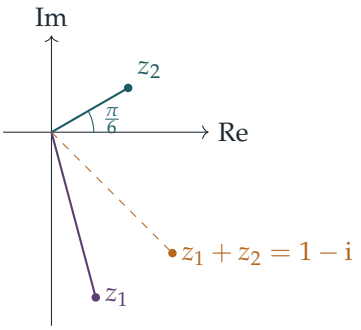
$$z_1 = \frac{(\sqrt{3}-\mathrm{i})(1-\mathrm{i})}{2} = \frac{(\sqrt{3}-1)-(\sqrt{3}+1)\mathrm{i}}{2},$$
$$z_2 = \frac{\left((2-\sqrt{3})+\mathrm{i}\right)(1-\mathrm{i})}{2} = \frac{(3-\sqrt{3})+(\sqrt{3}-1)\mathrm{i}}{2}.$$

Ask yourself Is this "as simple a form as possible"? Can either root be tidied further?

The second root factorises: $(3-\sqrt{3})+(\sqrt{3}-1)\mathrm{i} = (\sqrt{3}-1)(\sqrt{3}+\mathrm{i})$, and $\sqrt{3}+\mathrm{i} = 2\mathrm{e}^{\mathrm{i}\pi/6}$. For the first, $|z_1|^2 = \frac{1}{4}\big((4-2\sqrt{3})+(4+2\sqrt{3})\big) = 2$, and $z_1 = \frac{\sqrt{3}-\mathrm{i}}{1+\mathrm{i}}$ is a quotient of $2\mathrm{e}^{-\mathrm{i}\pi/6}$ by $\sqrt{2}\,\mathrm{e}^{\mathrm{i}\pi/4}$. So the solutions are

$$z_1 = \tfrac{1}{2}\big((\sqrt{3}-1)-(\sqrt{3}+1)\mathrm{i}\big) = \sqrt{2}\,\mathrm{e}^{-5\pi\mathrm{i}/12},$$
$$z_2 = \tfrac{1}{2}(\sqrt{3}-1)(\sqrt{3}+\mathrm{i}) = (\sqrt{3}-1)\,\mathrm{e}^{\pi\mathrm{i}/6}.$$

Either form (Cartesian or modulus–argument) is a perfectly good final answer.



Checks

- Sum and product of roots: for $az^2+bz+c=0$ we need $z_1+z_2 = -b/a = 2/(1+\mathrm{i}) = 1-\mathrm{i}$ and $z_1z_2 = c/a = (2\sqrt{3}-2)/(1+\mathrm{i}) = (\sqrt{3}-1)(1-\mathrm{i})$. From the Cartesian forms, $z_1+z_2 = \frac{1}{2}(2-2\mathrm{i}) = 1-\mathrm{i}$; from the polar forms, $z_1z_2 = \sqrt{2}(\sqrt{3}-1)\mathrm{e}^{-\mathrm{i}\pi/4} = (\sqrt{3}-1)(1-\mathrm{i})$. Both agree.
- Numerically, $z_1 \approx 0.366 - 1.366\mathrm{i}$ and $z_2 \approx 0.634 + 0.366\mathrm{i}$, and substituting either into $(1+\mathrm{i})z^2 - 2z + 2\sqrt{3} - 2$ gives 0 to machine precision.

Common slips

- In part (i), forgetting to say *why* we may compare coefficients of $\sqrt{3}$: it relies on r, s being rational and $\sqrt{3}$ irrational. Without it, there are infinitely many real solutions.
- In part (ii), keeping $p^2 = -1$, or pairing the signs of p and q wrongly: $pq = 1-\sqrt{3} < 0$ forces opposite signs.
- In part (iii), dividing the whole formula by 2 incorrectly, or leaving $1+\mathrm{i}$ in the denominator rather than realising it.

Beyond STEP

Part (i) works because 1 and $\sqrt{3}$ are linearly independent over $\mathbb{Q}$, so $\mathbb{Q}(\sqrt{3}) = \{r + s\sqrt{3}\}$ behaves like a two-dimensional space over the rationals; part (ii) uses the same idea with 1 and i over $\mathbb{R}$. This is the start of field extensions, which a first-year algebra course develops towards Galois theory. Allenby's *Rings, Fields and Groups* covers this, and Needham's *Visual Complex Analysis* gives a beautiful geometric view of complex square roots.

WORKED SOLUTION

STEP I 1996, Question 6

Pure • Trigonometric Sums and Integrals

STEP I 1996, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q6

The key idea

The given identity is built to telescope: if we multiply the sum $\sum \cos kx$ by $2\sin\frac{1}{2}x$, each term becomes a difference of two sines, and neighbouring differences cancel. Once f is written as a sum of cosines, the awkward-looking integrals in (ii) become integrals of single cosines, most of which vanish over $[0, \pi]$.

Solution

Throughout, n is a positive integer.

Part (i)

Put $S = \sum_{k=1}^{n} \cos kx$. Multiplying by $2\sin\frac{1}{2}x$ and using the given formula term by term,

$$\begin{aligned} 2\sin\tfrac{1}{2}x\, S &= \sum_{k=1}^{n} \Big(\sin(k+\tfrac{1}{2})x - \sin(k-\tfrac{1}{2})x\Big) \\ &= \big(\sin\tfrac{3}{2}x - \sin\tfrac{1}{2}x\big) + \big(\sin\tfrac{5}{2}x - \sin\tfrac{3}{2}x\big) + \cdots + \big(\sin(n+\tfrac{1}{2})x - \sin(n-\tfrac{1}{2})x\big) \\ &= \sin(n+\tfrac{1}{2})x - \sin\tfrac{1}{2}x, \end{aligned}$$

since every intermediate term appears once with each sign and cancels. Hence

$$\sin(n+\tfrac{1}{2})x = \sin\tfrac{1}{2}x + 2\sin\tfrac{1}{2}x \sum_{k=1}^{n} \cos kx.$$

For $0 < x \le \pi$ we have $0 < \frac{1}{2}x \le \frac{1}{2}\pi$, so $\sin\frac{1}{2}x > 0$ and we may divide by it:

$$\mathrm{f}(x) = \frac{\sin(n+\frac{1}{2})x}{\sin\frac{1}{2}x} = 1 + 2\sum_{k=1}^{n} \cos kx,$$

as required.

Part (ii)

Ask yourself f isn't defined at $x = 0$. Is $\int_0^\pi f(x)\,dx$ even a sensible integral?

Yes. By part (i), on $0 < x \leq \pi$ the function f agrees with the cosine polynomial $1 + 2\sum \cos kx$, which is continuous on all of $[0, \pi]$ (with value $2n + 1$ at 0). Changing or leaving undefined the value at a single point doesn't affect an integral, so we may integrate the cosine sum instead.

First integral. For each integer $k \geq 1$,

$$\int_0^\pi \cos kx\,dx = \left[\frac{\sin kx}{k}\right]_0^\pi = 0,$$

because $\sin k\pi = 0$. So only the constant term survives:

$$\int_0^\pi f(x)\,dx = \int_0^\pi 1\,dx + 2\sum_{k=1}^{n} \int_0^\pi \cos kx\,dx = \pi.$$

Second integral. Now

$$\int_0^\pi f(x)\cos x\,dx = \int_0^\pi \cos x\,dx + 2\sum_{k=1}^{n} \int_0^\pi \cos kx \cos x\,dx.$$

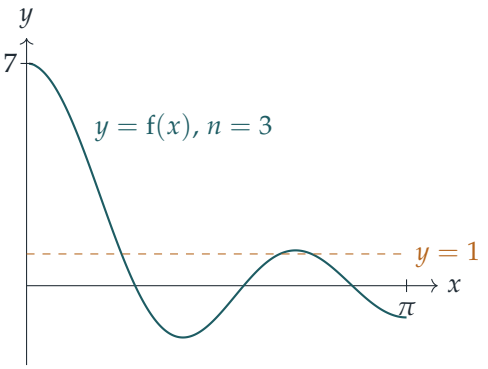
The first integral is $[\sin x]_0^\pi = 0$. For the others, use $2\cos kx \cos x = \cos(k+1)x + \cos(k-1)x$:

$$2\int_0^\pi \cos kx \cos x\,dx = \int_0^\pi \cos(k+1)x\,dx + \int_0^\pi \cos(k-1)x\,dx.$$

Here $k + 1 \geq 2$, so the first integral is 0 as above. The second is also 0 unless $k - 1 = 0$, when the integrand is 1 and the integral is π. So only the $k = 1$ term contributes, and

$$\int_0^\pi f(x)\cos x\,dx = \pi.$$

(If n were allowed to be 0, then $f \equiv 1$ and this integral would be 0; for every $n \geq 1$ it is π.)



The picture shows why the first answer doesn't depend on n: the tall spike near 0 (height $2n+1$) is balanced by the oscillations, and the net area is always that of the constant term, π.

Checks

- As $x \to 0$, $\sin(n+\frac{1}{2})x / \sin\frac{1}{2}x \to (2n+1)$, and the cosine sum at $x=0$ is $1+2n$. They agree.
- At $x=\pi$: $\mathrm{f}(\pi) = \sin(n+\frac{1}{2})\pi = (-1)^n$, and $1 + 2\sum_{k=1}^{n}(-1)^k = (-1)^n$ too.
- Midpoint-rule quadrature of the original quotient for $n = 1,\ldots,7$ gives $3.14159265\ldots$ for both integrals, matching π.

Common slips

- Forgetting to justify dividing by $\sin\frac{1}{2}x$: it's nonzero precisely because $0 < x \leq \pi$.
- Writing $\int_0^\pi \cos kx \cos x \,\mathrm{d}x = 0$ for all k, and so getting 0 for the second integral; the $k=1$ term gives $\int_0^\pi \cos^2 x \,\mathrm{d}x = \frac{1}{2}\pi$.

Beyond STEP

The function f is the *Dirichlet kernel*, the central object in the theory of Fourier series. The calculations in (ii) are instances of orthogonality: $\cos kx$ and $\cos mx$ integrate to zero against each other over $[0,\pi]$ unless $k=m$. That's exactly how Fourier coefficients are extracted, and integrating a function against the Dirichlet kernel gives the nth partial sum of its Fourier series. For a careful treatment of when those partial sums converge, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP I 1996, Question 7

Pure • First-Order Differential Equations

STEP I 1996, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q7

The key idea

"Flows out at a rate proportional to the amount" translates directly into a first-order differential equation, $\frac{\mathrm{d}y}{\mathrm{d}t} = -ky$ with $k > 0$, whose solution is an exponential. Writing e^{-k} as a single number b turns the exponential into b^t, and the same substitution $k = -\ln b$ rewrites the equation in part (ii). That equation is linear, so an integrating factor finishes it.

Solution

Part (i)

The water flows *out*, so y decreases at a rate proportional to y:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = -ky \qquad \text{for some constant } k > 0.$$

While there is water in the tank, $y > 0$, so we may separate the variables:

$$\int \frac{\mathrm{d}y}{y} = -\int k\,\mathrm{d}t \quad \Longrightarrow \quad \ln y = -kt + c.$$

At $t = 0$ we have $y = 1$, so $c = \ln 1 = 0$. Hence

$$y = \mathrm{e}^{-kt} = \left(\mathrm{e}^{-k}\right)^t = b^t, \qquad \text{where } b = \mathrm{e}^{-k}.$$

Since $k > 0$, we have $0 < b < 1$; in particular $b < 1$, as required. Note for later that

$$\ln b = -k, \qquad \text{so} \qquad k = -\ln b > 0.$$

Part (ii)

The outflow is exactly as before, at rate ky with the same constant k, and in addition water comes in at rate a. So the net rate of change is

$$\frac{\mathrm{d}y}{\mathrm{d}t} = a - ky.$$

? **Ask yourself** Where does $\ln b$ come from, when this equation only involves k?

The constant b was *defined* in part (i) by $b = \mathrm{e}^{-k}$, so it describes the outflow and nothing else; it is still the same number here. Substituting $k = -\ln b$ gives $\frac{\mathrm{d}y}{\mathrm{d}t} = a + y\ln b$, that is,

$$\frac{\mathrm{d}y}{\mathrm{d}t} - y\ln b = a,$$

which is the required equation.

This is linear in y with constant coefficients. The integrating factor is

$$\exp\left(-\int \ln b \,\mathrm{d}t\right) = \mathrm{e}^{-t\ln b} = b^{-t}.$$

Multiplying through by b^{-t}, the left-hand side becomes an exact derivative, because $\frac{\mathrm{d}}{\mathrm{d}t}b^{-t} = -b^{-t}\ln b$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(y\,b^{-t}\right) = a\,b^{-t}.$$

Integrating, and using $\int b^{-t}\,\mathrm{d}t = -\dfrac{b^{-t}}{\ln b}$ (valid since $\ln b \neq 0$),

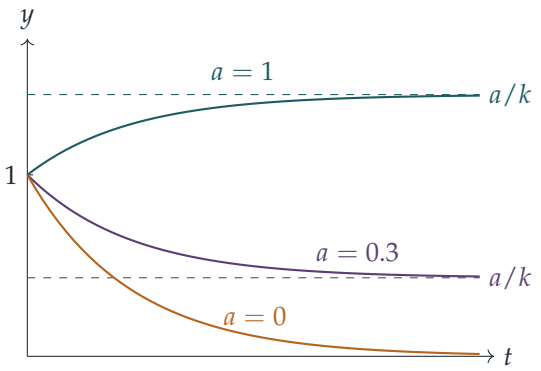
$$y\,b^{-t} = -\frac{a}{\ln b}\,b^{-t} + C.$$

At $t = 0$, $y = 1$, so $1 = -\dfrac{a}{\ln b} + C$, giving $C = 1 + \dfrac{a}{\ln b}$. Multiplying through by b^t,

$$\begin{aligned} y &= -\frac{a}{\ln b} + \left(1 + \frac{a}{\ln b}\right)b^t \\ &= b^t + \frac{a\,(1 - b^t)}{-\ln b}. \end{aligned}$$

Both forms are fine. The second is easier to read, because $-\ln b = \ln(1/b) > 0$: it is the part (i) solution b^t plus a positive term coming from the inflow.

As $t \to \infty$, $b^t \to 0$ (since $0 < b < 1$), so $y \to \dfrac{a}{-\ln b} = \dfrac{a}{k}$. This is the equilibrium level, at which the outflow ky exactly balances the inflow a. The tank fills towards this level if $a > k$ and drains towards it if $a < k$.



The sketch shows $b = \frac{1}{2}$ (so $k = \ln 2 \approx 0.693$) with $a = 1$, $a = 0.3$ and, for comparison, the part (i) curve $a = 0$.

Checks

- Put $a = 0$: the answer reduces to $y = b^t$, the part (i) result.
- At $t = 0$: $y = 1 + \frac{a(1-1)}{-\ln b} = 1$.
- Differentiate $y = b^t + \frac{a}{\ln b}(b^t - 1)$: we get $y' = b^t \ln b + ab^t$, while $y \ln b + a = b^t \ln b + a(b^t - 1) + a = b^t \ln b + ab^t$, which agrees.
- Numerically, with $b = \frac{1}{2}$, $a = 1$, $t = 2$, the formula gives $y \approx 1.3320$; stepping the differential equation with a Runge–Kutta method gives the same value, and in general agreement to about 10^{-13} for random a, b, t.

Common slips

- Writing $\frac{dy}{dt} = ky$ and then forgetting that k must be negative for an outflow; the sign is what forces $b < 1$.
- Treating b as a new unknown in part (ii). It is fixed by the outflow alone, so $\ln b = -k$ carries straight over.
- Getting the sign wrong in $\int b^{-t}\,dt = -b^{-t}/\ln b$; check by differentiating.

Beyond STEP

This is the simplest compartment model: one tank with a constant input and a proportional output. The same equation describes a drug infused into the bloodstream at a steady rate and cleared by the kidneys, or a capacitor charging through a resistor. Every solution relaxes exponentially to the equilibrium a/k, which is an example of a stable fixed point, an idea that becomes central when you meet systems of differential

equations. Boyce and DiPrima, *Elementary Differential Equations*, treats linear first-order equations and many such mixing problems.

WORKED SOLUTION

STEP I 1996, Question 8

Pure • Recurring Decimals

STEP I 1996, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q8

The key idea

A recurring block of digits is a geometric series in disguise: each repeat of the block is the previous one shifted k places to the right, so it is multiplied by 10^{-k}. Part (i) is the simplest case. In part (ii) we split the expansion into the part before the repetition starts (a finite decimal) and the recurring tail (a geometric series), and both pieces turn out to be integers divided by powers of ten or by $10^k - 1$.

Solution

Part (i)

The block 38 repeats every two places, so

$$\begin{aligned} 0.383838\ldots &= \frac{38}{100} + \frac{38}{100^2} + \frac{38}{100^3} + \cdots \\ &= \frac{38/100}{1 - 1/100} = \frac{38}{99}, \end{aligned}$$

summing the geometric series with ratio $\frac{1}{100}$, which is allowed because $\left|\frac{1}{100}\right| < 1$. Since $38 = 2 \times 19$ and $99 = 3^2 \times 11$ share no prime factor, this is in lowest terms:

$$0.383838\ldots = \frac{38}{99}.$$

Part (ii)

(The printed expansion $0 \cdot a_1 a_2 a_2 \cdots$ is a misprint for $0 \cdot a_1 a_2 a_3 \cdots$.) Each a_n is a digit from 0 to 9 and

$$x = \sum_{n=1}^{\infty} \frac{a_n}{10^n}.$$

We split this sum at $n = N$.

The first N digits. Let b be the integer whose decimal digits are $a_1 a_2 \ldots a_N$ (leading zeros allowed), that is,

$$b = \sum_{n=1}^{N} a_n 10^{N-n}.$$

Then

$$\sum_{n=1}^{N} \frac{a_n}{10^n} = \frac{1}{10^N} \sum_{n=1}^{N} a_n 10^{N-n} = \frac{b}{10^N}.$$

The tail. The condition $a_{n+k} = a_n$ for every $n > N$, applied repeatedly, gives $a_{N+mk+j} = a_{N+j}$ for every $m \geq 0$ and $1 \leq j \leq k$ (induction on m: the index $N + (m-1)k + j$ exceeds N, so we may add k to it). So after the Nth place the expansion consists of the block $a_{N+1} a_{N+2} \ldots a_{N+k}$ repeated for ever. Let c be the integer with these digits,

$$c = \sum_{j=1}^{k} a_{N+j} 10^{k-j}.$$

Group the tail into blocks of k terms; this is allowed because all the terms are non-negative and the series converges. The block for a given $m \geq 0$ is

$$\sum_{j=1}^{k} \frac{a_{N+mk+j}}{10^{N+mk+j}} = \frac{1}{10^{N+(m+1)k}} \sum_{j=1}^{k} a_{N+j} 10^{k-j} = \frac{c}{10^N} \left(\frac{1}{10^k} \right)^{m+1}.$$

? **Ask yourself** Why is each block exactly 10^{-k} times the one before it?

Moving from block m to block $m+1$ uses the same digits (by periodicity), but each sits k places further to the right. That divides each term by 10^k. This is the geometric series hidden in the recurring decimal.

Summing the geometric series with ratio $10^{-k} < 1$,

$$\sum_{n=N+1}^{\infty} \frac{a_n}{10^n} = \frac{c}{10^N} \cdot \frac{10^{-k}}{1 - 10^{-k}} = \frac{c}{10^N (10^k - 1)}.$$

Putting the pieces together. Adding the two parts,

$$x = \frac{b}{10^N} + \frac{c}{10^N (10^k - 1)},$$

where b is the integer formed by the first N digits $a_1 \ldots a_N$ and c is the integer formed by one period $a_{N+1} \ldots a_{N+k}$. This is the required form.

Rationality. Over a common denominator,

$$x = \frac{b(10^k - 1) + c}{10^N (10^k - 1)}.$$

The numerator is an integer, and the denominator is a positive integer because $k \geq 1$ gives $10^k - 1 \geq 9$. So x is a ratio of integers, which means x is rational.

Checks

- Part (i) is the case with no pre-period and $k = 2$, $c = 38$: the formula gives $38/(10^2 - 1) = 38/99$. (With $N = 1$ you get $b = 3$ and $c = 83$, and $\frac{3}{10} + \frac{83}{990} = \frac{380}{990} = \frac{38}{99}$, which agrees.)
- Take $0.12345345\ldots$, with $N = 2$, $k = 3$, $b = 12$, $c = 345$: $\frac{12}{100} + \frac{345}{100 \times 999} = \frac{4111}{33300} = 0.1234534534\ldots$
- We also compared the formula with the summed expansion, in exact arithmetic, for 500 random choices of pre-period and repeating block; all agreed.

Common slips

- Forgetting the factor 10^{-N} on the tail: the repeating block starts at place $N + 1$, not at place 1.
- Using 10^k instead of $10^k - 1$ in the denominator: the first term of the geometric series is $c\,10^{-N-k}$, not $c\,10^{-N}$.
- Not checking lowest terms in (i): say why $\gcd(38, 99) = 1$.

Beyond STEP

The converse is also true: every rational number has an eventually recurring decimal expansion, because long division by q has only q possible remainders, so one must repeat. So the irrationals are exactly the reals whose expansions never settle into a cycle. The length of the period of $1/q$ is linked to the order of 10 modulo q, which is a first taste of group theory in number theory. Davenport's *The Higher Arithmetic* develops these ideas about congruences and periods beautifully.

WORKED SOLUTION

STEP I 1996, Question 9

Mechanics • Elastic Strings and Energy

STEP I 1996, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q9

The key idea

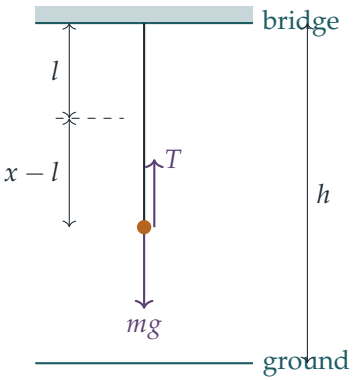
Everything here is energy. Gravity and the rope's tension are the only forces doing work, and both are conservative, so we never need the equation of motion in full. "Only just avoids the ground" means her speed is zero exactly when she reaches the ground. The maximum speed comes where the acceleration changes sign, which is where the tension balances her weight.

Solution

We model the jumper as a particle starting from rest at the bridge, the rope as light and obeying Hooke's law, and we ignore air resistance. Let x be the distance she has fallen. For $x \leq l$ the rope is slack and she is in free fall. For $x > l$ the extension is $x - l$, and Hooke's law gives tension

$$T = \frac{mg}{k} \cdot \frac{x - l}{l},$$

with elastic energy $\dfrac{mg}{k} \cdot \dfrac{(x-l)^2}{2l}$ stored in the rope.



The height of the bridge

At the lowest point she is momentarily at rest. She only just avoids the ground, so this lowest point is the ground itself, at $x = h$. Clearly $h > l$: if the rope never went taut she would be in free fall and would hit the ground at speed. Between leaving the bridge and the lowest point the kinetic energy starts and ends at zero, so the gravitational energy lost all goes into the rope:

$$mgh = \frac{mg}{k} \cdot \frac{(h-l)^2}{2l}.$$

Multiplying by $2kl/(mg)$ gives $2klh = (h-l)^2 = h^2 - 2hl + l^2$, so

$$h^2 - 2hl(k+1) + l^2 = 0,$$

as required. By the quadratic formula,

$$h = l(k+1) \pm l\sqrt{(k+1)^2 - 1} = l\left(k + 1 \pm \sqrt{k^2 + 2k}\right).$$

Ask yourself Which root do we want, and why does the other one appear at all?

Since $k > 0$ we have $\sqrt{k^2+2k} > \sqrt{k^2} = k$, so the minus sign gives $h < l(k+1-k) = l$. But the energy equation assumed the rope was stretched by $h - l$, which only makes sense for $h > l$. The spurious root comes from squaring: it gives the same value of $(h-l)^2$ for a "compression" the rope can't actually provide. (The two roots multiply to l^2, so one is above l and the other below.) Hence

$$h = l\left(k + 1 + \sqrt{k^2 + 2k}\right).$$

The maximum speed

While the rope is slack she accelerates downwards at g. Once it is taut, Newton's second law downwards gives

$$m\ddot{x} = mg - \frac{mg}{k} \cdot \frac{x-l}{l}.$$

This is positive while $x - l < kl$ and negative after that. So she speeds up until the extension is kl, the point where the tension equals her weight, and slows down from then on. Her maximum speed is therefore reached at $x = l + kl$. By conservation of energy from the bridge to this point,

$$\begin{aligned} \tfrac{1}{2}mv^2 &= mg(l+kl) - \frac{mg}{k} \cdot \frac{(kl)^2}{2l} \\ &= mgl(1+k) - \tfrac{1}{2}mgkl. \end{aligned}$$

Doubling and dividing by m,

$$v^2 = 2gl + 2kgl - kgl = (k+2)gl,$$

as required.

Checks

- As $k \to 0$ the rope becomes very stiff. Then $h \to l$ (she barely stretches it), and $v^2 \to 2gl$, the free-fall speed after a drop of l, which is right since the rope stops her almost at once.
- Stepping the equation of motion numerically for random k and l gives a lowest point and maximum speed agreeing with the formulae to five decimal places. For example, $k = 1.5$, $l = 10\,\text{m}$, $g = 9.8\,\text{m}\,\text{s}^{-2}$ gives $h = 10(2.5 + \sqrt{5.25}) \approx 47.9\,\text{m}$ and $v^2 = 343$, so $v \approx 18.5\,\text{m}\,\text{s}^{-1}$.

Common slips

- Taking the maximum speed at the moment the rope goes taut. She is still accelerating then, because the tension is zero.
- Using the full drop x instead of the extension $x - l$ in the elastic energy.
- Keeping both roots for h without checking that $h > l$.

Beyond STEP

Once the rope is taut the motion is simple harmonic about the equilibrium extension kl, so the second half of the fall is part of a sine curve with angular frequency $\sqrt{g/(kl)}$. Piecewise systems like this, which switch between free fall and an oscillator, are a nice introduction to how energy methods and phase-plane pictures handle non-smooth forces. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops energy conservation and oscillations carefully at this level.

WORKED SOLUTION

STEP I 1996, Question 10

Mechanics • Conservation of Momentum

STEP I 1996, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q10

The key idea

There are two conservation statements to use: momentum is conserved (it was zero before the explosion), and the kinetic energy afterwards is E. Together they fix both speeds once the split of the mass is known, so the relative speed becomes a function of how the mass is shared. Minimising it then comes down to a single inequality, and AM–GM is the natural tool because $m_1 + m_2$ is fixed.

Solution

We model the two parts as particles, assume no mass is lost in the explosion, so $m_1 + m_2 = M$, and assume no external forces act during it.



Momentum. The spaceship starts at rest, so the total momentum is zero before and therefore after the explosion. The two momenta $m_1\mathbf{v}_1$ and $m_2\mathbf{v}_2$ must cancel, so the parts move in exactly opposite directions along a line, and

$$m_1 v_1 = m_2 v_2 = p,$$

say, where $p > 0$ is the common magnitude of momentum. Because the parts move in opposite directions, their relative speed really is $v_1 + v_2$.

Energy. All of the kinetic energy afterwards comes from the explosion, so

$$E = \tfrac{1}{2}m_1 v_1^2 + \tfrac{1}{2}m_2 v_2^2 = \frac{p^2}{2m_1} + \frac{p^2}{2m_2} = \frac{p^2(m_1+m_2)}{2m_1 m_2} = \frac{p^2 M}{2m_1 m_2}.$$

Hence $p^2 = \dfrac{2Em_1m_2}{M}$.

Relative speed. Using $v_1 = p/m_1$ and $v_2 = p/m_2$,

$$v_1 + v_2 = p\left(\frac{1}{m_1} + \frac{1}{m_2}\right) = \frac{pM}{m_1m_2},$$

so

$$(v_1 + v_2)^2 = \frac{p^2M^2}{m_1^2m_2^2} = \frac{2Em_1m_2}{M} \cdot \frac{M^2}{m_1^2m_2^2} = \frac{2EM}{m_1m_2}.$$

Ask yourself E and M are fixed. What is the only thing left that we are free to vary, and which way do we want it to go?

Only the split of the mass can change, and to make $v_1 + v_2$ as small as possible we need the product m_1m_2 as large as possible, subject to $m_1 + m_2 = M$. By the AM–GM inequality (both masses are positive),

$$m_1m_2 \le \left(\frac{m_1 + m_2}{2}\right)^2 = \frac{M^2}{4},$$

with equality exactly when $m_1 = m_2 = \frac{1}{2}M$. (Equivalently, $m_1m_2 = \frac{1}{4}M^2 - \frac{1}{4}(m_1 - m_2)^2$.) Therefore

$$(v_1 + v_2)^2 = \frac{2EM}{m_1m_2} \ge \frac{2EM}{M^2/4} = \frac{8E}{M},$$

and since speeds are non-negative we may take square roots:

$$v_1 + v_2 \ge \left(\frac{8E}{M}\right)^{1/2}.$$

The bound is attained when the spaceship splits into two equal halves, each then moving at speed $\sqrt{2E/M}$. So the minimum possible relative speed is $(8E/M)^{1/2}$, as required.

Checks

- Equal halves: each has kinetic energy $\frac{1}{2}E = \frac{1}{2} \cdot \frac{1}{2}Mv^2$, so $v = \sqrt{2E/M}$ and $2v = \sqrt{8E/M}$, agreeing with the bound.
- Extreme split: as $m_1 \to 0$, $m_1m_2 \to 0$ and the relative speed blows up, which makes sense, because a tiny fragment carrying a share of the energy must move very fast.
- Units: E/M has units of $\mathrm{m^2\,s^{-2}}$, so its square root is a speed.
- Numerically, with $M = 7$ and $E = 3$, scanning m_1 over $(0, 7)$ gives a least relative speed of $1.85164\ldots$ at $m_1 = 3.5$, matching $\sqrt{24/7} = 1.85164\ldots$.

Common slips

- Forgetting to say why the parts move in opposite directions; that is what makes the relative speed $v_1 + v_2$ rather than $|v_1 - v_2|$.
- Minimising over m_1 without using $m_1 + m_2 = M$; the constraint is what makes the problem one-dimensional.
- Showing only that $(8E/M)^{1/2}$ is a lower bound, without noting it is actually attained (by $m_1 = m_2$).

Beyond STEP

This is the simplest example of working in the centre-of-mass frame: in that frame a two-body break-up always produces equal and opposite momenta, and the kinetic energy is $p^2/(2\mu)$, where $\mu = m_1 m_2/M$ is the *reduced mass*. Here $(v_1 + v_2)^2 = 2E/\mu$, and the reduced mass is largest when the masses are equal. The same idea underlies nuclear decays and two-body orbits; Kleppner and Kolenkow, *An Introduction to Mechanics*, develops it carefully.

WORKED SOLUTION

STEP I 1996, Question 11

Mechanics • Projectile Motion

STEP I 1996, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

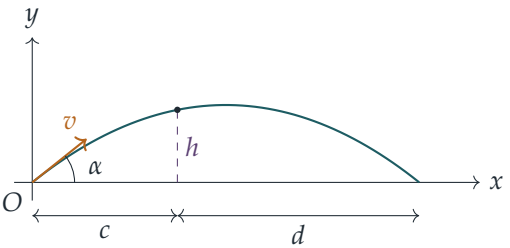
step.maths.org/questions/96-s1-q11

The key idea

The path of a projectile is a parabola, and we know both places where it meets the ground: $x = 0$ and $x = c + d$. A quadratic is pinned down by its roots up to one constant, so the trajectory must be $y = kx(c + d - x)$ for some k. The point (c, h) fixes k, and comparing with the standard trajectory equation turns k into $\tan\alpha$ and then into v^2.

Solution

We model the particle as a point mass moving freely under gravity alone (no air resistance), and take axes with O at the origin, x horizontal in the direction of motion and y vertically upwards. Write $R = c + d$ for the range and $T = \tan\alpha$.



The angle of projection

With initial velocity components $v\cos\alpha$ and $v\sin\alpha$, the position after time t is $x = vt\cos\alpha$, $y = vt\sin\alpha - \frac{1}{2}gt^2$. Eliminating $t = x/(v\cos\alpha)$ gives the trajectory

$$y = x\tan\alpha - \frac{gx^2}{2v^2\cos^2\alpha}.$$

The particle lands at $x = R$, so $y = 0$ there. Since $R \neq 0$ we may divide by R:

$$\frac{g}{2v^2 \cos^2 \alpha} = \frac{\tan \alpha}{R}. \qquad (*)$$

Substituting this back into the trajectory,

$$y = Tx - \frac{Tx^2}{R} = \frac{T\,x(R-x)}{R}.$$

> **Ask yourself** Why is it worth replacing $g/(2v^2 \cos^2 \alpha)$ before using the point (c, h)?

Because $(*)$ removes v completely, leaving an equation in the single unknown T. Geometrically, we have written the parabola in factorised form with its roots 0 and R showing. Now the point (c, h) lies on the path, and $R - c = d$, so

$$h = \frac{T\,cd}{c+d}, \qquad \text{that is,} \qquad \tan \alpha = \frac{h(c+d)}{cd},$$

as required.

The speed of projection

Rearranging $(*)$ and using $\sec^2 \alpha = 1 + \tan^2 \alpha$,

$$v^2 = \frac{gR}{2T \cos^2 \alpha} = \frac{gR(1+T^2)}{2T} = \frac{g}{2}\left(\frac{R}{T} + RT\right).$$

From the first result, $\dfrac{R}{T} = \dfrac{cd}{h}$ and $RT = \dfrac{(c+d)^2 h}{cd}$. Hence

$$v^2 = \frac{g}{2}\left(\frac{cd}{h} + \frac{(c+d)^2 h}{cd}\right),$$

which completes the question.

Checks

- By the AM–GM inequality, $\frac{R}{T} + RT \geq 2R$, so $v^2 \geq gR$, with equality exactly when $T = 1$. This is the familiar fact that the greatest range for a given speed is v^2/g, achieved at 45°.
- If the point (c, h) is the top of the flight, then $c = d$ and $\tan \alpha = 2h/c$, which matches the standard result that the maximum height is a quarter of the range times $\tan \alpha$.
- Numerically, with $g = 9.81$, $c = 3$, $d = 5$ and $h = 2$ the formulae give $\tan \alpha = \frac{16}{15}$ and $v^2 \approx 78.64$. Stepping the motion forward in small time steps, the particle passes through height 2.000 at $x = 3$ and lands at $x = 8.000$.

Common slips

- Forgetting the $\cos^2\alpha$ in the trajectory equation, or replacing $\sec^2\alpha$ by $1+\tan\alpha$ instead of $1+\tan^2\alpha$.
- Trying to solve the two equations "height h at $x=c$" and "height 0 at $x=c+d$" for v and α directly. This works, but the algebra is much heavier than using $(*)$ to eliminate v first.

Beyond STEP

The trick of writing a quadratic in terms of its roots is a small case of a big idea: a polynomial is determined by its zeros up to a constant, which leads on to interpolation and to the factor theorem over the complex numbers. In mechanics, letting α vary for fixed v gives a family of parabolas whose envelope is the "parabola of safety", the boundary of every point the projectile can reach. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats projectile motion and its extensions (including air resistance) at first-year university level.

WORKED SOLUTION

STEP I 1996, Question 12

Probability and Statistics • Maximum of Random Sample

STEP I 1996, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q12

The key idea

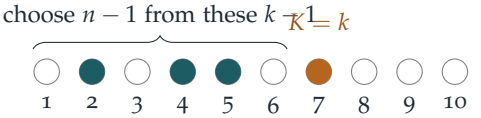
The examiner's choice comes down to choosing a *set* of n different marks from $\{1, 2, \ldots, m\}$, and every such set is equally likely. So probabilities are counts of sets divided by $\binom{m}{n}$. Saying that the largest mark is k pins down one mark and leaves the other $n-1$ to be chosen from below it. The identity then comes free, because probabilities add up to 1. For the expectation, the trick is to turn $k\binom{k-1}{n-1}$ back into a single binomial coefficient so that the same identity can be used again.

Solution

The distribution of K

The examiner gives n distinct marks, and by symmetry every subset of $\{1, \ldots, m\}$ of size n is equally likely to be the set of marks used. (Which script gets which mark has no effect on K.) There are $\binom{m}{n}$ such subsets.

The event $K = k$ happens exactly when the set contains k and its other $n-1$ marks all lie in $\{1, \ldots, k-1\}$. There are $\binom{k-1}{n-1}$ ways to choose those $n-1$ marks. This needs $k - 1 \geq n - 1$, that is $k \geq n$: the largest of n distinct positive integers is at least n. Also $k \leq m$, of course.



(The picture shows $m = 10$, $n = 4$, $k = 7$.) Hence

$$\mathrm{P}(K = k) = \binom{k-1}{n-1} \Big/ \binom{m}{n}, \qquad n \leq k \leq m.$$

The identity

K must take exactly one of the values $n, n+1, \ldots, m$, so these probabilities add up to 1:

$$\sum_{k=n}^{m} \binom{k-1}{n-1} \Big/ \binom{m}{n} = 1 \quad \Longrightarrow \quad \sum_{k=n}^{m} \binom{k-1}{n-1} = \binom{m}{n}.$$

This holds for *all* integers $1 \le n \le m$, which we will need next.

The expected value

By definition,

$$\mathrm{E}(K) = \frac{1}{\binom{m}{n}} \sum_{k=n}^{m} k \binom{k-1}{n-1}.$$

Ask yourself How can we get rid of the factor k so that the identity applies again?

Absorb it into the binomial coefficient:

$$k\binom{k-1}{n-1} = \frac{k\,(k-1)!}{(n-1)!\,(k-n)!} = n \cdot \frac{k!}{n!\,(k-n)!} = n\binom{k}{n}.$$

Now $\binom{k}{n} = \binom{(k+1)-1}{(n+1)-1}$. As k runs from n to m, $j = k+1$ runs from $n+1$ to $m+1$. So the identity, used with $n+1$ and $m+1$ in place of n and m (allowed, since $1 \le n+1 \le m+1$), gives

$$\sum_{k=n}^{m} \binom{k}{n} = \sum_{j=n+1}^{m+1} \binom{j-1}{(n+1)-1} = \binom{m+1}{n+1}.$$

Therefore

$$\begin{aligned} \mathrm{E}(K) &= \frac{n\binom{m+1}{n+1}}{\binom{m}{n}} = n \cdot \frac{(m+1)!}{(n+1)!\,(m-n)!} \cdot \frac{n!\,(m-n)!}{m!} \\ &= \frac{n(m+1)}{n+1}. \end{aligned}$$

Checks

- $n = m$: every mark is used, so $K = m$ for certain, and indeed $\frac{m(m+1)}{m+1} = m$.
- $n = 1$: K is a single mark, uniform on $\{1, \ldots, m\}$, with mean $\frac{m+1}{2}$, which matches.
- $m = 6$, $n = 2$: listing all 15 pairs from $\{1, \ldots, 6\}$ gives $\mathrm{E}(K) = \frac{70}{15} = \frac{14}{3}$, and the formula gives $\frac{2 \cdot 7}{3} = \frac{14}{3}$. We checked the distribution, the identity and the mean by listing every subset for all $1 \le n \le m \le 9$.
- A nice way to see the answer: the n chosen marks split the remaining $m-n$ marks into $n+1$ gaps of equal expected size $\frac{m-n}{n+1}$. Then $\mathrm{E}(K) = m - \frac{m-n}{n+1} = \frac{n(m+1)}{n+1}$.

Common slips

- Counting ordered assignments of marks to scripts in the numerator but unordered sets in the denominator (or the other way round). Either way works if you are consistent, because the factor $n!$ cancels.
- Using $\binom{k}{n}$ instead of $\binom{k-1}{n-1}$, which counts sets whose marks are all *at most* k, not sets whose largest mark *equals* k.

Beyond STEP

The identity is the "hockey-stick" identity of Pascal's triangle, and here we proved it by counting the same thing in two ways, a standard combinatorial technique. K is the largest of a random sample taken without replacement, an *order statistic*. The "German tank problem" turns this question round and estimates m from the observed K: solving $\mathrm{E}(K) = \frac{n(m+1)}{n+1}$ for m gives the unbiased estimator $\frac{n+1}{n}K - 1$. For more on order statistics and counting arguments, see Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP I 1996, Question 13

Probability and Statistics • Discrete Distributions and Expectation

STEP I 1996, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q13

This question has a full guided treatment in *Beyond STEP*, Section 12.11, *Stamps, Cups and e*: https://beyondstep.keithmcnulty.org/ch12/stamps-and-e.html

The key idea

The game stops at the first wrong guess, so $N = k$ means "right with $2, 3, \ldots, k-1$ cups, then wrong with k cups". Multiplying these probabilities gives a factorial in the denominator, and factorials in a sum are a strong hint that the exponential series $e^x = \sum x^n/n!$ is about to appear. The trick for the expectations is to cancel factors of k against $k!$ so that each sum becomes a shifted copy of $\sum 1/n!$.

Solution

The distribution of N

With c cups jumbled, Jim is effectively guessing at random, so he is right with probability $1/c$ and wrong with probability $1 - 1/c$. Each round uses a fresh jumble, so the rounds are independent.

Since there are always at least two cups, $N \geq 2$. For $k \geq 2$, the event $N = k$ happens exactly when Jim guesses correctly with $2, 3, \ldots, k-1$ cups and then fails with k cups. By independence,

$$\begin{aligned} \mathrm{P}(N = k) &= \frac{1}{2} \cdot \frac{1}{3} \cdots \frac{1}{k-1} \cdot \left(1 - \frac{1}{k}\right) = \frac{1}{(k-1)!} \cdot \frac{k-1}{k} \\ &= \frac{k-1}{k!}, \qquad k = 2, 3, 4, \ldots \end{aligned}$$

(For $k = 2$ the product of correct guesses is empty and we get $\frac{1}{2}$, as we should.)

It is worth noticing a neat equivalent form: $N > k$ means Jim was right with $2, \ldots, k$

cups, so

$$\mathrm{P}(N > k) = \frac{1}{k!} \qquad (k \geq 1).$$

Indeed $\frac{1}{(k-1)!} - \frac{1}{k!} = \frac{k-1}{k!}$, matching the formula above. As $k \to \infty$ this tends to 0, so the game ends with probability 1 and the probabilities sum to 1.

The expectation

Cancelling k against $k!$,

$$\begin{aligned}\mathrm{E}(N) &= \sum_{k=2}^{\infty} k \cdot \frac{k-1}{k!} = \sum_{k=2}^{\infty} \frac{k(k-1)}{k!} = \sum_{k=2}^{\infty} \frac{1}{(k-2)!} \\ &= \sum_{j=0}^{\infty} \frac{1}{j!} = \mathrm{e},\end{aligned}$$

putting $j = k-2$ and using $\mathrm{e}^x = \sum_{j \geq 0} x^j / j!$ at $x = 1$. This is the required result.

The variance

> **Ask yourself** Which second moment is easiest to sum here: $\mathrm{E}(N^2)$ or $\mathrm{E}\big(N(N-1)\big)$?

Either works, because the factor $k(k-1)$ in $k^2\,\mathrm{P}(N=k) = k \cdot \dfrac{k(k-1)}{k!}$ already cancels cleanly. We find

$$\begin{aligned}\mathrm{E}(N^2) &= \sum_{k=2}^{\infty} \frac{k^2(k-1)}{k!} = \sum_{k=2}^{\infty} \frac{k}{(k-2)!} = \sum_{j=0}^{\infty} \frac{j+2}{j!} \\ &= \sum_{j=1}^{\infty} \frac{1}{(j-1)!} + 2\sum_{j=0}^{\infty} \frac{1}{j!} = \mathrm{e} + 2\mathrm{e} = 3\mathrm{e},\end{aligned}$$

where in the middle step the $j = 0$ term of $\sum j/j!$ is zero, and $j/j! = 1/(j-1)!$ for $j \geq 1$. Therefore

$$\mathrm{Var}(N) = \mathrm{E}(N^2) - \big(\mathrm{E}(N)\big)^2 = 3\mathrm{e} - \mathrm{e}^2 = \mathrm{e}(3-\mathrm{e}) \approx 0.766.$$

Checks

- The probabilities $\frac{1}{2}, \frac{1}{3}, \frac{1}{8}, \frac{1}{30}, \ldots$ add up to 1, as the tail form $\mathrm{P}(N > k) = 1/k!$ shows.
- Using the tail formula, $\mathrm{E}(N) = \sum_{k \geq 0} \mathrm{P}(N > k) = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots = \mathrm{e}$, an independent route to the same answer.
- The variance is positive, as it must be, since $\mathrm{e} < 3$. Exact sums give $\mathrm{E}(N) = 2.71828\ldots$ and $\mathrm{Var}(N) = 0.76579\ldots$, and a simulation of 400 000 games gave a mean of 2.720 and a variance of 0.767.

Common slips

- Starting the distribution at $k = 1$: there are two cups from the start, so $N \geq 2$.
- Forgetting the final factor $1 - \frac{1}{k}$ for the failed guess, which gives $1/(k-1)!$, and these "probabilities" sum to $\mathrm{e} - 1$, not 1.
- Writing $\mathrm{Var}(N) = \mathrm{E}(N^2) - \mathrm{E}(N)$ instead of subtracting $\big(\mathrm{E}(N)\big)^2$.

Beyond STEP

The tail-sum formula $\mathrm{E}(N) = \sum_{k \geq 0} \mathrm{P}(N > k)$, valid for any random variable taking values in $\{0, 1, 2, \dots\}$, is a standard first-year tool, and it often turns a messy expectation into a one-line calculation, as it did here. A cleaner way to package all the moments is the probability generating function $G(s) = \mathrm{E}(s^N)$. Here it works out to $G(s) = 1 + (s-1)\mathrm{e}^s$, from which $G'(1) = \mathrm{e}$ and $G''(1) = 2\mathrm{e}$ give the mean and variance at once. Grimmett and Welsh, *Probability: An Introduction*, develops generating functions in exactly this spirit.

WORKED SOLUTION

STEP I 1996, Question 14

Probability and Statistics • Runs and Conditional Probability

STEP I 1996, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q14

The key idea

Condition on how the current run ends. Once the coin changes face, everything that happened before is irrelevant: the old run is broken, and the new run has length one. So after a head the game "restarts" the moment a tail appears, in exactly the state it would be in if the very first toss had been a tail, and vice versa. That gives two linear equations in two unknowns.

Solution

Write $x = \mathrm{P}(A \mid H)$ and $y = \mathrm{P}(A \mid T)$. Tosses are independent throughout. We take $r, s \geq 1$.

The equation for $\mathrm{P}(A \mid H)$

Suppose the first toss is a head. We now have a run of one head, and no run of tails has started. Look at the next $r-1$ tosses (tosses 2 to r).

- If all of them are heads, which has probability p^{r-1}, a run of r heads is completed at toss r. No tail has appeared at all, so certainly no run of s tails came first, and A occurs.
- Otherwise, which has probability $1 - p^{r-1}$, a tail appears among tosses $2, \ldots, r$ before the run of heads reaches length r. At the first such tail, the run of heads is broken without reaching r, and no earlier tail has appeared, so no run of s tails has happened yet either. We are now holding a run of exactly one tail, and nothing earlier can contribute to any future run.

? **Ask yourself** Why is the probability of A from that tail onwards exactly $\mathrm{P}(A \mid T)$?

After the first toss is a tail, the situation is: a run of one tail, no run of heads in progress,

and the future tosses independent of the past. That is precisely the situation at the first tail in the second case above. Since later tosses are independent of earlier ones, and which of the two runs comes first depends only on the current run and the future tosses, the conditional probability of A from that point is y. Hence, by the law of total probability,

$$\mathrm{P}(A \mid H) = p^{r-1} + \left(1 - p^{r-1}\right)\mathrm{P}(A \mid T),$$

so $\alpha = r - 1$.

The equation for $\mathrm{P}(A \mid T)$

Suppose the first toss is a tail. Look at tosses 2 to s.

- If all are tails (probability q^{s-1}), a run of s tails is completed before any head appears, so A fails.
- Otherwise (probability $1 - q^{s-1}$), a head appears before the tails reach s in a row. At that head we have a run of one head and nothing in progress for tails, which is exactly the situation after a first-toss head. So from there A occurs with probability x.

Therefore

$$\mathrm{P}(A \mid T) = \left(1 - q^{s-1}\right)\mathrm{P}(A \mid H),$$

so $\beta = s - 1$.

(The edge cases fit too. If $r = 1$ the first equation says $x = 1$, which is right because a single head is already a run of length r. If $s = 1$ the second says $y = 0$.)

Solving

Substitute the second equation into the first:

$$x = p^{r-1} + \left(1 - p^{r-1}\right)\left(1 - q^{s-1}\right)x.$$

Now

$$1 - \left(1 - p^{r-1}\right)\left(1 - q^{s-1}\right) = p^{r-1} + q^{s-1} - p^{r-1}q^{s-1},$$

which is positive (it is at least $p^{r-1} > 0$ when $0 < p < 1$, since $q^{s-1}(1 - p^{r-1}) \geq 0$). So we may divide by it. Writing $D = p^{r-1} + q^{s-1} - p^{r-1}q^{s-1}$,

$$\mathrm{P}(A \mid H) = \frac{p^{r-1}}{D}, \qquad \mathrm{P}(A \mid T) = \frac{p^{r-1}\left(1 - q^{s-1}\right)}{D}.$$

Finally, by the law of total probability with $\mathrm{P}(H) = p$ and $\mathrm{P}(T) = q$:

$$\begin{aligned}\mathrm{P}(A) &= p\,\mathrm{P}(A \mid H) + q\,\mathrm{P}(A \mid T)\\ &= \frac{p^{r-1}\left(p + q - q^{s}\right)}{D}\\ &= \frac{p^{r-1}\left(1 - q^{s}\right)}{p^{r-1} + q^{s-1} - p^{r-1}q^{s-1}},\end{aligned}$$

using $p + q = 1$.

Checks

- Symmetry: for a fair coin with $r = s$, the answer should be $\frac{1}{2}$. With $p = q = \frac{1}{2}$ and $r = s = n$, the numerator is $2^{1-n}(1 - 2^{-n})$ and the denominator is $2^{2-n} - 2^{2-2n} = 2 \cdot 2^{1-n}(1 - 2^{-n})$, giving $\frac{1}{2}$.
- Small case: $r = 1$ gives $\mathrm{P}(A) = 1 - q^s$, the chance that the first s tosses are not all tails, which is correct.
- For $p = \frac{1}{2}$, $r = 3$, $s = 2$ the formulae give $\mathrm{P}(A \mid H) = \frac{2}{5}$, $\mathrm{P}(A \mid T) = \frac{1}{5}$ and $\mathrm{P}(A) = \frac{3}{10}$. We confirmed this by solving the full Markov chain exactly, and a simulation of 200 000 games gave 0.298.

Common slips

- Using p^r instead of p^{r-1}: the first head has already been given, so only $r - 1$ more are needed.
- Forgetting that $\mathrm{P}(A)$ needs the final weighting by p and q. The conditional probabilities alone are not the answer.

Beyond STEP

The argument is a *first-step analysis* of a Markov chain: the states are "current face and run length", and the restart property is the Markov property. The same method finds absorption probabilities and expected waiting times for any pattern of heads and tails, and leads on to Penney's game, where some patterns surprisingly beat others. Grimmett and Welsh, *Probability: An Introduction*, covers Markov chains and first-step analysis at the right level.

WORKED SOLUTION

STEP II 1996, Question 1

Pure • Binomial Series Coefficients

STEP II 1996, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q1

The key idea

A coefficient in a product of three brackets is a sum over all ways of choosing one term from each bracket so that the powers add up to the target. In (i) that means organising a finite count carefully. In (ii) the coefficients $1, -2, 3, -4, \ldots$ are recognisable as those of $(1+x)^{-2}$, and the extra terms in the bracket are exactly the ones needed to make the truncation invisible at x^6.

Solution

Part (i)

Write the bracket as $P(x) = \sum_{k=0}^{4} a_k x^k$ with

$$a_k = (-1)^k(k+1), \qquad k = 0, 1, 2, 3, 4.$$

Choosing $a_p x^p$, $a_q x^q$, $a_r x^r$ from the three factors contributes to x^6 exactly when $p+q+r=6$, so the coefficient we want is

$$\sum_{\substack{p+q+r=6 \\ 0\le p,q,r\le 4}} a_p a_q a_r.$$

Every term has sign $(-1)^{p+q+r} = (-1)^6 = +1$, so we may ignore signs and add up the products $(p+1)(q+1)(r+1)$.

Ask yourself How do we make sure no choice of exponents is missed or counted twice?

List the unordered triples $\{p, q, r\}$ with sum 6 and every entry at most 4, then multiply by the number of orders in which each can be taken from the three (distinguishable) brackets: 6 orders if the entries are all different, 3 if exactly two are equal, 1 if all three are equal.

exponents	product $(p+1)(q+1)(r+1)$	orders	contribution
$4,2,0$	$5\cdot 3\cdot 1 = 15$	6	90
$4,1,1$	$5\cdot 2\cdot 2 = 20$	3	60
$3,3,0$	$4\cdot 4\cdot 1 = 16$	3	48
$3,2,1$	$4\cdot 3\cdot 2 = 24$	6	144
$2,2,2$	$3\cdot 3\cdot 3 = 27$	1	27

These are all the possibilities: the largest entry is 4, 3 or 2 (it can't be less, since $3\times 1 < 6$, and it can't exceed 4), and for each largest entry we have listed every way of splitting the remainder into two smaller parts. Adding,

$$90 + 60 + 48 + 144 + 27 = 369,$$

so the coefficient of x^6 is $\boxed{369}$.

Part (ii)

Now the bracket is $Q(x) = \sum_{k=0}^{6}(-1)^k(k+1)x^k$. The binomial expansion, valid for $|x|<1$, gives

$$(1+x)^{-2} = \sum_{k=0}^{\infty}\binom{-2}{k}x^k = \sum_{k=0}^{\infty}(-1)^k(k+1)x^k,$$

since $\binom{-2}{k} = \frac{(-2)(-3)\cdots(-k-1)}{k!} = (-1)^k(k+1)$. So $Q(x)$ is the expansion of $(1+x)^{-2}$ cut off after the x^6 term:

$$(1+x)^{-2} = Q(x) + x^7R(x)$$

for some power series $R(x)$.

Ask yourself Why doesn't the missing tail $x^7R(x)$ affect the coefficient of x^6 in the cube?

Cubing,

$$(1+x)^{-6} = Q(x)^3 + 3Q(x)^2\,x^7R(x) + 3Q(x)\,x^{14}R(x)^2 + x^{21}R(x)^3.$$

Every term after the first is x^7 times a power series, so contains no power of x below x^7. Hence the coefficient of x^6 in $Q(x)^3$ equals the coefficient of x^6 in $(1+x)^{-6}$. (Put another way: a product of exponents adding to 6 can never use a term of degree 7 or more from any factor.)

By the binomial expansion again,

$$\binom{-6}{6} = \frac{(-6)(-7)(-8)(-9)(-10)(-11)}{6!} = \frac{6\cdot 7\cdot 8\cdot 9\cdot 10\cdot 11}{720} = \binom{11}{5} = 462.$$

So the coefficient of x^6 in $Q(x)^3$ is $\boxed{462}$.

Checks

Part (ii) gives an independent check on part (i). The two brackets differ only in the terms $-6x^5$ and $7x^6$, so $462 - 369$ should be the total of all contributions that use one of them. A term using $7x^6$ must take 1 from the other two brackets: 3 positions, each giving 7, total 21. A term using $-6x^5$ must take $-2x$ and 1 from the other two: $3 \times 2 = 6$ arrangements, each giving $(-6)(-2)(1) = 12$, total 72. Indeed $21 + 72 = 93 = 462 - 369$. Multiplying out the cubes by computer gives 369 and 462 directly.

Common slips

- Forgetting to count orders: the exponents $4, 2, 0$ can come from the three brackets in 6 ways, not 1.
- Including triples with an entry of 5 or 6 in part (i), where the bracket stops at x^4.
- Worrying about signs term by term: since the exponents always add to 6, every contribution is positive.

Beyond STEP

The argument in (ii) is the start of the theory of *formal power series*: series can be multiplied coefficient by coefficient without any thought of convergence, and the coefficient of x^n in a product depends only on the terms up to x^n in each factor. Treating a sequence through its generating function in this way is a powerful counting tool; for example, the coefficient of x^n in $(1 - x)^{-m}$ counts the ways of writing n as an ordered sum of m non-negative integers. Siklos's *Advanced Problems in Mathematics* has more STEP-style practice with expansions of this kind.

WORKED SOLUTION

STEP II 1996, Question 2

Pure • Nonlinear Simultaneous Equations

STEP II 1996, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q2

The key idea

The equations are not linear in x, y and z, but every term is one of only three products: yz, zx and xy. Treat those three products as the unknowns and the system becomes three *linear* equations, which we can solve by elimination. Multiplying the three products together gives $(xyz)^2$, and dividing xyz by each product in turn recovers the individual variables.

Solution

A linear system in disguise

Put

$$a = yz, \qquad b = zx, \qquad c = xy.$$

Then the three equations read

$$2a + b - 5c = 2, \tag{1}$$
$$a - b + 2c = 1, \tag{2}$$
$$a - 2b + 6c = 3. \tag{3}$$

From (2), $a = 1 + b - 2c$. Substituting into (1):

$$2(1 + b - 2c) + b - 5c = 2 \quad \Longrightarrow \quad 3b - 9c = 0 \quad \Longrightarrow \quad b = 3c.$$

Substituting into (3):

$$(1 + b - 2c) - 2b + 6c = 3 \quad \Longrightarrow \quad -b + 4c = 2.$$

With $b = 3c$ this gives $c = 2$, so $b = 6$ and $a = 1 + 6 - 4 = 3$. The coefficient matrix has determinant $-3 \neq 0$, so this is the *only* solution of the linear system:

$$yz = 3, \qquad zx = 6, \qquad xy = 2.$$

Showing $xyz = \pm 6$

Multiplying the three products together,

$$(yz)(zx)(xy) = x^2y^2z^2 = (xyz)^2 = 3 \times 6 \times 2 = 36,$$

so $xyz = 6$ or $xyz = -6$, as required.

> **Ask yourself** We now know xyz and each of yz, zx, xy. How do we get x on its own?

None of the products is zero, so none of x, y, z is zero and we may divide:

$$x = \frac{xyz}{yz}, \qquad y = \frac{xyz}{zx}, \qquad z = \frac{xyz}{xy}.$$

The possible values

If $xyz = 6$, then $x = 6/3 = 2$, $y = 6/6 = 1$ and $z = 6/2 = 3$.

If $xyz = -6$, then $x = -2$, $y = -1$ and $z = -3$.

Each step above was forced, so these are the only candidates; both satisfy the original equations (the equations involve only products of pairs, so changing the sign of all three variables leaves them unchanged). Hence

$$(x, y, z) = (2, 1, 3) \quad \text{or} \quad (x, y, z) = (-2, -1, -3).$$

Checks

Substitute $(2, 1, 3)$: $yz = 3$, $zx = 6$, $xy = 2$, so the left-hand sides are $6 + 6 - 10 = 2$, $3 - 6 + 4 = 1$ and $3 - 12 + 12 = 3$. All three match. A computer search over all triples of fractions p/q with $|p| \leq 12$, $1 \leq q \leq 4$ found exactly the two solutions above, and exact arithmetic confirms that the linear system has the unique solution $(a, b, c) = (3, 6, 2)$.

Common slips

- Taking square roots of $(xyz)^2 = 36$ separately for each variable, for example writing $x = \pm 2$, $y = \pm 1$, $z = \pm 3$ independently. That gives eight triples, but only two work: the signs are linked because xyz has a single sign.
- Forgetting the negative root, and giving only $(2, 1, 3)$.

Beyond STEP

The trick of renaming awkward expressions so that a nonlinear problem becomes a linear one is used everywhere, from fitting curves like $y = Ae^{kx}$ by taking logarithms to solving differential equations by substitution. Once the system is linear, matrices

and determinants tell you at once whether the solution is unique, which is the starting point of a first course in linear algebra. Strang's *Introduction to Linear Algebra* develops elimination and determinants in exactly this spirit.

WORKED SOLUTION

STEP II 1996, Question 3

Pure • Fibonacci Identities by Induction

STEP II 1996, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q3

The key idea

Everything here follows from the recurrence, so induction is the natural tool. For the product $F_{n+1}F_{n-1} - F_n^2$ the trick is to compare consecutive values: one use of the recurrence turns each value into minus the previous one. For the addition formula, the neat move is to induct on k while letting the hypothesis hold for *every* n at once, so that we may apply it with $n+1$ in place of n.

Solution

The first few values

Using $F_{n+1} = F_n + F_{n-1}$ with $n = 1, 2, 3$:

$$F_2 = F_1 + F_0 = 1 + 0 = 1, \qquad F_3 = F_2 + F_1 = 1 + 1 = 2, \qquad F_4 = F_3 + F_2 = 2 + 1 = 3,$$

as required. Continuing,

$$F_5 = 3 + 2 = 5, \qquad F_6 = 5 + 3 = 8, \qquad F_7 = 8 + 5 = 13.$$

The product $F_{n+1}F_{n-1} - F_n^2$

Write $C_n = F_{n+1}F_{n-1} - F_n^2$ for $n \geq 1$. From the table of values:

n	1	2	3	4	5	6
$F_{n+1}F_{n-1}$	$1 \cdot 0$	$2 \cdot 1$	$3 \cdot 1$	$5 \cdot 2$	$8 \cdot 3$	$13 \cdot 5$
F_n^2	1	1	4	9	25	64
C_n	-1	1	-1	1	-1	1

The values alternate, so we guess

$$F_{n+1}F_{n-1} - F_n^2 = (-1)^n \qquad \text{for all } n \geq 1.$$

Ask yourself How can we get from C_n to C_{n+1} using only the recurrence?

Expand the larger index in C_{n+1}. For $n \geq 1$,

$$\begin{aligned} C_{n+1} &= F_{n+2}F_n - F_{n+1}^2 \\ &= (F_{n+1} + F_n)F_n - F_{n+1}^2 && \text{(recurrence for } F_{n+2}) \\ &= F_n^2 - F_{n+1}(F_{n+1} - F_n) \\ &= F_n^2 - F_{n+1}F_{n-1} && \text{(since } F_{n+1} - F_n = F_{n-1} \text{ for } n \geq 1) \\ &= -C_n. \end{aligned}$$

Now the induction is immediate. The base case is $C_1 = F_2F_0 - F_1^2 = 0 - 1 = -1 = (-1)^1$. If $C_n = (-1)^n$ for some $n \geq 1$, then $C_{n+1} = -C_n = (-1)^{n+1}$. Hence by induction

$$F_{n+1}F_{n-1} - F_n^2 = (-1)^n \qquad \text{for every } n \geq 1.$$

The addition formula

For each positive integer k let $P(k)$ be the statement

$$F_{n+k} = F_kF_{n+1} + F_{k-1}F_n \qquad \text{for } \textit{all} \text{ positive integers } n.$$

Base case $k = 1$. The right-hand side is $F_1F_{n+1} + F_0F_n = 1 \cdot F_{n+1} + 0 = F_{n+1}$, which is the left-hand side. So $P(1)$ holds.

Inductive step. Suppose $P(k)$ holds for some $k \geq 1$, and let n be any positive integer.

Ask yourself The hypothesis is about F_{n+k}, but we need F_{n+k+1}. Which n should we feed it?

Since $P(k)$ holds for every positive integer, we may apply it with $n+1$ in place of n, because $(n+1)+k = n+(k+1)$:

$$\begin{aligned} F_{n+(k+1)} = F_{(n+1)+k} &= F_kF_{n+2} + F_{k-1}F_{n+1} && \text{(by } P(k) \text{ with } n+1) \\ &= F_k(F_{n+1} + F_n) + F_{k-1}F_{n+1} && \text{(recurrence for } F_{n+2}) \\ &= (F_k + F_{k-1})F_{n+1} + F_kF_n \\ &= F_{k+1}F_{n+1} + F_kF_n && \text{(recurrence, as } k \geq 1). \end{aligned}$$

This is exactly $P(k+1)$ for this n, and n was arbitrary. So by induction $P(k)$ holds for every positive integer k, that is,

$$F_{n+k} = F_kF_{n+1} + F_{k-1}F_n \qquad \text{for all positive integers } n, k,$$

as required.

(An alternative is two-step induction on k with n fixed: check $k = 1$ and $k = 2$, then add the formulae for $k-1$ and k. Both routes are fine; the one above needs only one base case.)

Checks

- Cassini with $n = 6$: $F_7F_5 - F_6^2 = 13 \cdot 5 - 64 = 1 = (-1)^6$.
- Addition formula with $n = 3$, $k = 4$: $F_7 = 13$, and $F_4F_4 + F_3F_3 = 9 + 4 = 13$.
- Putting $k = n$ gives $F_{2n} = F_n(F_{n+1} + F_{n-1})$; for $n = 3$ this is $F_6 = 2 \cdot (3 + 1) = 8$.

A computer confirmed the product formula for $1 \leq n < 150$ and the addition formula for all $1 \leq n, k < 90$.

Common slips

- Getting the sign the wrong way round: $C_1 = -1$, so the answer is $(-1)^n$, not $(-1)^{n+1}$.
- In the addition formula, inducting on k with n fixed and then using the hypothesis at $n + 1$: that is only legitimate if the hypothesis is stated for all n.
- Using $F_{n+1} - F_n = F_{n-1}$ without noting it needs $n \geq 1$.

Beyond STEP

Both identities are shadows of one matrix fact: $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$. Taking determinants gives the product formula (Cassini's identity) at once, and writing $M^{n+k} = M^k M^n$ gives the addition formula. Cassini's identity also shows that consecutive Fibonacci numbers are coprime, and with the addition formula one can prove the striking result $\gcd(F_m, F_n) = F_{\gcd(m,n)}$. Davenport's *The Higher Arithmetic* is a good place to explore this kind of number theory, and Strang's *Introduction to Linear Algebra* treats the Fibonacci numbers through eigenvalues.

WORKED SOLUTION

STEP II 1996, Question 4

Pure • Integration by Substitution

STEP II 1996, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q4

The key idea

The square roots $\sqrt{1+x}$ and $\sqrt{1-x}$ are begging for $x = \cos t$, because the half-angle formulae turn $1 \pm \cos t$ into perfect squares. Once the roots are gone, the sum $\cos\theta + \sin\theta$ collapses into a single cosine, and the whole denominator becomes $1 + \cos(\text{something})$, which is again a perfect square. The final substitution recentres the interval so that the numerator becomes $\cos 4u$, which is exactly why the identity is proved first.

Solution

The identity

Using $\cos 2\theta = 2\cos^2\theta - 1$ twice, with $c = \cos u$:

$$\begin{aligned}\cos 4u &= 2\cos^2 2u - 1 = 2\left(2c^2 - 1\right)^2 - 1 \\ &= 2\left(4c^4 - 4c^2 + 1\right) - 1 = 8\cos^4 u - 8\cos^2 u + 1,\end{aligned}$$

as required.

The first substitution

Put $x = \cos t$, so $\mathrm{d}x = -\sin t\,\mathrm{d}t$. As x runs from -1 to 1, t runs from π down to 0; the minus sign flips the limits back, giving

$$I = \int_0^{\pi} \frac{\sin t}{\sqrt{1+\cos t} + \sqrt{1-\cos t} + 2}\,\mathrm{d}t.$$

Now $1 + \cos t = 2\cos^2\frac{t}{2}$ and $1 - \cos t = 2\sin^2\frac{t}{2}$.

Ask yourself Is $\sqrt{2\cos^2\frac{t}{2}}$ really $\sqrt{2}\cos\frac{t}{2}$, and not $\sqrt{2}\left|\cos\frac{t}{2}\right|$?

In general it is the modulus, but here $0 \le t \le \pi$, so $0 \le \frac{t}{2} \le \frac{\pi}{2}$ and both $\cos\frac{t}{2}$ and $\sin\frac{t}{2}$ are non-negative. This is precisely why the choice $t \in [0, \pi]$ (rather than, say, $[-\pi, 0]$) matters. Hence

$$\begin{aligned}\sqrt{1+\cos t}+\sqrt{1-\cos t} &= \sqrt{2}\left(\cos\tfrac{t}{2}+\sin\tfrac{t}{2}\right)\\ &= 2\left(\cos\tfrac{t}{2}\cos\tfrac{\pi}{4}+\sin\tfrac{t}{2}\sin\tfrac{\pi}{4}\right) = 2\cos\left(\tfrac{t}{2}-\tfrac{\pi}{4}\right).\end{aligned}$$

Writing $v = \frac{t}{4} - \frac{\pi}{8}$, so that $\frac{t}{2} - \frac{\pi}{4} = 2v$, the denominator is

$$2 + 2\cos 2v = 2(1+\cos 2v) = 4\cos^2 v = 4\cos^2\left(\tfrac{t}{4}-\tfrac{\pi}{8}\right),$$

and therefore

$$I = \int_0^{\pi} \frac{\sin t}{4\cos^2\left(\frac{t}{4}-\frac{\pi}{8}\right)}\,\mathrm{d}t,$$

as required.

The second substitution

Let $u = \frac{t}{4} - \frac{\pi}{8}$, so $t = 4u + \frac{\pi}{2}$ and $\mathrm{d}t = 4\,\mathrm{d}u$. The limits $t = 0$ and $t = \pi$ become $u = -\frac{\pi}{8}$ and $u = \frac{\pi}{8}$. The numerator is

$$\sin t = \sin\left(4u + \tfrac{\pi}{2}\right) = \cos 4u,$$

which is where the opening identity comes in. So

$$\begin{aligned}I &= \int_{-\pi/8}^{\pi/8} \frac{\cos 4u}{4\cos^2 u}\cdot 4\,\mathrm{d}u = \int_{-\pi/8}^{\pi/8} \frac{8\cos^4 u - 8\cos^2 u + 1}{\cos^2 u}\,\mathrm{d}u\\ &= \int_{-\pi/8}^{\pi/8} \left(8\cos^2 u - 8 + \sec^2 u\right)\mathrm{d}u.\end{aligned}$$

Dividing by $\cos^2 u$ is fine because $\cos u \ge \cos\frac{\pi}{8} > 0$ on the interval. Using $8\cos^2 u = 4 + 4\cos 2u$, the integrand is $4\cos 2u - 4 + \sec^2 u$, so

$$I = \Big[\,2\sin 2u - 4u + \tan u\,\Big]_{-\pi/8}^{\pi/8}.$$

The bracket is an odd function of u, so its value is twice its value at $\frac{\pi}{8}$:

$$\begin{aligned}I &= 2\left(2\sin\tfrac{\pi}{4} - \tfrac{\pi}{2} + \tan\tfrac{\pi}{8}\right) = 2\left(\sqrt{2} - \tfrac{\pi}{2} + \sqrt{2} - 1\right)\\ &= 4\sqrt{2} - \pi - 2,\end{aligned}$$

using $\sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}$ and the given $\tan\frac{\pi}{8} = \sqrt{2} - 1$. This is the required result.

Checks

- Numerically, $4\sqrt{2} - \pi - 2 \approx 0.51526$. Simpson's rule applied to the original integral in x, to the integral in t, and to the integral in u all give 0.515262.
- Size check: on $[-1, 1]$ the sum $\sqrt{1+x} + \sqrt{1-x}$ lies between $\sqrt{2}$ (at $x = \pm 1$) and 2 (at $x = 0$), so the integrand lies between $\frac{1}{4}$ and $\frac{1}{2+\sqrt{2}} \approx 0.293$. Hence $0.5 \leq I \leq 0.586$, and 0.515 fits.
- At $t = 0$ the denominator $4\cos^2(-\frac{\pi}{8})$ should equal $\sqrt{2} + 0 + 2$; indeed $4\cos^2\frac{\pi}{8} = 2(1 + \cos\frac{\pi}{4}) = 2 + \sqrt{2}$.

Common slips

- Forgetting the minus sign in $\mathrm{d}x = -\sin t\,\mathrm{d}t$, or forgetting to swap the limits, and so getting $-I$.
- Writing $\sqrt{1+\cos t} = \sqrt{2}\cos\frac{t}{2}$ without noting that $\cos\frac{t}{2} \geq 0$ on $[0, \pi]$.
- Dropping the factor 4 from $\mathrm{d}t = 4\,\mathrm{d}u$; it exactly cancels the 4 in the denominator.

Beyond STEP

Substitutions like $x = \cos t$ that "rationalise" square roots are a special case of a general fact: an integrand that is a rational function of x and $\sqrt{1-x^2}$ (or similar quadratic surds) can always be turned into a rational function by a trigonometric or the $t = \tan\frac{\theta}{2}$ substitution, and then integrated in elementary terms. At university you meet the systematic version of this, and also its limits: with cubic or quartic surds you get elliptic integrals, which are not elementary. For many more questions in this style, try Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP II 1996, Question 5

Pure • Roots of Unity

STEP II 1996, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q5

The key idea

The quartic is *palindromic*: its coefficients read the same forwards and backwards. Dividing by z^2 groups the terms into $z^2 + z^{-2}$ and $z + z^{-1}$, and both are expressible in terms of $u = z + z^{-1}$, so a quartic in z collapses to a quadratic in u. The second half recognises the quartic as the factor left over when $z - 1$ is taken out of $z^5 - 1$, so its roots are fifth roots of unity; comparing the two descriptions of the same four numbers gives exact values of trigonometric functions.

Solution

The values of u

Since $z = 0$ does not satisfy $(*)$, we may divide by z^2:

$$z^2 + z + 1 + z^{-1} + z^{-2} = 0.$$

Now $u^2 = (z + z^{-1})^2 = z^2 + 2 + z^{-2}$, so $z^2 + z^{-2} = u^2 - 2$. The equation becomes

$$u^2 - 2 + u + 1 = 0, \qquad \text{that is} \qquad u^2 + u - 1 = 0,$$

so

$$u = \frac{-1 + \sqrt{5}}{2} \quad \text{or} \quad u = \frac{-1 - \sqrt{5}}{2}.$$

The values of z

From $z + z^{-1} = u$ we get $z^2 - uz + 1 = 0$, so

$$z = \frac{u \pm \sqrt{u^2 - 4}}{2}.$$

Using $u^2 = 1 - u$, the discriminant is $u^2 - 4 = -3 - u$, which is negative for both values of u (as $|u| < 2$ in each case), so the roots are non-real:

$$z = \frac{u \pm i\sqrt{3+u}}{2}.$$

For $u = \frac{-1+\sqrt{5}}{2}$ we have $3 + u = \frac{5+\sqrt{5}}{2}$, and for $u = \frac{-1-\sqrt{5}}{2}$ we have $3 + u = \frac{5-\sqrt{5}}{2}$. Writing $\sqrt{\frac{5\pm\sqrt{5}}{2}} = \frac{1}{2}\sqrt{10 \pm 2\sqrt{5}}$, the four solutions of $(*)$ are

$$z = \frac{-1 + \sqrt{5} \pm i\sqrt{10 + 2\sqrt{5}}}{4} \qquad \text{and} \qquad z = \frac{-1 - \sqrt{5} \pm i\sqrt{10 - 2\sqrt{5}}}{4}.$$

Each value of u gives two values of z (a pair z, z^{-1}, which here are complex conjugates), and a quartic has exactly four roots, so these are all of them.

The link with $z^5 = 1$

Multiplying out,

$$(z-1)(z^4 + z^3 + z^2 + z + 1) = z^5 - 1,$$

since the middle terms telescope. So if z satisfies $(*)$, the second factor is zero and $z^5 - 1 = 0$.

The solutions of $z^5 = 1$ are $z = \cos\frac{2k\pi}{5} + i\sin\frac{2k\pi}{5}$ for $k = 0, 1, 2, 3, 4$. The value $k = 0$ gives $z = 1$, which does *not* satisfy $(*)$ (the left-hand side is 5). The other four are distinct and all satisfy $(*)$, because for them $z - 1 \neq 0$ and so the factor $z^4 + z^3 + z^2 + z + 1$ must vanish. Hence the solutions of $(*)$ are

$$z = \cos\theta + i\sin\theta, \qquad r = 1, \quad \theta = \pm\frac{2\pi}{5}, \pm\frac{4\pi}{5}.$$

The deduction

We now have two descriptions of the same four numbers, and we must match them up.

Ask yourself Which of the four surd expressions is $\cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}$?

Look at signs. Since $0 < \frac{2\pi}{5} < \frac{\pi}{2}$, the number $\cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}$ has positive real part and positive imaginary part. Among the surd expressions, the real parts are $\frac{-1+\sqrt{5}}{4} > 0$ and $\frac{-1-\sqrt{5}}{4} < 0$, so exactly one of them lies in the first quadrant:

$$\cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5} = \frac{-1+\sqrt{5}}{4} + i\,\frac{\sqrt{10+2\sqrt{5}}}{4}.$$

(The other angle in $(0, \pi)$, namely $\frac{4\pi}{5}$, has negative cosine, so it is matched with the other root with positive imaginary part.) Equating real and imaginary parts,

$$\sin\frac{2\pi}{5} = \frac{\sqrt{10+2\sqrt{5}}}{4} \qquad \text{and} \qquad \cos\frac{2\pi}{5} = \frac{-1+\sqrt{5}}{4},$$

as required. As a by-product, $\cos\frac{4\pi}{5} = \frac{-1-\sqrt{5}}{4}$ and $\sin\frac{4\pi}{5} = \frac{\sqrt{10-2\sqrt{5}}}{4}$.

Checks

- $\cos\frac{2\pi}{5} = \cos 72^\circ \approx 0.30902$ and $\frac{-1+\sqrt{5}}{4} \approx 0.30902$; $\sin 72^\circ \approx 0.95106$ and $\frac{1}{4}\sqrt{10+2\sqrt{5}} \approx 0.95106$.
- The sum of the four roots should be -1 (minus the coefficient of z^3): the real parts add to $\frac{2(-1+\sqrt{5})+2(-1-\sqrt{5})}{4} = -1$. Equivalently, the two values of u add to -1.
- Each root should have modulus 1: for example $\frac{(-1+\sqrt{5})^2+(10+2\sqrt{5})}{16} = \frac{16}{16} = 1$.

We checked numerically that all four surd expressions satisfy $(*)$ and $z^5 = 1$ and coincide with $e^{\pm 2\pi i/5}$, $e^{\pm 4\pi i/5}$.

Common slips

- Forgetting to exclude $z = 1$: it satisfies $z^5 = 1$ but not $(*)$, so $(*)$ has four roots, not five.
- Matching the surds to the angles without a reason. The sign of the real part is what pins down which root is $\cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}$.
- Writing $z^2 + z^{-2} = u^2$ and losing the -2.

Beyond STEP

The substitution $u = z + z^{-1}$ halves the degree of any palindromic polynomial. The fact that $\cos\frac{2\pi}{5}$ is reached by solving quadratics only is no accident: Gauss showed that $\cos\frac{2\pi}{n}$ can be written with square roots alone exactly when a regular n-gon can be constructed with ruler and compass, and this question is the algebra behind constructing the regular pentagon. The full story is Galois theory, met in a second-year algebra course; Needham's *Visual Complex Analysis* gives a beautiful geometric view of roots of unity.

WORKED SOLUTION

STEP II 1996, Question 6

Pure • Counting Divisors

STEP II 1996, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q6

The key idea

Every factor of N is built by choosing, independently for each prime p_i, how many copies of it to take, anywhere from 0 to m_i. Counting those choices counts all the factors, and then we throw away 1 and N. After that, both search questions become questions about exponents: which exponent patterns give the right count, and how to place those exponents on primes so that N is as small as possible.

Solution

The opening examples

The factors of 12 are $1, 2, 3, 4, 6, 12$. Removing 1 and 12 leaves $2, 3, 4, 6$, which is 4 proper factors. The factors of 16 are $1, 2, 4, 8, 16$. Removing 1 and 16 leaves $2, 4, 8$, which is 3 proper factors.

The general count

Suppose M divides N. Every prime factor of M is then a prime factor of N, and it cannot appear in M more often than in N. So

$$M = p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r}, \qquad 0 \leq a_i \leq m_i.$$

Conversely, every such product divides N. By uniqueness of prime factorisation, different choices of $(a_1, \ldots, a_r)$ give different integers M. So the factors of N correspond exactly to these choices. There are $m_i + 1$ choices for each a_i, made independently, so N has

$$d(N) = (m_1 + 1)(m_2 + 1) \cdots (m_r + 1)$$

factors in total. Since $r \geq 1$, we have $N > 1$, so 1 and N are two *different* factors. Removing them, the number of proper factors is

$$(m_1 + 1)(m_2 + 1) \cdots (m_r + 1) - 2.$$

Check: $12 = 2^2 \cdot 3$ gives $3 \cdot 2 - 2 = 4$, and $16 = 2^4$ gives $5 - 2 = 3$.

A principle for making N small

In both parts we fix the exponents and ask for the smallest N they allow. Two observations settle this.

- Use the smallest primes. If some prime q is used and a smaller prime p is not, replacing q by p keeps every exponent, so it keeps the count, and it makes N smaller. So the primes should be $2, 3, 5, \ldots$ with no gaps.
- Put the biggest exponents on the smallest primes. Suppose $p < q$, with p carrying exponent a and q carrying exponent b, where $a < b$. Swapping the two exponents multiplies N by $p^{b-a}/q^{b-a} < 1$, and it does not change the count.

So, for exponents $e_1 \geq e_2 \geq \cdots \geq e_r$, the smallest N is $2^{e_1}3^{e_2}5^{e_3}\cdots$.

Part (i)

We need exactly 12 proper factors, so $d(N) = 14$. Each factor $m_i + 1$ is at least 2, and $14 = 2 \cdot 7$ with 7 prime. So 14 can be written as a product of factors that are each at least 2 in only two ways:

- 14 itself: $N = p^{13}$, which is smallest at $2^{13} = 8192$;
- 7×2: $N = p^6 q$, which is smallest at $2^6 \cdot 3 = 192$.

So the smallest positive integer with exactly 12 proper factors is

$$\boxed{192 = 2^6 \cdot 3}.$$

Its factors are $1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 48, 64, 96, 192$, and there are 14 of them.

Part (ii)

Now we need $d(N) \geq 14$. Since $d(N)$ is not fixed any more, we sort by the number r of distinct primes. Write the exponents in decreasing order, $e_1 \geq e_2 \geq \cdots$.

? **Ask yourself** Why can't we just reuse part (i) and look at $d(N) = 14$ only?

Because having *more* factors does not mean N has to be bigger. Exponents that are spread over several small primes give many factors cheaply. For example, $2^3 \cdot 3 \cdot 5$ has $4 \cdot 2 \cdot 2 = 16$ factors and is much smaller than 192. So we must look at every value $d(N) \geq 14$.

- $r \geq 4$: then $N \geq 2 \cdot 3 \cdot 5 \cdot 7 = 210$.
- $r = 3$: here $d(N) = (e_1 + 1)(e_2 + 1)(e_3 + 1) \geq 14$. If $e_2 = e_3 = 1$, we need $4(e_1 + 1) \geq 14$, so $e_1 \geq 3$ and $N \geq 2^3 \cdot 3 \cdot 5 = 120$. Otherwise $e_2 \geq 2$, so $e_1 \geq 2$ too, and $N \geq 2^2 \cdot 3^2 \cdot 5 = 180$.

- $r = 2$: here $(e_1 + 1)(e_2 + 1) \geq 14$ with $e_1 \geq e_2$. If $e_2 = 1$, then $e_1 \geq 6$ and $N \geq 2^6 \cdot 3 = 192$. If $e_2 = 2$, then $3(e_1 + 1) \geq 14$ gives $e_1 \geq 4$, so $N \geq 2^4 \cdot 3^2 = 144$. If $e_2 \geq 3$, then $e_1 \geq 3$ and $N \geq 2^3 \cdot 3^3 = 216$.
- $r = 1$: $e_1 + 1 \geq 14$, so $N \geq 2^{13} = 8192$.

Every case gives $N \geq 120$, and $N = 120 = 2^3 \cdot 3 \cdot 5$ actually works: it has $4 \cdot 2 \cdot 2 - 2 = 14$ proper factors. So the smallest positive integer with at least 12 proper factors is

$$\boxed{120},$$

which in fact has 14 proper factors.

Checks

A brute-force count of proper factors for every N from 2 to 9000 agrees with the formula $d(N) - 2$. It also confirms that 192 is the first N with exactly 12 proper factors, and that 120 is the first with at least 12 (it has 14). For a quick hand check, the numbers just below 120 with lots of factors, namely $60, 72, 84, 90, 96$ and 108, all have 12 factors and so only 10 proper ones.

Common slips

- Setting $d(N) = 12$ instead of 14. Remember that 1 and N are excluded.
- Answering 192 in part (ii) too. The smallest N with at least 12 proper factors need not have exactly 12.
- Putting the big exponent on the big prime: $2 \cdot 3^6$ has the same count as $2^6 \cdot 3$ but is far larger.

Beyond STEP

The divisor function $d(N)$ is *multiplicative*: $d(ab) = d(a)d(b)$ whenever a and b are coprime. That is exactly what the product formula expresses. Numbers with more factors than every smaller number are called *highly composite*, and Ramanujan studied them in depth. 120 is one of them, and the exponent-ordering argument used here is the starting point of his analysis. Davenport's *The Higher Arithmetic* is a lovely introduction to divisor functions and unique factorisation.

WORKED SOLUTION

STEP II 1996, Question 7

Pure • Loci in Coordinate Geometry

STEP II 1996, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q7

The key idea

Put the square in coordinates so that it's symmetric about both axes. Then each distance is the modulus of a linear expression, and the two products pr and qs become $|y^2 - a^2|$ and $|x^2 - a^2|$. An equation of the form $|U| = |V|$ splits into $U = V$ or $U = -V$. One case gives the two lines and the other gives the circle.

Solution

Let the square have side $2a$ (with $a > 0$) and centre at the origin, with sides parallel to the axes:

$$A = (-a, -a), \quad B = (a, -a), \quad C = (a, a), \quad D = (-a, a).$$

Then AB lies on the line $y = -a$, BC on $x = a$, CD on $y = a$ and DA on $x = -a$. The perpendicular distance from $P = (x, y)$ to a horizontal line $y = c$ is $|y - c|$, and to a vertical line $x = c$ it is $|x - c|$. So

$$p = |y + a|, \qquad q = |x - a|, \qquad r = |y - a|, \qquad s = |x + a|.$$

These formulae hold for every P in the plane, inside or outside the square. The moduli are what make the distances non-negative, so we mustn't drop them.

Multiplying in pairs, and using $|u|\,|v| = |uv|$,

$$pr = |(y + a)(y - a)| = |y^2 - a^2|, \qquad qs = |(x - a)(x + a)| = |x^2 - a^2|.$$

Ask yourself How do we remove the moduli from $|y^2 - a^2| = |x^2 - a^2|$ without splitting into lots of regions?

For real numbers, $|U| = |V|$ holds if and only if $U = V$ or $U = -V$, because both sides are non-negative and $|U| = |V| \iff U^2 = V^2 \iff (U - V)(U + V) = 0$. Since we only ever use "if and only if" steps, we get the necessary and the sufficient conditions

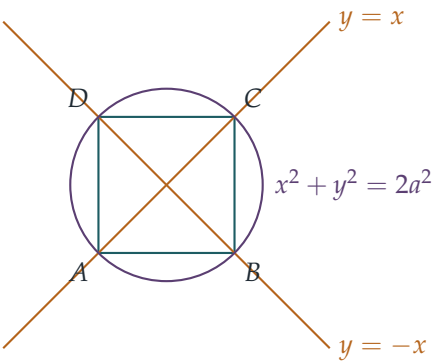
at the same time. So $pr = qs$ holds if and only if

$$y^2 - a^2 = x^2 - a^2 \qquad \text{or} \qquad y^2 - a^2 = -(x^2 - a^2).$$

First case. $y^2 = x^2$, that is, $(y - x)(y + x) = 0$, so $y = x$ or $y = -x$. These are the two (infinite) lines through the centre of the square that contain the diagonals AC and BD.

Second case. $x^2 + y^2 = 2a^2$. This is the circle with centre the centre of the square and radius $\sqrt{2}\,a$, which is half the diagonal. So it is the circumcircle of the square, passing through A, B, C and D.

Hence $pr = qs$ forces P to lie on one of the two diagonal lines AC, BD (extended indefinitely) or on the circumcircle of $ABCD$. Conversely, every step above was reversible. If P is on either line, then $x^2 = y^2$, so $|y^2 - a^2| = |x^2 - a^2|$. If P is on the circle, then $y^2 - a^2 = a^2 - x^2$, which again gives $|y^2 - a^2| = |x^2 - a^2|$. Either way $pr = qs$, so every point of the two lines and the circle is a possible position of P. This completes the proof.



Ask yourself Is a point outside the square really allowed, and do the lines stop at the vertices?

The question allows any point in the plane, and our distance formulae are valid everywhere. So the locus is the whole of each line, not just the diagonals as segments. For example, $P = (3a, 3a)$ has $p = 4a$, $r = 2a$, $q = 2a$, $s = 4a$, so $pr = qs = 8a^2$.

Checks

- The centre $(0, 0)$ is on both lines: $p = q = r = s = a$, so $pr = qs$.
- Take the point of the circle directly below the midpoint of AB, namely $(0, -\sqrt{2}\,a)$. Then $p = (\sqrt{2} - 1)a$, $r = (\sqrt{2} + 1)a$, so $pr = a^2$, and $q = s = a$, so $qs = a^2$.
- A point not on the locus, such as $(\frac{1}{2}a, 0)$, gives $pr = a^2$ but $qs = \frac{3}{4}a^2$, so it fails.
- Numerically (with $a = 1.7$), $pr = qs$ held to 10^{-9} at 30 000 random points on the three curves. A sign-change scan of $pr - qs$ over a 600×600 grid on $[-5, 5]^2$ found no zeros away from the two lines and the circle.

Common slips

- Dropping the moduli and writing $pr = a^2 - y^2$, $qs = a^2 - x^2$. This only works inside the square and loses the circle, giving just $y = \pm x$.
- Proving only that P must lie on the curves. The question also asks for the converse, so say why every point of the lines and the circle works.
- Describing the lines as the diagonals *segments* AC and BD. They are full lines.

Beyond STEP

The locus is the zero set of the polynomial $(y^2 - a^2)^2 - (x^2 - a^2)^2$, which factorises as $(y - x)(y + x)(x^2 + y^2 - 2a^2)$. A curve given by a polynomial equation that splits into simpler pieces like this is a first example of a reducible *algebraic curve*. Choosing coordinates to match a figure's symmetry is the habit that makes such factorisations visible. For more geometry in this style, see Siklos, *Advanced Problems in Mathematics*, and for the algebra of curves, Reid, *Undergraduate Algebraic Geometry*.

WORKED SOLUTION

STEP II 1996, Question 8

Pure • Second-Order Differential Equations

STEP II 1996, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q8

The key idea

The equation mixes $\mathrm{f}(x)$ with $\mathrm{f}(-x)$, so it isn't an ordinary differential equation for f at all. The trick is to write the equation down twice, once at x and once at $-x$, and then add and subtract. The sum g and the difference h each satisfy an honest constant-coefficient equation that we can solve with a complementary function and a particular integral, and f is recovered as $\frac{1}{2}(\mathrm{g}+\mathrm{h})$.

Solution

The values of $\mathrm{g}(0)$ and $\mathrm{g}'(0)$

Putting $x=0$ in $\mathrm{g}(x)=\mathrm{f}(x)+\mathrm{f}(-x)$ gives $\mathrm{g}(0)=2\mathrm{f}(0)=2$.

Differentiating, with the chain rule giving a minus sign on the second term,

$$\mathrm{g}'(x)=\mathrm{f}'(x)-\mathrm{f}'(-x),$$

so $\mathrm{g}'(0)=\mathrm{f}'(0)-\mathrm{f}'(0)=0$, as required. (Notice that we didn't need the value of $\mathrm{f}'(0)$: g is an even function, so its derivative is odd and vanishes at 0.)

The equation for g

Differentiating again, the two minus signs from the chain rule cancel:

$$\mathrm{g}''(x)=\mathrm{f}''(x)+\mathrm{f}''(-x).$$

? **Ask yourself** What is $\mathrm{f}''(-x)$? The given equation only tells us about $\mathrm{f}''(x)$.

The given equation holds for every x, so we may replace x by $-x$ throughout. Since cos is even, this gives

$$\mathrm{f}''(-x)+\mathrm{f}(x)=-x+3\cos 2x.$$

Adding this to the original equation,

$$\mathrm{f}''(x)+\mathrm{f}''(-x)+\mathrm{f}(-x)+\mathrm{f}(x)=6\cos 2x, \qquad \text{that is,} \qquad \mathrm{g}''(x)+\mathrm{g}(x)=6\cos 2x,$$

as required.

Solving for g

The auxiliary equation $m^2+1=0$ has roots $\pm\mathrm{i}$, so the complementary function is $A\cos x+B\sin x$. For a particular integral we try $\mathrm{g}=k\cos 2x$: then $\mathrm{g}''+\mathrm{g}=-4k\cos 2x+k\cos 2x=-3k\cos 2x$, so $k=-2$. Hence

$$\mathrm{g}(x)=A\cos x+B\sin x-2\cos 2x.$$

Now $\mathrm{g}(0)=A-2=2$ gives $A=4$, and $\mathrm{g}'(0)=B=0$. So

$$\mathrm{g}(x)=4\cos x-2\cos 2x.$$

Solving for h

For $\mathrm{h}(x)=\mathrm{f}(x)-\mathrm{f}(-x)$ we have

$$\mathrm{h}(0)=0, \qquad \mathrm{h}'(x)=\mathrm{f}'(x)+\mathrm{f}'(-x), \quad \text{so} \quad \mathrm{h}'(0)=2\mathrm{f}'(0)=-2,$$

and $\mathrm{h}''(x)=\mathrm{f}''(x)-\mathrm{f}''(-x)$. This time we subtract the equation at $-x$ from the equation at x:

$$\mathrm{f}''(x)-\mathrm{f}''(-x)+\mathrm{f}(-x)-\mathrm{f}(x)=2x, \qquad \text{that is,} \qquad \mathrm{h}''(x)-\mathrm{h}(x)=2x.$$

Ask yourself Why has the sign in front of h changed from + to −?

In the subtraction the f terms appear as $\mathrm{f}(-x)-\mathrm{f}(x)$, which is $-\mathrm{h}(x)$, not $+\mathrm{h}(x)$. So the solutions are now exponential rather than trigonometric: the auxiliary equation $m^2-1=0$ gives the complementary function $C\mathrm{e}^{x}+D\mathrm{e}^{-x}$. Trying $\mathrm{h}=px+q$ as a particular integral gives $-(px+q)=2x$, so $p=-2$, $q=0$. Hence

$$\mathrm{h}(x)=C\mathrm{e}^{x}+D\mathrm{e}^{-x}-2x.$$

The conditions give $\mathrm{h}(0)=C+D=0$ and $\mathrm{h}'(0)=C-D-2=-2$, so $C=D=0$ and

$$\mathrm{h}(x)=-2x.$$

Recovering f

Adding the definitions of g and h gives $\mathrm{g}+\mathrm{h}=2\mathrm{f}$, so

$$\mathrm{f}(x)=\tfrac{1}{2}\big(\mathrm{g}(x)+\mathrm{h}(x)\big)=\tfrac{1}{2}\big(4\cos x-2\cos 2x-2x\big)=2\cos x-\cos 2x-x,$$

which is the required result.

Checks

- $f(0) = 2 - 1 - 0 = 1$ and $f'(x) = -2\sin x + 2\sin 2x - 1$, so $f'(0) = -1$.
- Substituting back: $f''(x) = -2\cos x + 4\cos 2x$ and $f(-x) = 2\cos x - \cos 2x + x$, which add to $x + 3\cos 2x$.
- The parities are right: g came out even and h odd, as they must.
- Stepping the pair $f(x)$, $f(-x)$ numerically from $x = 0$ to $x = 3$ gives $f(3) \approx -5.940155$, agreeing with $2\cos 3 - \cos 6 - 3$.

Common slips

- Forgetting the chain-rule sign: $\frac{d}{dx}f(-x) = -f'(-x)$, but $\frac{d^2}{dx^2}f(-x) = +f''(-x)$.
- Replacing x by $-x$ on only one side of the equation: the x on the right becomes $-x$ too.
- Getting $h'' + h = 2x$ and ending up with trigonometric terms in h.

Beyond STEP

This is a functional differential equation, in which the unknown is evaluated at more than one point. Splitting into even and odd parts works because reflection $x \mapsto -x$ is a linear map that squares to the identity, so its eigenvalues are ± 1 and every function splits uniquely into an even part (eigenvalue 1) and an odd part (eigenvalue -1). On each part the equation becomes an ordinary one. This idea of decomposing by symmetry runs all through linear algebra and Fourier analysis. For the constant-coefficient equations themselves, see Boyce and DiPrima, *Elementary Differential Equations*; for the linear-algebra viewpoint, Strang, *Introduction to Linear Algebra*.

WORKED SOLUTION

STEP II 1996, Question 9

Mechanics • Stability and Centre of Mass

STEP II 1996, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q9

The key idea

Everything hinges on where the centre of mass G of the toy lies on its axis of symmetry. On the hemisphere, the table always touches the toy directly below the centre O of the hemisphere's flat face, so whether G is below or above O decides between "rights itself" and "topples". On the cone, the toy rests along a whole generator, and it stays put provided the vertical through G meets that line of contact.

Solution

Model the toy as a rigid uniform solid of density ρ, symmetric about its axis. Let O be the centre of the common circular face and measure distances along the axis from O, positive towards the vertex V of the cone.

Centre of mass

The cone has volume $\frac{1}{3}\pi a^2(\lambda a) = \frac{1}{3}\lambda\pi a^3$ and its centre of mass is at $+\frac{1}{4}\lambda a$. The hemisphere has volume $\frac{2}{3}\pi a^3$ and its centre of mass is at $-\frac{3}{8}a$. So G is at

$$h = \frac{\frac{1}{3}\lambda \cdot \frac{1}{4}\lambda a - \frac{2}{3}\cdot\frac{3}{8}a}{\frac{1}{3}\lambda + \frac{2}{3}} = \frac{\frac{1}{12}\lambda^2 a - \frac{1}{4}a}{\frac{1}{3}(\lambda+2)} = \frac{a(\lambda^2-3)}{4(\lambda+2)}.$$

Hence G is below O if $\lambda < \sqrt{3}$, at O if $\lambda = \sqrt{3}$, and above O if $\lambda > \sqrt{3}$.

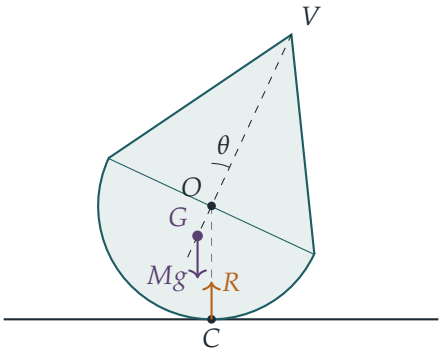
Resting on the hemisphere

Suppose the axis is tilted at angle θ to the upward vertical, with $0 \le \theta < \frac{1}{2}\pi$, so that the curved surface of the hemisphere is on the table.

? **Ask yourself** Where exactly does the hemisphere touch the table?

The table is tangent to the sphere of which the hemisphere is part, and a tangent plane

is perpendicular to the radius at the point of contact. So the contact point C is always *directly below* O, and O is always at height a, whatever θ is.



(The figure shows G below O, the case $\lambda < \sqrt{3}$.)

Take moments about C. The normal reaction and any friction act at C, so only the weight Mg at G has a moment. The horizontal displacement of G from C (that is, from the vertical through O) is $h\sin\theta$, measured towards the side to which the vertex leans.

- If $\lambda < \sqrt{3}$, then $h < 0$. For every θ in $0 < \theta < \frac{1}{2}\pi$, G lies on the opposite side of the vertical through C from the vertex. So the weight turns the toy back towards $\theta = 0$. Equivalently, G is at height $a + h\cos\theta = a - |h|\cos\theta$, which is least when $\theta = 0$, so the upright position is the position of least potential energy. Either way, wherever it is placed on its hemisphere, the toy moves to the upright position (rocking about it until friction and air resistance settle it there).
- If $\lambda > \sqrt{3}$, then $h > 0$. For any tilt $\theta > 0$, G lies on the same side as the vertex, and the weight turns the toy *further* from upright. The height $a + h\cos\theta$ of G is now greatest at $\theta = 0$, so upright is an unstable equilibrium. The slightest tilt grows until the rim reaches the table and the toy falls onto its conical surface: it overbalances.

(At $\lambda = \sqrt{3}$, $G = O$ and the toy is in neutral equilibrium at every tilt.) This is the required result.

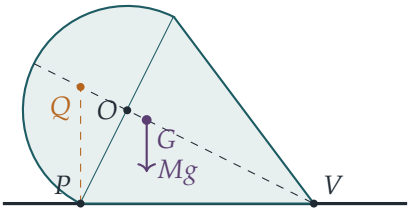
Resting on the cone

Now the toy lies with a generator VP of the cone on the table, where P is on the rim. Work in the plane through the axis and VP. Take axes in this plane with O at the origin, $V = (0, \lambda a)$ and $P = (a, 0)$, so the hemisphere is on the side $y < 0$. The tangent plane to the cone along VP is perpendicular to this plane, and its trace is the line

$$\frac{x}{a} + \frac{y}{\lambda a} = 1.$$

Ask yourself Could the hemisphere poke through the table instead, so that the cone can't lie flat on a generator?

No. Every point of the toy projects into this plane at a point of the cone's triangular section or of the half-disc $x^2 + y^2 \leq a^2$, $y \leq 0$. On the triangle, $\frac{x}{a} + \frac{y}{\lambda a} \leq 1$. On the half-disc, $x \leq a$ and $y \leq 0$, so $\frac{x}{a} + \frac{y}{\lambda a} \leq \frac{x}{a} \leq 1$, with equality only at P. So the whole toy lies on one side of the plane, touching it along the segment VP and nowhere else.



(The figure shows $\lambda = 2$. The perpendicular to VP at P is dashed.)

The reactions all act along VP and are perpendicular to the table. So the toy is in equilibrium on the table, and can't tip about P or V, exactly when the vertical through G meets the segment VP, i.e. when the foot of the perpendicular from G to VP lies between V and P.

The perpendicular to VP at V meets the axis only at V. Let the perpendicular to VP at P meet the axis at Q. In the triangle VPQ, the angle at P is a right angle and PO is the altitude to the hypotenuse VQ, so $OP^2 = OV \cdot OQ$, giving

$$OQ = \frac{a^2}{\lambda a} = \frac{a}{\lambda}.$$

So Q is a distance a/λ below O. A point of the axis projects onto the segment VP precisely when it lies between Q and V. So we need

$$-\frac{a}{\lambda} < h < \lambda a.$$

The right-hand inequality holds because G is inside the toy, below V. For the left-hand one, multiplying by $4\lambda(\lambda+2)/a > 0$,

$$\frac{a(\lambda^2-3)}{4(\lambda+2)} > -\frac{a}{\lambda} \iff \lambda^3 - 3\lambda + 4\lambda + 8 > 0 \iff \lambda^3 + \lambda + 8 > 0,$$

which is true for every $\lambda > 0$. So the vertical through G always meets the line of contact, and the toy stays where it is. By the rotational symmetry, it doesn't matter which generator is in contact either. So whatever the value of λ, the toy remains on its conical surface, as required.

Checks

- At $\lambda = \sqrt{3}$ the formula gives $h = 0$: the cone's moment $\frac{1}{12}\lambda^2 a$ exactly balances the hemisphere's $\frac{1}{4}a$. Slicing the toy into thin discs and summing numerically reproduces h to $10^{-5}a$ for several λ.
- With $\lambda = 1$: $h = -\frac{1}{6}a$ and $OQ = a$, and the foot of the perpendicular from G lies 0.583 of the way from V to P. With $\lambda = 10$ it lies 0.790 of the way along. Testing 20,000 random values of λ in $(0, 50)$, it always fell strictly inside VP.
- Limiting cases: as $\lambda \to 0$ the toy is almost a hemisphere ($h \to -\frac{3}{8}a$), which famously rights itself. As $\lambda \to \infty$ it is a long cone, and G moves up the axis with the cone's own centre of mass.

Common slips

- Taking moments about the lowest point of the *upright* toy rather than the actual contact point, which moves as the toy rolls but is always directly below O.
- In the last part, assuming the hemisphere touches the table too. It doesn't, since the half-disc lies on the side $y \leq 0$.

Beyond STEP

The rocking toy is a first look at stability of equilibrium. In general, a body rolling on a plane is stable when its centre of mass lies below the centre of curvature of its surface at the contact point (here, O). The same idea governs the metacentre of a floating ship. In a first-year course this becomes the study of minima of potential energy and small oscillations about them. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers centres of mass and rigid-body equilibrium in this spirit.

WORKED SOLUTION

STEP II 1996, Question 10

Mechanics • Projectile from Rotating Wheel

STEP II 1996, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q10

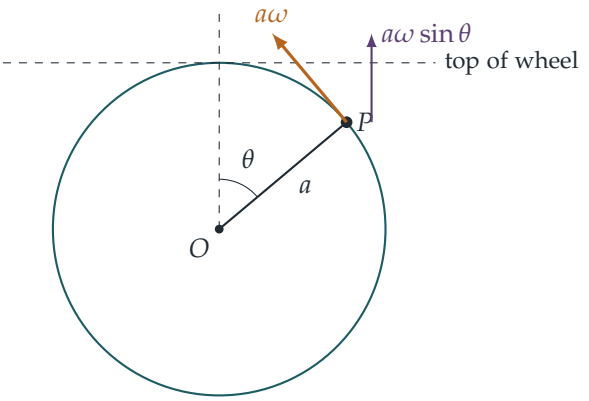
The key idea

At the moment she lets go, the heroine is a projectile whose initial position and velocity are fixed by where on the wheel she is. Only the vertical motion matters for the height, so the greatest height is her release height plus the extra height bought by the vertical component of her velocity. That gives the height as a function of a single release angle, and the question becomes: when can this function beat the top of the wheel, and what is its maximum?

Solution

Setting up

Model the heroine as a particle moving with the rim, and ignore air resistance. Let O be the centre of the wheel and suppose she lets go at the point P where OP makes an angle θ with the upward vertical, $0 \leq \theta \leq \pi$. Her speed on the rim is $a\omega$, directed along the tangent at P.



The tangent at P is perpendicular to OP, so it makes an angle θ with the horizontal. If

she is moving towards the top of the wheel when she lets go, her velocity has upward component $a\omega\sin\theta$ (shown in plum). If she is moving away from the top, her vertical velocity is downward or zero, so she never rises above her release point, which is no higher than the top; we may ignore those releases.

Her height above O at release is $a\cos\theta$. Under gravity alone her vertical velocity falls from $a\omega\sin\theta$ to zero, rising a further $(a\omega\sin\theta)^2/(2g)$, by $v^2 = u^2 - 2gs$. So her greatest height above the top of the wheel is

$$H(\theta) = a\cos\theta + \frac{a^2\omega^2\sin^2\theta}{2g} - a.$$

Write $c = \cos\theta \in [-1, 1]$ and $k = \dfrac{a\omega^2}{2g}$. Then $\sin^2\theta = 1 - c^2$ and

$$\begin{aligned}\frac{H}{a} &= c - 1 + k(1 - c^2)\\ &= (1 - c)\big[k(1 + c) - 1\big].\end{aligned}$$

Finding ω_0

At $c = 1$ (letting go at the top) $H = 0$, as we'd expect. For $c < 1$ the factor $1 - c$ is positive, so $H > 0$ exactly when $k(1 + c) > 1$.

> **Ask yourself** Over the allowed values of c, how large can $k(1 + c)$ be?

Since $c < 1$, $k(1 + c)$ takes every value in $[0, 2k)$, but never reaches $2k$. So:

- if $2k > 1$, we can choose $c < 1$ close enough to 1 that $k(1 + c) > 1$, and she rises above the top;
- if $2k \le 1$, then $k(1 + c) < 2k \le 1$ for every $c < 1$, so $H \le 0$ for every release point and she can never rise above the top.

The condition $2k \le 1$ is $a\omega^2 \le g$. Note that the boundary case $\omega^2 = g/a$ belongs to the "not possible" side, so the largest such ω exists and is

$$\omega_0 = \sqrt{\frac{g}{a}}.$$

(This is also the angular speed at which a particle on top of the wheel would just lose contact if it were resting on the outside: the centripetal acceleration $a\omega^2$ equals g.)

The greatest height when $\omega > \omega_0$

Now $2k = a\omega^2/g = \omega^2/\omega_0^2 > 1$. As a function of c,

$$\frac{H}{a} = -kc^2 + c + k - 1$$

is a downward-opening quadratic, maximised where its derivative $1 - 2kc$ vanishes, at

$$c = \frac{1}{2k} = \frac{\omega_0^2}{\omega^2}.$$

This lies strictly between 0 and 1, so it is an allowed release point (a little before the top of the wheel), and it gives the maximum over all of $[-1, 1]$. Substituting,

$$\frac{H_{\max}}{a} = -\frac{1}{4k} + \frac{1}{2k} + k - 1 = k - 1 + \frac{1}{4k}.$$

With $k = \frac{1}{2}\omega^2/\omega_0^2$ and so $\frac{1}{4k} = \frac{1}{2}\omega_0^2/\omega^2$,

$$H_{\max} = \frac{a}{2}\left(\frac{\omega^2}{\omega_0^2} - 2 + \frac{\omega_0^2}{\omega^2}\right) = \frac{a}{2}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)^2,$$

which is the required result. She achieves it by letting go where $\cos\theta = \omega_0^2/\omega^2$, while moving towards the top.

Checks

- At $\omega = \omega_0$ the formula gives 0, matching the boundary case; and it grows like $a\omega^2/(2\omega_0^2) = a^2\omega^2/(2g)$ for large ω, which is the height reached by a particle thrown straight up at speed $a\omega$ from the level of O. Indeed, for large ω the best release is near $\theta = \frac{\pi}{2}$, where the velocity is vertical.
- Units: ω/ω_0 is dimensionless, so $H_{\max}$ is a length.
- Numerically, with $a = 2\,\text{m}$ and $\omega = 2\omega_0$ the formula gives $\frac{a}{2}\left(2 - \frac{1}{2}\right)^2 = 2.25\,\text{m}$. Stepping the projectile from the release point $\cos\theta = \frac{1}{4}$ gives the same value, and a search over all release angles confirms that no other release does better.

Common slips

- Measuring the height from the centre of the wheel rather than from the top: the $-a$ in H matters.
- Assuming the best release is at $\theta = \frac{\pi}{2}$, where her velocity is vertical. That maximises the extra rise, not the total height.
- Forgetting to check that the optimal $c = \omega_0^2/\omega^2$ actually lies in $[-1, 1]$.

Beyond STEP

The method here, writing a quantity as a function of one control variable and optimising, is the simplest form of an optimal-control problem. A close cousin is the *envelope of trajectories* (the "parabola of safety"): the boundary of all points reachable from release

points on the wheel, which you can find by the same discriminant-style argument. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops projectile and rotational motion at first-year level.

WORKED SOLUTION

STEP II 1996, Question 11

Mechanics • Simple Harmonic Motion

STEP II 1996, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q11

The key idea

Work with the stretch of the string, which is the particle's position *relative to the lift*. Write Newton's second law for the particle in the ground frame and subtract the lift's known acceleration: what's left is simple harmonic motion about a shifted centre. The string stays taut as long as the stretch stays positive, and an SHM solution tells you exactly how low the stretch can go. When the lift stops accelerating, only the centre of the oscillation moves; the particle's position and velocity relative to the lift carry straight over, and they give the new amplitude.

Solution

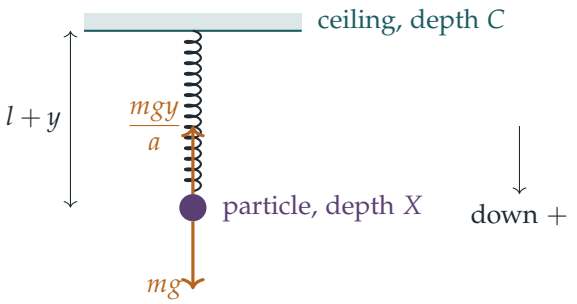
We model the particle as having mass m and the string as light and elastic, with modulus λ, obeying Hooke's law while it is taut. Throughout, y is the amount by which the string's length exceeds its natural length l, so $y = a$ at the start. This is the only reading that fits the stated differential equation. If y were measured from the old equilibrium length $l + a$, the right-hand side would be $-f$.

The equation of motion

In the initial equilibrium the tension $\lambda a/l$ balances the weight, so

$$\frac{\lambda a}{l} = mg \quad \Longrightarrow \quad \frac{\lambda}{l} = \frac{mg}{a},$$

and when the extension is $y > 0$ the tension is mgy/a.



Measure depths downwards from a point fixed to the ground: the ceiling is at depth C and the particle at depth X, so $X - C = l + y$. The lift descends with acceleration f, so $\ddot{C} = f$. Resolving vertically (downwards) for the particle, while the string is taut,

$$m\ddot{X} = mg - \frac{mgy}{a}.$$

Since $y = X - C - l$, we have $\ddot{y} = \ddot{X} - \ddot{C} = \ddot{X} - f$, and therefore

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + \frac{g}{a}y = g - f,$$

as required.

The string stays taut; amplitude af/g

Put $\omega^2 = g/a$. The constant particular solution is $y = (g-f)/\omega^2 = a(g-f)/g$, so the general solution is

$$y = a - \frac{af}{g} + A\cos\omega t + B\sin\omega t.$$

At $t = 0$ the particle is in its old equilibrium and moving with the lift, so $y(0) = a$ and $\dot{y}(0) = 0$. These give $A = af/g$ and $B = 0$:

$$y = a - \frac{af}{g} + \frac{af}{g}\cos\omega t.$$

This is SHM about $y = a - af/g$ with amplitude af/g. The particle's position relative to the lift is $l + y$, so the particle oscillates relative to the lift with amplitude af/g.

> **Ask yourself** We used Hooke's law, which assumes the string is taut. Isn't that circular?

No. We assume the string is taut and solve, then check that the solution never leaves the region where the assumption holds. The smallest value of y is

$$y_{\min} = a - \frac{2af}{g} = a\left(1 - \frac{2f}{g}\right) > 0,$$

because $f < g/2$. So y stays positive, the string is always taut and the solution is self-consistent. This is exactly where the condition $f < g/2$ is needed.

After the lift stops accelerating

For $t > T$ we have $\ddot{C} = 0$, and the same argument gives

$$\ddot{y} + \omega^2 y = g,$$

which is SHM about $y = a$, the natural equilibrium extension. Write $z = y - a$, so that $\ddot{z} + \omega^2 z = 0$.

? **Ask yourself** What are the starting conditions for this new motion?

The lift's acceleration jumps at $t = T$, but its velocity $\dot{C}$ doesn't, and nor do the particle's position and velocity. So $y = X - C - l$ and $\dot{y} = \dot{X} - \dot{C}$ are continuous at $t = T$. From the first stage, at $t = T$,

$$z = \frac{af}{g}(\cos\omega T - 1), \qquad \dot{z} = -\frac{af}{g}\,\omega \sin\omega T.$$

For SHM $\ddot{z} = -\omega^2 z$ the amplitude R satisfies $R^2 = z^2 + \dot{z}^2/\omega^2$ at any instant (this is energy conservation, or the identity $\cos^2 + \sin^2 = 1$ applied to $z = R\cos(\omega t + \varepsilon)$). Hence

$$\begin{aligned} R^2 &= \frac{a^2 f^2}{g^2}\left[(1 - \cos\omega T)^2 + \sin^2\omega T\right] \\ &= \frac{a^2 f^2}{g^2}\,(2 - 2\cos\omega T) \\ &= \frac{a^2 f^2}{g^2} \cdot 4\sin^2 \tfrac{1}{2}\omega T, \end{aligned}$$

using $1 - \cos\theta = 2\sin^2\frac{1}{2}\theta$. Therefore the new amplitude is

$$R = \frac{2af}{g}\left|\sin \tfrac{1}{2}\omega T\right|,$$

as required.

The string never becomes slack: the least extension is now

$$a - R \;\geq\; a - \frac{2af}{g} = a\left(1 - \frac{2f}{g}\right) > 0,$$

again because $f < g/2$. So $y > 0$ throughout and the equation of motion stays valid.

Checks

- If T is a whole number of periods, $\omega T = 2k\pi$, the particle is back at $y = a$ and at rest relative to the lift just when the lift stops accelerating. The formula gives amplitude 0: the particle hangs still, as it should.

- If ωT is an odd multiple of π, the particle is at its highest point, $y = a - 2af/g$, and momentarily at rest relative to the lift. The new centre is $y = a$, so the amplitude is $2af/g$, which is the maximum value of the formula.
- Numerically, we stepped the full motion in the ground frame (lift and particle, with tension switched off if the string went slack) for random a, $f < g/2$ and T. For example, with $a = 0.546$, $f/g = 0.337$ and $\omega T = 15.27$, the amplitudes came out as 0.18380 and 0.35864, matching af/g and $\frac{2af}{g}|\sin\frac{1}{2}\omega T|$ to six figures, and the extension never fell below $a - 2af/g$.

Common slips

- Measuring y from the old equilibrium length $l + a$. Then the right-hand side would be $-f$, not $g - f$.
- Restarting the second stage with $\dot{y} = 0$. The particle is generally moving relative to the lift at $t = T$, and that velocity contributes to the amplitude.
- Getting the sign of f wrong. A lift accelerating *downwards* reduces the effective gravity to $g - f$, so the centre of oscillation rises.

Beyond STEP

Subtracting the lift's acceleration is the same as working in an accelerating frame, where a uniform acceleration f acts like an extra gravitational field $-f$. That idea leads to fictitious forces and, eventually, to Einstein's equivalence principle. The second stage is a simple case of an oscillator driven by a step change in force. Its response depends on when the step happens relative to the oscillation, and the general tool for this is Green's functions (Duhamel's principle). Kleppner and Kolenkow, *An Introduction to Mechanics*, treats non-inertial frames and oscillators in this spirit.

WORKED SOLUTION

STEP II 1996, Question 12

Probability and Statistics • Maxima and Minima Distributions

STEP II 1996, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q12

The key idea

The maximum of several independent variables is awkward to handle directly, but the event "the maximum is small" is easy: it happens exactly when *every* variable is small, and independence lets us multiply. So we find the cumulative distribution function first and differentiate to get the density. Part (ii) is the mirror image: the *minimum* is large exactly when every variable is large.

Solution

Part (i)

Each X_j is uniform on $[0,1]$, so $\mathrm{P}(X_j < y) = y$ for $0 \le y \le 1$. For y in this range, $Y < y$ happens if and only if $X_j < y$ for every j, and since the X_j are independent,

$$F(y) = \mathrm{P}(Y < y) = \prod_{j=1}^{n} \mathrm{P}(X_j < y) = y^n, \qquad 0 \le y \le 1.$$

Of course $F(y) = 0$ for $y < 0$ and $F(y) = 1$ for $y > 1$. Differentiating, the probability density function of Y is

$$f(y) = \begin{cases} ny^{n-1}, & 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

(As a sanity check, $\int_0^1 ny^{n-1}\,\mathrm{d}y = 1$.)

Now the moments:

$$\mathrm{E}(Y) = \int_0^1 y \cdot ny^{n-1}\,\mathrm{d}y = \frac{n}{n+1},$$
$$\mathrm{E}(Y^2) = \int_0^1 y^2 \cdot ny^{n-1}\,\mathrm{d}y = \frac{n}{n+2}.$$

Hence

$$\mathrm{Var}(Y) = \frac{n}{n+2} - \frac{n^2}{(n+1)^2} = \frac{n\left[(n+1)^2 - n(n+2)\right]}{(n+2)(n+1)^2} = \frac{n}{(n+2)(n+1)^2},$$

since $(n+1)^2 - n(n+2) = n^2 + 2n + 1 - n^2 - 2n = 1$. This is the required result.

Ask yourself Why work with $\mathrm{P}(Y < y)$ rather than trying to write down $\mathrm{P}(Y = y)$ or a density for Y straight away?

For a continuous variable the density isn't a probability, and "the largest equals y" mixes up which X_j is largest with what the others do. The event $\{Y < y\}$ is an intersection of n independent events, so its probability is a simple product. Getting the distribution function and then differentiating is the standard route for maxima and minima.

Part (ii)

Let T_j be the failure time of the jth light, so $\mathrm{P}(T_j \le t) = 1 - \mathrm{e}^{-t/\lambda}$ and

$$\mathrm{P}(T_j > t) = \mathrm{e}^{-t/\lambda}, \qquad t > 0.$$

The lights are independent, so the T_j are independent. Let $T = \min(T_1, \ldots, T_n)$ be the time of the first failure. This time the convenient event is $T > t$, which happens if and only if every light is still working at time t:

$$\mathrm{P}(T > t) = \prod_{j=1}^{n} \mathrm{P}(T_j > t) = \left(\mathrm{e}^{-t/\lambda}\right)^n = \mathrm{e}^{-nt/\lambda}.$$

So $\mathrm{P}(T \le t) = 1 - \mathrm{e}^{-nt/\lambda}$, and differentiating gives the density

$$g(t) = \frac{n}{\lambda}\,\mathrm{e}^{-nt/\lambda}, \qquad t > 0.$$

In other words, T has the same kind of distribution as a single light, with λ replaced by λ/n. Integrating by parts with $u = t$,

$$\begin{aligned}\mathrm{E}(T) &= \int_0^\infty t\,\frac{n}{\lambda}\,\mathrm{e}^{-nt/\lambda}\,\mathrm{d}t = \Big[-t\,\mathrm{e}^{-nt/\lambda}\Big]_0^\infty + \int_0^\infty \mathrm{e}^{-nt/\lambda}\,\mathrm{d}t \\ &= 0 + \Big[-\frac{\lambda}{n}\,\mathrm{e}^{-nt/\lambda}\Big]_0^\infty = \frac{\lambda}{n},\end{aligned}$$

where the boundary term vanishes because $t\,\mathrm{e}^{-nt/\lambda} \to 0$ as $t \to \infty$. So the expected time until the first failure is λ/n, as required.

Checks

- With $n = 1$, Y is just uniform on $[0, 1]$, and the formula gives $\frac{1}{3\cdot 4} = \frac{1}{12}$, the familiar variance of a uniform variable.

- As n grows, the variance behaves like $1/n^2$: the maximum of many uniforms is squeezed up against 1, as you'd expect.
- With $n = 1$ in part (ii), the expected lifetime of one light is λ, which is the mean of an exponential distribution with parameter $1/\lambda$.
- A simulation of 200 000 maxima of five uniforms gave variance 0.01993, against $\frac{5}{7 \cdot 36} \approx 0.01984$; simulating four lights with $\lambda = 3$ gave a mean first failure time of 0.749, against $\frac{3}{4}$.

Common slips

- Differentiating $\mathrm{P}(T > t)$ and forgetting the minus sign, which gives a negative "density".
- Writing $\mathrm{Var}(Y) = \mathrm{E}(Y^2) - \mathrm{E}(Y)$ instead of subtracting $(\mathrm{E}(Y))^2$.
- Thinking that n lights should fail on average at time λ each, so the first fails at λ. The minimum of n variables is much smaller than any one of them.

Beyond STEP

The maximum and minimum are the extreme cases of *order statistics*: if you sort a sample, the kth smallest of n uniforms has a beta distribution, and Y here is the case $k = n$. Part (ii) shows the key fact that the minimum of independent exponential variables is again exponential, with the rates adding. This drives the theory of Poisson processes and continuous-time Markov chains. Grimmett and Welsh, *Probability: An Introduction*, and Blitzstein and Hwang, *Introduction to Probability*, both develop these ideas.

WORKED SOLUTION

STEP II 1996, Question 13

Probability and Statistics • Random Walks on a Grid

STEP II 1996, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q13

The key idea

The identity is Vandermonde's: compare coefficients of t^n on the two sides. For the walk, the useful quantity is a single number that every step changes by exactly one: how many blocks north plus how many blocks west you are of my hotel. Each of my steps raises it by 1, and each of my friend's steps lowers it by 1. So we can only meet at one moment, along one diagonal line of intersections, and the meeting probability becomes exactly the sum in the identity.

Solution

The identity

The coefficient of t^r in $(1+t)^n$ is $\binom{n}{r}$. A term in t^n in the product $(1+t)^n(1+t)^n$ comes from choosing t^r from the first factor and t^{n-r} from the second, for some $r = 0, 1, \ldots, n$. So the coefficient of t^n on the left is

$$\sum_{r=0}^{n} \binom{n}{r}\binom{n}{n-r},$$

and on the right it is $\binom{2n}{n}$. Equating the two gives the required identity. The same argument applied to $(1+t)^a(1+t)^b = (1+t)^{a+b}$ gives the more general version

$$\sum_{r} \binom{a}{r}\binom{b}{n-r} = \binom{a+b}{n}, \qquad (*)$$

which we shall need later.

Where we can meet

Put my hotel at the origin, with x measured east and y north in blocks, so my friend's hotel is at $(-n, n)$. For any position define

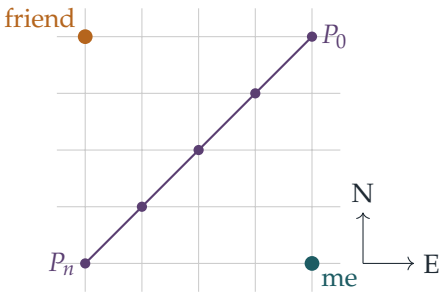
$$D = y - x,$$

the number of blocks north plus the number of blocks west of my hotel. Going north or going west raises D by 1; going south or going east lowers it by 1. So after walking for t minutes, I have $D = t$, and my friend, who starts at $D = 2n$, has $D = 2n - t$. This holds at every instant, mid-block as well, because D changes steadily as we walk.

To be at the same place at the same time we need equal values of D, so $t = 2n - t$, giving $t = n$. Hence we can meet only after exactly n minutes, and then at an intersection with $y - x = n$. My n steps keep me inside the region $-n \le x \le 0$, $0 \le y \le n$, so the possible meeting points are

$$P_r = (-r,\, n - r), \qquad r = 0, 1, \ldots, n,$$

which are the $n + 1$ intersections on the diagonal of the square joining the intersection n blocks north of my hotel to the one n blocks west of it. (In particular we can never pass each other in the middle of a block.)



The sketch shows $n = 4$: the meeting points are the dots on the diagonal.

The probability of meeting

I am at P_r after n minutes exactly when r of my n steps were west and $n - r$ were north. Each sequence of n steps has probability 2^{-n}, so

$$\mathrm{P}(\text{I am at } P_r) = \binom{n}{r}\frac{1}{2^n}.$$

My friend reaches $P_r = (-r, n - r)$ from $(-n, n)$ after n steps exactly when $n - r$ of them were east and r were south, which has probability $\binom{n}{n-r}2^{-n}$. Our walks are independent, and the events "we are both at P_r" for different r are mutually exclusive, so

$$\mathrm{P}(\text{meet}) = \sum_{r=0}^{n} \binom{n}{r}\binom{n}{n-r}\frac{1}{4^n} = \frac{1}{4^n}\binom{2n}{n},$$

by the identity.

Leaving $2k$ minutes late

Now measure time T from when my friend sets off. For $T \ge 2k$ I have walked for $T - 2k$ minutes, so I have $D = T - 2k$, while my friend has $D = 2n - T$. These are equal only

when $T = n + k$. Then I have taken $n - k$ steps (here $k \le n$, so this is a non-negative whole number) and my friend has taken $n + k$. Since $n + k \ge 2k$, my friend cannot reach my side of the diagonal while I am still asleep in my hotel. Again we can only meet at an intersection, never mid-block.

After $n - k$ steps, r of them west, I am at $(-r, n - k - r)$, with probability $\binom{n-k}{r}2^{-(n-k)}$. My friend is at the same point if $n - r$ of her $n + k$ steps were east and $k + r$ were south, which has probability $\binom{n+k}{n-r}2^{-(n+k)}$. Summing over r as before,

$$\mathrm{P}(\text{meet}) = \frac{1}{2^{n-k}\,2^{n+k}} \sum_{r=0}^{n-k} \binom{n-k}{r}\binom{n+k}{n-r} = \frac{1}{4^n}\binom{2n}{n},$$

by $(*)$ with $a = n - k$ and $b = n + k$.

Ask yourself Why should oversleeping make no difference at all?

Look at the walk from my friend's point of view, measuring my position relative to hers. Each of her steps (south or east) and each of my steps (north or west) moves me, relative to her, one block north or one block west, with probability $\frac{1}{2}$ each, independently. We start n blocks apart in each direction, and we meet exactly when the total of $2n$ steps (mine plus hers) consists of n "north" and n "west" relative moves. It doesn't matter who makes which steps, so the answer is $\binom{2n}{n}/2^{2n}$ however the $2n$ steps are shared out, whether n and n, or $n - k$ and $n + k$. The condition on k just ensures that I really do get to walk my share. For large n the probability is about $1/\sqrt{\pi n}$, which is small, and it is the same whether or not I oversleep by an even number of minutes.

$n = 1$, one minute late

Now I have $D = T - 1$ and my friend has $D = 2 - T$, which are equal only at $T = \frac{3}{2}$, a time when we are both halfway along a block. So we can *never* be at the same intersection at the same time, and if meeting means meeting at an intersection, the probability is 0. The parity has changed: the total number of steps is now odd.

If we count passing each other in the middle of a block (as two real people would), we meet at $T = \frac{3}{2}$ when we walk along the same block in opposite directions. At $T = 1$ I am still at my hotel $(0, 0)$ and my friend is at $(0, 1)$ or $(-1, 0)$. We then meet if she is at $(0, 1)$ and goes south while I go north, or she is at $(-1, 0)$ and goes east while I go west. Each of these has probability $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}$, so

$$\mathrm{P}(\text{meet mid-block}) = \tfrac{1}{8} + \tfrac{1}{8} = \tfrac{1}{4}.$$

Checks

- For $n = 1$ with no delay we meet if I go north and she goes east, or I go west and she goes south: probability $\frac{1}{4} + \frac{1}{4} = \frac{1}{2} = \binom{2}{1}/4$.
- By computer, we enumerated every pair of walks for $n \le 5$ and all allowed k, tracking both walkers every half minute. The meeting probabilities were $\frac{1}{2}, \frac{3}{8}, \frac{5}{16}, \frac{35}{128}, \frac{63}{256}$

whatever the even delay, always at the single predicted time. With $n = 1$ and a one-minute delay, the only meetings were mid-block at $T = \frac{3}{2}$, with total probability $\frac{1}{4}$.

- For $n = 100$, $\binom{200}{100}/4^{100} \approx 0.0563$, close to $1/\sqrt{100\pi} \approx 0.0564$.

Common slips

- Assuming we could meet anywhere in the square at any time. The quantity $y - x$ pins the meeting down to one time and one diagonal.
- Squaring the probabilities without summing over the meeting points, or forgetting that my friend's step counts at P_r are swapped (she needs $n - r$ east steps, not r).

Beyond STEP

The relative-position argument is a coupling: two independent random walks are replaced by one walk of combined length, which is why $\binom{2n}{n}/4^n$ keeps reappearing. The same quantity is the probability that a simple symmetric random walk on a line is back at the origin after $2n$ steps, and its $1/\sqrt{\pi n}$ decay (from Stirling's formula) is the key to Pólya's theorem that such walks in one and two dimensions return to the start with probability 1. Grimmett and Welsh, *Probability: An Introduction*, treats random walks and these return probabilities.

WORKED SOLUTION

STEP II 1996, Question 14

Probability and Statistics • Moments and Central Limit Theorem

STEP II 1996, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q14

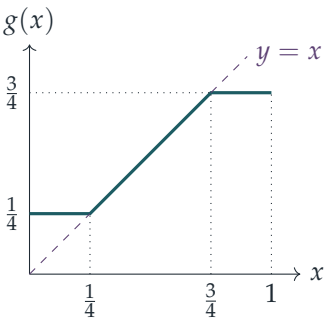
The key idea

Y is X with its two ends "clipped": all the probability below $\frac{1}{4}$ is piled up at the single value $\frac{1}{4}$, and all the probability above $\frac{3}{4}$ at $\frac{3}{4}$. So Y is a mixed random variable, with two point masses and a continuous middle piece, and any expectation $\mathrm{E}(g(Y))$ splits into three pieces in the same way. The final part has a sum of a very large number of independent copies, which calls for the central limit theorem. For that we need only the mean and variance of Y, and the earlier parts have already done that work.

Solution

The moments of Y

Since $X \sim \mathrm{U}[0,1]$, we have $\mathrm{P}(X \leq \frac{1}{4}) = \mathrm{P}(X \geq \frac{3}{4}) = \frac{1}{4}$, and on $[\frac{1}{4}, \frac{3}{4}]$ the density of X is 1. Writing $Y = g(X)$, where g is the clipping function sketched below, we get $\mathrm{E}(Y^n) = \int_0^1 g(x)^n \, \mathrm{d}x$.



Splitting the integral at $\frac{1}{4}$ and $\frac{3}{4}$,

$$\begin{aligned}\mathrm{E}(Y^n) &= \left(\tfrac{1}{4}\right)^n \cdot \tfrac{1}{4} + \int_{1/4}^{3/4} x^n \,\mathrm{d}x + \left(\tfrac{3}{4}\right)^n \cdot \tfrac{1}{4} \\ &= \frac{1}{4^{n+1}} + \frac{3^{n+1}-1}{(n+1)4^{n+1}} + \frac{3^n}{4^{n+1}} \\ &= \frac{(n+1)(1+3^n)+3^{n+1}-1}{(n+1)4^{n+1}}.\end{aligned}$$

The numerator is $n+1+(n+1)3^n+3\cdot 3^n-1 = n+(n+4)3^n$, so

$$\boxed{\mathrm{E}(Y^n) = \frac{n+(n+4)\,3^n}{(n+1)\,4^{n+1}}} \qquad (n \geq 1).$$

Comparing with X

For the uniform variable, $\mathrm{E}(X^n) = \int_0^1 x^n \,\mathrm{d}x = \dfrac{1}{n+1}$.

With $n=1$, the formula gives $\mathrm{E}(Y) = \dfrac{1+5\cdot 3}{2\cdot 16} = \dfrac{1}{2} = \mathrm{E}(X)$, as required. (This is no surprise: the clipping moves the mass at the two ends inwards by matching amounts, symmetrically about $\frac{1}{2}$.)

With $n=2$, $\mathrm{E}(Y^2) = \dfrac{2+6\cdot 9}{3\cdot 64} = \dfrac{56}{192} = \dfrac{7}{24}$, while $\mathrm{E}(X^2) = \frac{1}{3} = \frac{8}{24}$. Hence

$$\mathrm{E}(X^2) - \mathrm{E}(Y^2) = \frac{8}{24} - \frac{7}{24} = \frac{1}{24},$$

as required.

For general $n \geq 2$, write both moments over the common denominator $(n+1)4^{n+1}$:

$$\mathrm{E}(X^n) = \frac{4^{n+1}}{(n+1)4^{n+1}}.$$

So we must show that $4^{n+1} > n+(n+4)3^n$. Using $4^{n+1} = 4(3+1)^n$ and the binomial theorem, and keeping the top two terms separate,

$$4^{n+1} = 4\cdot 3^n + 4n\cdot 3^{n-1} + 4\sum_{k=0}^{n-2}\binom{n}{k}3^k.$$

Subtracting $n+(n+4)3^n = 4\cdot 3^n + 3n\cdot 3^{n-1} + n$ gives

$$4^{n+1} - \left(n+(n+4)3^n\right) = n\,3^{n-1} - n + 4\sum_{k=0}^{n-2}\binom{n}{k}3^k.$$

Ask yourself Why is this difference positive?

For $n \geq 2$ we have $3^{n-1} \geq 3$, so $n\,3^{n-1} - n \geq 2n > 0$. The sum is also positive, since it contains the term $k = 0$, which equals 1, and every term is non-negative. Hence the difference is strictly positive, and

$$\mathrm{E}(X^n) > \mathrm{E}(Y^n) \qquad \text{for all } n \geq 2.$$

(For $n = 1$ the difference is $3^0 - 1 + 0 = 0$, which matches $\mathrm{E}(X) = \mathrm{E}(Y)$.)

The sum of 240 000 copies

From the previous part,

$$\mathrm{Var}(Y) = \mathrm{E}(Y^2) - \big(\mathrm{E}(Y)\big)^2 = \frac{7}{24} - \frac{1}{4} = \frac{1}{24}.$$

Let $S = Y_1 + \cdots + Y_N$ with $N = 240\,000$. Since the Y_i are independent, expectations and variances add:

$$\mathrm{E}(S) = \frac{N}{2} = 120\,000, \qquad \mathrm{Var}(S) = \frac{N}{24} = 10\,000,$$

so S has standard deviation 100. The Y_i are independent and identically distributed with finite variance and N is huge, so by the central limit theorem S is approximately $\mathrm{N}(120\,000,\, 100^2)$. Hence

$$\mathrm{P}(S < K) \approx \Phi\Big(\frac{K - 120\,000}{100}\Big) = \frac{3}{4} \quad \Longrightarrow \quad \frac{K - 120\,000}{100} = \Phi^{-1}(0.75) \approx 0.6745,$$

so

$$K \approx 120\,000 + 67.45 \approx 120\,067.$$

No continuity correction is needed: Y is not integer-valued, and with a standard deviation of 100 any such correction would make no difference to this accuracy anyway.

Checks

- The formula for $\mathrm{E}(Y^n)$ lies between $(\frac{1}{4})^n$ and $(\frac{3}{4})^n$, as it must, since $\frac{1}{4} \leq Y \leq \frac{3}{4}$. For $n = 3$ it gives $\frac{3+7\cdot 27}{4\cdot 256} = \frac{192}{1024} = \frac{3}{16} = 0.1875$, and a midpoint-rule integral of $g(x)^3$ agrees to ten decimal places.
- $\mathrm{Var}(Y) = \frac{1}{24} < \frac{1}{12} = \mathrm{Var}(X)$: clipping pulls values towards the mean, so the spread should drop.
- A simulation of 20 000 sums of 2400 copies of Y gave a standardised upper quartile of about 0.678, close to 0.6745, which supports the normal approximation.

Common slips

- Treating Y as uniform on $[\frac{1}{4}, \frac{3}{4}]$, and so forgetting the point masses of $\frac{1}{4}$ at each end. A uniform variable on that interval has a density of 2, not 1.

- Using $\mathrm{Var}(S) = N^2 \mathrm{Var}(Y)$ rather than $N \mathrm{Var}(Y)$. Independence makes variances add, so the standard deviation grows like $\sqrt{N}$.
- Getting the sign wrong in the last step. $\mathrm{P}(S < K) = \frac{3}{4} > \frac{1}{2}$, so K must be above the mean.

Beyond STEP

Y is a clipped (or "Winsorised") version of X, and it has a distribution that is neither discrete nor continuous. Measure-theoretic probability handles such distributions uniformly through the distribution function, and every expectation becomes a single integral $\int g \, \mathrm{d}F$. The inequality $\mathrm{E}(X^n) > \mathrm{E}(Y^n)$ with equal means is an instance of convex ordering: X is a "mean-preserving spread" of Y, so $\mathrm{E}(\phi(X)) \geq \mathrm{E}(\phi(Y))$ for every convex ϕ. The central limit theorem used at the end is proved in Grimmett and Welsh, *Probability: An Introduction*.

WORKED SOLUTION

STEP III 1996, Question 1

Pure • Powers of Hyperbolic Functions

STEP III 1996, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q1

The key idea

Everything follows from writing hyperbolic functions as exponentials. A power $\cosh^n x$ becomes $2^{-n}(e^x + e^{-x})^n$, which the binomial theorem expands into exponentials $e^{(n-2r)x}$; these come in matching pairs $e^{\pm jx}$, and each pair is a single $\cosh jx$. The same expansion for $\sinh^{2m} x$ differs only by alternating signs, so adding or subtracting kills every other term.

Solution

Definitions and the two identities

We define

$$\cosh x = \frac{e^x + e^{-x}}{2}, \qquad \sinh x = \frac{e^x - e^{-x}}{2}.$$

Two facts follow at once. Since $\cosh x + \sinh x = e^x$ and $\cosh x - \sinh x = e^{-x}$,

$$\cosh^2 x - \sinh^2 x = (\cosh x + \sinh x)(\cosh x - \sinh x) = e^x e^{-x} = 1.$$

Squaring the definitions, $\cosh^2 x = \frac{1}{4}(e^{2x} + 2 + e^{-2x})$ and $\sinh^2 x = \frac{1}{4}(e^{2x} - 2 + e^{-2x})$, so

$$\cosh^2 x + \sinh^2 x = \tfrac{1}{2}(e^{2x} + e^{-2x}) = \cosh 2x,$$

and, adding or subtracting the identity $\cosh^2 x - \sinh^2 x = 1$,

$$\cosh^2 x = \tfrac{1}{2}(\cosh 2x + 1), \qquad \sinh^2 x = \tfrac{1}{2}(\cosh 2x - 1).$$

The last two hold for any argument, so we may replace x by $2x$ in them.

Now factorise the difference of squares:

$$\cosh^4 x - \sinh^4 x = (\cosh^2 x - \sinh^2 x)(\cosh^2 x + \sinh^2 x) = 1 \cdot \cosh 2x = \cosh 2x,$$

as required. For the sum, complete the square and use $\cosh x \sinh x = \frac{1}{4}(e^{2x} - e^{-2x}) = \frac{1}{2}\sinh 2x$:

$$\begin{aligned}\cosh^4 x + \sinh^4 x &= (\cosh^2 x + \sinh^2 x)^2 - 2\cosh^2 x \sinh^2 x \\ &= \cosh^2 2x - \tfrac{1}{2}\sinh^2 2x \\ &= \tfrac{1}{2}(\cosh 4x + 1) - \tfrac{1}{4}(\cosh 4x - 1) \\ &= \tfrac{1}{4}\cosh 4x + \tfrac{3}{4},\end{aligned}$$

which is the second identity.

The coefficients a_k

By the binomial theorem,

$$\cosh^n x = \frac{1}{2^n}\left(e^x + e^{-x}\right)^n = \frac{1}{2^n}\sum_{r=0}^{n}\binom{n}{r}e^{(n-2r)x}.$$

Pair the term with index r with the term with index $n-r$. They have the same coefficient, since $\binom{n}{r} = \binom{n}{n-r}$, and opposite exponents $\pm(n-2r)x$, so together they give $2\binom{n}{r}\cosh\big((n-2r)x\big)$.

Ask yourself Which multiples $\cosh kx$ actually appear, and what happens to the middle term?

The exponents $n-2r$ are $n, n-2, n-4, \ldots$, so only k with the same parity as n occur. Writing $k = n-2r$, the coefficient of $\cosh kx$ for $1 \le k \le n$ is $2\cdot 2^{-n}\binom{n}{r}$ with $r = \frac{1}{2}(n-k)$. When n is even there is also an unpaired middle term $r = \frac{n}{2}$, giving the constant $2^{-n}\binom{n}{n/2}$. Hence

$$a_k = \begin{cases} \dfrac{1}{2^{n-1}}\dbinom{n}{\frac{1}{2}(n-k)} & \text{if } 1 \le k \le n \text{ and } n-k \text{ is even,} \\ 0 & \text{if } n-k \text{ is odd,}\end{cases} \qquad a_0 = \begin{cases} \dfrac{1}{2^n}\dbinom{n}{\frac{1}{2}n} & n \text{ even,} \\ 0 & n \text{ odd.}\end{cases}$$

For example, $\cosh^5 x = \frac{5}{8}\cosh x + \frac{5}{16}\cosh 3x + \frac{1}{16}\cosh 5x$.

The final part

Take $n = 2m$ and expand $\sinh^{2m} x$ in the same way:

$$\sinh^{2m} x = \frac{1}{2^{2m}}\sum_{r=0}^{2m}(-1)^r\binom{2m}{r}e^{(2m-2r)x}.$$

The terms r and $2m-r$ still pair up, because $(-1)^{2m-r} = (-1)^r$, so

$$\sinh^{2m} x = \frac{1}{2^{2m}}\sum_{r=0}^{2m}(-1)^r\binom{2m}{r}\cosh\big(2(m-r)x\big),$$

which is the expansion of $\cosh^{2m} x$ (written the same way, with cosh even) with alternating signs. Subtracting and adding,

$$\cosh^{2m} x \mp \sinh^{2m} x = \frac{1}{2^{2m}} \sum_{r=0}^{2m} \left(1 \mp (-1)^r\right) \binom{2m}{r} \cosh\bigl(2(m-r)x\bigr).$$

So the difference keeps only odd r (each doubled) and the sum keeps only even r. Collecting the pairs r, $2m-r$ as before, and writing $k = 2j = 2(m-r)$, we get

$$\cosh^{2m} x - \sinh^{2m} x = b_0 + \sum_{\substack{1 \le j \le m \\ m-j \text{ odd}}} \frac{1}{2^{2m-2}} \binom{2m}{m-j} \cosh 2jx,$$

$$\cosh^{2m} x + \sinh^{2m} x = c_0 + \sum_{\substack{1 \le j \le m \\ m-j \text{ even}}} \frac{1}{2^{2m-2}} \binom{2m}{m-j} \cosh 2jx,$$

where the constant terms come from the unpaired middle term $r = m$, which survives in the difference when m is odd and in the sum when m is even:

$$b_0 = \begin{cases} \dfrac{1}{2^{2m-1}} \dbinom{2m}{m} & m \text{ odd}, \\ 0 & m \text{ even}, \end{cases} \qquad c_0 = \begin{cases} \dfrac{1}{2^{2m-1}} \dbinom{2m}{m} & m \text{ even}, \\ 0 & m \text{ odd}. \end{cases}$$

In words: the coefficient of $\cosh kx$ is 0 for odd k. For even $k = 2j \ge 2$ it is $2^{2-2m}\binom{2m}{m-j}$ in the difference when $m-j$ is odd, and in the sum when $m-j$ is even, and 0 otherwise. With $m = 2$ the difference is $\frac{1}{4}\binom{4}{1}\cosh 2x = \cosh 2x$ and the sum is $\frac{1}{8}\binom{4}{2} + \frac{1}{4}\binom{4}{0}\cosh 4x = \frac{3}{4} + \frac{1}{4}\cosh 4x$, recovering the identities we proved first.

Checks

Put $x = 0$: every $\cosh kx$ becomes 1, so the a_k must sum to 1 (they do: the binomial coefficients sum to 2^n), and both final expansions must also sum to 1. For $m = 3$ the formulas give $\cosh^6 x - \sinh^6 x = \frac{5}{8} + \frac{3}{8}\cosh 4x$ and $\cosh^6 x + \sinh^6 x = \frac{15}{16}\cosh 2x + \frac{1}{16}\cosh 6x$. At $x = 1$ both sides of the first are 10.87 to four significant figures, which a calculator confirms. We checked all the formulas numerically for $n \le 12$ and $m \le 8$ at random values of x.

Common slips

- Doubling the middle term: when n is even, the $r = \frac{n}{2}$ term has no partner, so a_0 is $2^{-n}\binom{n}{n/2}$, not twice that.
- Forgetting that $a_k = 0$ whenever k and n have opposite parity.
- Assuming the constant term is always present in the final part. It appears in the difference only for odd m and in the sum only for even m.

Beyond STEP

Writing powers of $\cosh x$ (or $\cos x$) as combinations of $\cosh kx$ is a first taste of Fourier analysis: e^{ikx} are the natural building blocks, and $\cos^n x$ is a finite Fourier series whose coefficients are binomial. Reversing the process, expressing $\cos nx$ as a polynomial in $\cos x$, gives the Chebyshev polynomials, which matter in approximation theory. Needham's *Visual Complex Analysis* shows why the exponential form makes all these identities natural.

WORKED SOLUTION

STEP III 1996, Question 2

Pure • Simultaneous Linear Equations

STEP III 1996, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q2

The key idea

Three linear equations in three unknowns have exactly one solution unless the determinant of the coefficients vanishes. So the first job is to find the values of a that make the determinant zero; for every other a we simply solve. The real content of the question is in the exceptional values, where we must decide, depending on b, whether the equations are inconsistent or have infinitely many solutions.

Solution

Label the equations

$$x + y + az = 2, \tag{1}$$
$$x + ay + z = 2, \tag{2}$$
$$2x + y + z = 2b. \tag{3}$$

When is there a unique solution?

The coefficient matrix has determinant

$$\begin{vmatrix} 1 & 1 & a \\ 1 & a & 1 \\ 2 & 1 & 1 \end{vmatrix} = 1(a-1) - 1(1-2) + a(1-2a)$$
$$= 2a - 2a^2 = 2a(1-a).$$

So there is exactly one solution, whatever b is, when $a \neq 0$ and $a \neq 1$. The cases $a = 0$ and $a = 1$ need separate treatment.

The general case: $a \neq 0, 1$

Subtracting (2) from (1) gives

$$(1-a)y - (1-a)z = 0,$$

and since $a \neq 1$ we may divide by $1-a$ to get $y = z$. Putting $z = y$ into (1) and (3):

$$x + (1+a)y = 2, \qquad 2x + 2y = 2b, \quad \text{i.e.} \quad x + y = b.$$

Subtracting the second from the first gives $ay = 2 - b$, and since $a \neq 0$,

$$y = z = \frac{2-b}{a}, \qquad x = b - \frac{2-b}{a} = \frac{(a+1)b-2}{a}.$$

Substituting back into (1), (2) and (3) confirms that this is a solution, and by the argument above it is the only one.

The case $a = 0$

The equations become

$$x + y = 2, \qquad x + z = 2, \qquad 2x + y + z = 2b.$$

Adding the first two gives $2x + y + z = 4$. So the left-hand side of the third equation must equal both 4 and $2b$.

> **Ask yourself** Does a zero determinant mean "no solution"?

Not by itself: it means the equations are dependent, so either they contradict each other or one of them adds no new information. Which happens depends on the right-hand sides, here on b.

- If $b \neq 2$, then $4 = 2b$ is false, so there is **no solution**.
- If $b = 2$, the third equation is just the sum of the first two, so only $x + y = 2$ and $x + z = 2$ remain. Writing $x = t$, the solutions are

$$(x, y, z) = (t,\ 2-t,\ 2-t), \qquad t \text{ any real number.}$$

The case $a = 1$

Now (1) and (2) are the same equation, $x + y + z = 2$, and the system is

$$x + y + z = 2, \qquad 2x + y + z = 2b.$$

Subtracting the first from the second gives $x = 2b - 2$, and then $y + z = 2 - x = 4 - 2b$. These two equations never contradict each other, so for *every* value of b there are infinitely many solutions:

$$(x, y, z) = (2b-2,\ s,\ 4-2b-s), \qquad s \text{ any real number.}$$

Summary

- $a \neq 0, 1$: the unique solution $x = \dfrac{(a+1)b-2}{a}$, $y = z = \dfrac{2-b}{a}$.

- $a = 0$, $b \neq 2$: no solution.
- $a = 0$, $b = 2$: $(t, 2 - t, 2 - t)$ for any t.
- $a = 1$, any b: $(2b - 2, s, 4 - 2b - s)$ for any s.

Checks

Take $a = 2$, $b = 4$: the formula gives $y = z = -1$ and $x = 5$, and indeed $5 - 1 - 2 = 2$, $5 - 2 - 1 = 2$ and $10 - 1 - 1 = 8 = 2b$. As a structural check, the general formula has a in the denominator, so it must break down at $a = 0$; it does, and our separate analysis shows why. At $a = 1$ the formula still makes sense and gives $(2b - 2, 2 - b, 2 - b)$, which is the member $s = 2 - b$ of our family: it is one solution among many, because $y = z$ is no longer forced. We verified all of the cases with exact rational arithmetic on many random values of a, b and the parameters.

Common slips

- Dividing by $1 - a$ or by a without saying that $a \neq 1$ or $a \neq 0$. Those divisions are exactly where the special cases hide.
- Assuming that a zero determinant means no solution. For $a = 1$ there are always infinitely many solutions, and for $a = 0$ there are some when $b = 2$.
- Forgetting to describe the infinite families explicitly: "infinitely many solutions" is not the same as solving the equations.

Beyond STEP

Each equation describes a plane in three dimensions, and the question asks how three planes can meet: in a single point, along a common line, or not at all (for $a = 0$, $b \neq 2$ the planes meet in pairs along three parallel lines, like the faces of a prism). The systematic tool is Gaussian elimination and the idea of the rank of a matrix: a system is consistent exactly when appending the right-hand side does not raise the rank. Strang's *Introduction to Linear Algebra* develops this picture beautifully.

WORKED SOLUTION

STEP III 1996, Question 3

Pure • Half-Angle Substitution

STEP III 1996, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q3

The key idea

An integrand built from $1/(\text{constant} + \text{constant} \cdot \cos x)$ is the classic case for the half-angle substitution $t = \tan \frac{1}{2}x$, which turns any rational function of $\sin x$ and $\cos x$ into a rational function of t. Here it produces a standard $\int \mathrm{d}t/(k^2 + t^2)$, giving an inverse tangent. The two "hence" results are then just a matter of rescaling each integrand into the form $1/(1 - a\cos x)$ and choosing a and θ.

Solution

The general integral

Put $t = \tan \frac{1}{2}x$. As x runs from 0 to θ, t runs from 0 to $T = \tan \frac{1}{2}\theta$, which is finite and positive because $0 < \frac{1}{2}\theta < \frac{1}{2}\pi$. The standard half-angle formulae give

$$\cos x = \frac{1 - t^2}{1 + t^2}, \qquad \mathrm{d}x = \frac{2\,\mathrm{d}t}{1 + t^2},$$

the second because $\mathrm{d}t/\mathrm{d}x = \frac{1}{2}\sec^2 \frac{1}{2}x = \frac{1}{2}(1 + t^2)$. Then

$$1 - a\cos x = \frac{(1 + t^2) - a(1 - t^2)}{1 + t^2} = \frac{(1 - a) + (1 + a)t^2}{1 + t^2},$$

so the factors of $1 + t^2$ cancel and

$$\int_0^\theta \frac{1}{1 - a\cos x}\,\mathrm{d}x = \int_0^T \frac{2}{(1 - a) + (1 + a)t^2}\,\mathrm{d}t = \frac{2}{1 + a}\int_0^T \frac{1}{k^2 + t^2}\,\mathrm{d}t, \qquad k^2 = \frac{1 - a}{1 + a}.$$

Since $-1 < a < 1$, both $1 - a$ and $1 + a$ are positive, so $k = \sqrt{(1 - a)/(1 + a)}$ is a genuine positive real number. (This is exactly where the condition on a is used: if $|a| \geq 1$ the denominator could vanish and the integral would be improper or the answer a logarithm.) Using $\int \mathrm{d}t/(k^2 + t^2) = \frac{1}{k}\tan^{-1}(t/k)$,

$$\int_0^\theta \frac{1}{1 - a\cos x}\,\mathrm{d}x = \frac{2}{(1 + a)k}\tan^{-1}\frac{T}{k}.$$

Finally $(1+a)k = \sqrt{(1+a)^2}\sqrt{\frac{1-a}{1+a}} = \sqrt{1-a^2}$, so

$$\boxed{\int_0^\theta \frac{1}{1-a\cos x}\,\mathrm{d}x = \frac{2}{\sqrt{1-a^2}}\tan^{-1}\left(\sqrt{\frac{1+a}{1-a}}\tan\tfrac{1}{2}\theta\right).}$$

Ask yourself Why do we need $\theta < \pi$?

The substitution $t = \tan\frac{1}{2}x$ breaks down at $x = \pi$, where t would be infinite. Keeping $\theta < \pi$ means the whole range $[0, \theta]$ maps to the finite interval $[0, \tan\frac{1}{2}\theta]$, and the formula needs no special care. (Letting $\theta \to \pi$ gives $\pi/\sqrt{1-a^2}$, since $\tan^{-1}$ tends to $\frac{1}{2}\pi$.)

The first "hence"

Take out a factor of 2:

$$\frac{1}{2-a\cos x} = \frac{1}{2}\cdot\frac{1}{1-\frac{1}{2}a\cos x}.$$

For the integral to be finite we need $-2 < a < 2$, and then $b = \frac{1}{2}a$ satisfies $-1 < b < 1$, so the general result applies with a replaced by b and $\theta = \frac{1}{2}\pi$, where $\tan\frac{1}{4}\pi = 1$:

$$\begin{aligned}\int_0^{\frac{1}{2}\pi}\frac{1}{2-a\cos x}\,\mathrm{d}x &= \frac{1}{2}\cdot\frac{2}{\sqrt{1-\frac{1}{4}a^2}}\tan^{-1}\sqrt{\frac{1+\frac{1}{2}a}{1-\frac{1}{2}a}}\\ &= \frac{2}{\sqrt{4-a^2}}\tan^{-1}\sqrt{\frac{2+a}{2-a}},\end{aligned}$$

since $\sqrt{1-\frac{1}{4}a^2} = \frac{1}{2}\sqrt{4-a^2}$, and doubling the top and bottom of the fraction inside the inverse tangent. This is the required result.

The second "hence"

Here we take out $\sqrt{2}$:

$$\frac{1}{\sqrt{2}+\cos x} = \frac{1}{\sqrt{2}}\cdot\frac{1}{1-\left(-\frac{1}{\sqrt{2}}\right)\cos x},$$

so we use $a = -\frac{1}{\sqrt{2}}$ (which lies in $(-1,1)$) and $\theta = \frac{3}{4}\pi$ (which lies in $(0,\pi)$). Then $\sqrt{1-a^2} = \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}}$, and

$$\frac{1+a}{1-a} = \frac{1-\frac{1}{\sqrt{2}}}{1+\frac{1}{\sqrt{2}}} = \frac{\sqrt{2}-1}{\sqrt{2}+1} = (\sqrt{2}-1)^2,$$

on multiplying top and bottom by $\sqrt{2}-1$ and using $(\sqrt{2}+1)(\sqrt{2}-1)=1$. So the square root is $\sqrt{2}-1$, and the general formula gives

$$\int_0^{\frac{3}{4}\pi} \frac{1}{\sqrt{2}+\cos x}\,\mathrm{d}x = \frac{1}{\sqrt{2}}\cdot\frac{2}{1/\sqrt{2}}\tan^{-1}\Big((\sqrt{2}-1)\tan\tfrac{3}{8}\pi\Big) = 2\tan^{-1}\Big((\sqrt{2}-1)\tan\tfrac{3}{8}\pi\Big).$$

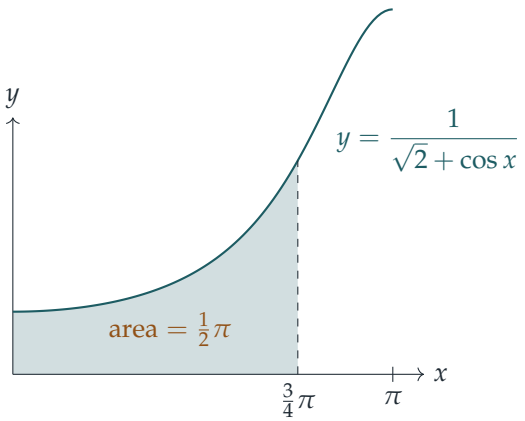
Ask yourself What is $\tan\frac{3}{8}\pi$?

Use the double-angle formula with $u=\tan\frac{1}{8}\pi>0$: $1=\tan\frac{1}{4}\pi=\dfrac{2u}{1-u^2}$, so $u^2+2u-1=0$ and $u=\sqrt{2}-1$ (the positive root). Since $\frac{3}{8}\pi=\frac{1}{2}\pi-\frac{1}{8}\pi$, we get $\tan\frac{3}{8}\pi=\cot\frac{1}{8}\pi=\dfrac{1}{\sqrt{2}-1}=\sqrt{2}+1$.

Hence $(\sqrt{2}-1)\tan\frac{3}{8}\pi=(\sqrt{2}-1)(\sqrt{2}+1)=1$, and

$$\int_0^{\frac{3}{4}\pi} \frac{1}{\sqrt{2}+\cos x}\,\mathrm{d}x = 2\tan^{-1}1 = 2\cdot\frac{\pi}{4}=\frac{\pi}{2},$$

as required.



Checks

- With $a=0$ the general formula gives $2\tan^{-1}(\tan\frac{1}{2}\theta)=\theta$, which is $\int_0^\theta 1\,\mathrm{d}x$. Similarly, $a=0$ in the first "hence" gives $\frac{2}{2}\tan^{-1}1=\frac{1}{4}\pi=\int_0^{\pi/2}\frac{1}{2}\,\mathrm{d}x$.
- Differentiating the boxed answer with respect to θ should give back $1/(1-a\cos\theta)$; it is a good exercise in the chain rule.
- Simpson's rule on the final integral gives 1.5707963…, matching $\frac{1}{2}\pi$; the general formula and the first "hence" also agree with Simpson's rule to within 10^{-10} for hundreds of random choices of a and θ.

Common slips

- Forgetting the factor $\frac{1}{2}$ (or $\frac{1}{\sqrt{2}}$) taken outside when rescaling the integrand to the form $1/(1 - a\cos x)$.
- Using $a = +\frac{1}{\sqrt{2}}$ in the last part: the sign in $\sqrt{2} + \cos x$ means a is negative.
- Losing track of k: the answer is $\frac{1}{k}\tan^{-1}(t/k)$, not $\tan^{-1}(t/k)$ or $\frac{1}{k}\tan^{-1}(kt)$.

Beyond STEP

The substitution $t = \tan\frac{1}{2}x$ (often called the Weierstrass substitution) works because it parametrises the unit circle by rational functions, so it reduces any rational function of $\sin x$ and $\cos x$ to a rational function of t, which can always be integrated by partial fractions. Integrals like $\int_0^{2\pi} \mathrm{d}x/(1 - a\cos x) = 2\pi/\sqrt{1 - a^2}$ are also standard first examples of contour integration in complex analysis, where the substitution $z = \mathrm{e}^{\mathrm{i}x}$ and the residue theorem do the work. For that viewpoint, see Needham, *Visual Complex Analysis*.

WORKED SOLUTION

STEP III 1996, Question 4

Pure • Maximal Product Partitions

STEP III 1996, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q4

The key idea

Instead of searching through every way of splitting N, ask what an *optimal* splitting cannot contain. Any part that is too big can be broken into two pieces without lowering the product, and that is exactly what the opening inequality measures. Parts equal to 1 are wasteful, and too many 2s can be traded for 3s. After these moves only a handful of shapes survive, and the sum N picks out one of them.

Solution

The inequality

We have

$$k \leq 2(k-2) \iff k \leq 2k-4 \iff k \geq 4.$$

So the integers satisfying the inequality are $k = 4, 5, 6, \ldots$. Moreover there is equality only when $k = 4$, and for $k \geq 5$ the inequality is strict: $k < 2(k-2)$.

What an optimal splitting looks like

Fix $N \geq 2$. There are only finitely many ways of writing N as a sum of positive integers, so a largest product $P(N)$ exists. The order of the parts doesn't affect the sum or the product, so we treat a splitting as a collection of parts. Call a splitting *optimal* if its product is $P(N)$. We make three observations, each of which changes a splitting while keeping its sum equal to N.

(a) **No part is 5 or more.** If a part $m \geq 5$ is replaced by the two parts 2 and $m-2$ (both positive), the product is multiplied by $2(m-2)/m$, which is greater than 1 by the strict form of the inequality. So a splitting with a part $m \geq 5$ is not optimal.

A part equal to 4 can be replaced by $2+2$ with no change to the product, since $2 \times 2 = 4$ (the equality case $k = 4$). So among the optimal splittings there is one with no part bigger than 3. From now on we work with such an optimal splitting.

(b) **No part is 1.** Since $N \geq 2$, a part 1 is not the only part, so there is some other part a. Replacing the two parts 1 and a by the single part $a+1$ keeps the sum and changes the factor $1 \times a = a$ into $a+1 > a$. That increases the product, which is impossible for an optimal splitting.

(c) **At most two parts are 2.** If there were three 2s, replacing $2+2+2$ by $3+3$ would change the factor 8 into 9, again increasing the product.

So some optimal splitting consists of a threes and b twos with $b \in \{0,1,2\}$, and

$$N = 3a + 2b, \qquad P(N) = 3^a\, 2^b.$$

Ask yourself Could two different choices of (a,b) both fit the same N, so that we'd still have to compare them?

No. Reducing modulo 3 gives $2b \equiv N \pmod 3$, and the values $b = 0,1,2$ give $2b \equiv 0,2,1 \pmod 3$, three different residues. So N fixes b, and then $a = (N-2b)/3$ is fixed as well. Hence

$$P(N) = \begin{cases} 3^{N/3} & \text{if } N \equiv 0 \pmod 3, \\ 2 \times 3^{(N-2)/3} & \text{if } N \equiv 2 \pmod 3, \\ 2^2 \times 3^{(N-4)/3} & \text{if } N \equiv 1 \pmod 3,\ N \geq 4. \end{cases}$$

The values for $N = 5, \ldots, 9$

Applying this to each N in turn:

$$\begin{aligned} 5 &= 3+2, & P(5) &= 2 \times 3, \\ 6 &= 3+3, & P(6) &= 3^2, \\ 7 &= 3+2+2, & P(7) &= 2^2 \times 3, \\ 8 &= 3+3+2, & P(8) &= 2 \times 3^2, \\ 9 &= 3+3+3, & P(9) &= 3^3, \end{aligned}$$

as required. (For $N = 7$ the splitting $4+3$ is also optimal, which is why observation (a) only lets us *replace* a 4 rather than rule it out.)

Finding $P(1000)$

Since $1000 = 3 \times 332 + 4$, we have $1000 \equiv 1 \pmod 3$, so $b = 2$ and $a = (1000-4)/3 = 332$. The optimal splitting is 332 threes and two 2s, and

$$P(1000) = 2^2 \times 3^{332}.$$

Equivalently, the optimal splittings are 332 threes with two 2s, or 332 threes with one 4. These are the only ones, because the moves in (a)–(c) show that any other splitting can be strictly improved.

Checks

- A brute-force search over every splitting of N for $N \leq 40$ agrees with the formula, including the five given values.
- An exact computer calculation using $P(n) = \max_{1 \leq p \leq n} p\,P(n-p)$ gives $P(1000) = 4 \times 3^{332}$, a number with 160 digits.
- As a sanity check, compare with a non-integer "split": with N/x equal parts the product is $x^{N/x}$, which is largest at $x = \mathrm{e} \approx 2.72$. That is why 3s (and the odd 2) win.

Common slips

- Writing $1000 = 3 \times 333 + 1$ and answering 3^{333}: that splitting contains a 1, and $3 + 1$ should be replaced by $2 + 2$, since $4 > 3$.
- Saying that a part 4 "cannot occur". It can occur in an optimal splitting; it just never does better than $2 + 2$.
- Listing the values for $N = 5, \ldots, 9$ without showing that nothing beats them. The structure argument is what proves they are the largest.

Beyond STEP

The argument here is a typical *extremal* or *exchange* argument: assume a best object exists, then show that it must avoid certain local patterns. The same idea turns up in combinatorics, in proving that greedy algorithms are optimal, and in the theory of integer partitions. The continuous version (maximising $x^{N/x}$) is a pleasant calculus exercise. For more problems in this spirit see Siklos, *Advanced Problems in Mathematics*, and for congruences and elementary number theory, Davenport, *The Higher Arithmetic*.

WORKED SOLUTION

STEP III 1996, Question 5

Pure • Multiple Angle Formulae

STEP III 1996, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q5

The key idea

Expanding $(\cos\theta + \mathrm{i}\sin\theta)^7$ by the binomial theorem and comparing with de Moivre's theorem gives $\cos 7\theta$ and $\sin 7\theta$ as polynomials; dividing through by $\cos^7\theta$ turns everything into powers of $t = \tan\theta$. Once $\tan 7\theta$ is a ratio of polynomials in t, the angles where $\tan 7\theta$ is zero make the numerator vanish, and the angles where $\tan 7\theta$ is undefined make the denominator vanish. Both polynomials are even apart from a factor t, so they become cubics in t^2, and Vieta's formulae do the rest.

Solution

The formula for $\tan 7\theta$

By de Moivre's theorem and the binomial theorem, with $c = \cos\theta$ and $s = \sin\theta$,

$$\cos 7\theta + \mathrm{i}\sin 7\theta = (c + \mathrm{i}s)^7 = \sum_{k=0}^{7} \binom{7}{k} \mathrm{i}^k c^{7-k} s^k.$$

Suppose $\cos\theta \neq 0$, so that $t = \tan\theta$ is defined. Taking out c^7 gives $(c+\mathrm{i}s)^7 = c^7(1+\mathrm{i}t)^7$. Using $\mathrm{i}^2 = -1$, $\mathrm{i}^3 = -\mathrm{i}$, $\mathrm{i}^4 = 1$ and the binomial coefficients $1, 7, 21, 35, 35, 21, 7, 1$,

$$(1+\mathrm{i}t)^7 = (1 - 21t^2 + 35t^4 - 7t^6) + \mathrm{i}\,(7t - 35t^3 + 21t^5 - t^7).$$

Equating real and imaginary parts (everything except i is real),

$$\cos 7\theta = \cos^7\theta\,(1 - 21t^2 + 35t^4 - 7t^6),$$
$$\sin 7\theta = \cos^7\theta\,(7t - 35t^3 + 21t^5 - t^7).$$

Dividing, the factor $\cos^7\theta$ cancels, and multiplying the top and bottom by -1,

$$\tan 7\theta = \frac{7t - 35t^3 + 21t^5 - t^7}{1 - 21t^2 + 35t^4 - 7t^6} = \frac{t(t^6 - 21t^4 + 35t^2 - 7)}{7t^6 - 35t^4 + 21t^2 - 1},$$

as required.

For later, note what the two displayed lines say. Write $N(t) = t(t^6 - 21t^4 + 35t^2 - 7)$ and $D(t) = 7t^6 - 35t^4 + 21t^2 - 1$. Then $\sin 7\theta = -\cos^7\theta\, N(t)$ and $\cos 7\theta = -\cos^7\theta\, D(t)$, so when $\cos\theta \neq 0$,

$$\sin 7\theta = 0 \iff N(t) = 0, \qquad \cos 7\theta = 0 \iff D(t) = 0.$$

Part (i)

Take $\theta = \frac{k\pi}{7}$ with $k = 1, 2, 3$. Then $\sin 7\theta = \sin k\pi = 0$ and $\cos\theta \neq 0$, so $N(\tan\theta) = 0$. Also $\tan\theta \neq 0$, since $0 < \theta < \frac{\pi}{2}$, so we may cancel the factor t:

$$t^6 - 21t^4 + 35t^2 - 7 = 0.$$

So $x = t^2$ satisfies

$$x^3 - 21x^2 + 35x - 7 = 0$$

for $x = \tan^2\frac{\pi}{7},\ \tan^2\frac{2\pi}{7},\ \tan^2\frac{3\pi}{7}$.

Ask yourself How do we know these three numbers are *all* the roots, not one root counted several times?

The angles $\frac{\pi}{7} < \frac{2\pi}{7} < \frac{3\pi}{7}$ all lie in $(0, \frac{\pi}{2})$, where tan is positive and strictly increasing. So the three values of $\tan^2$ are different. A cubic has at most three roots, so these are exactly its roots, and this is the cubic we want.

By Vieta's formulae the product of the roots is 7:

$$\tan^2\tfrac{\pi}{7}\,\tan^2\tfrac{2\pi}{7}\,\tan^2\tfrac{3\pi}{7} = 7.$$

Each tangent is positive, because each angle lies in $(0, \frac{\pi}{2})$, so we take the positive square root:

$$\tan\tfrac{\pi}{7}\,\tan\tfrac{2\pi}{7}\,\tan\tfrac{3\pi}{7} = \sqrt{7}.$$

Part (ii)

Now take $\theta = \frac{\pi}{14}, \frac{3\pi}{14}, \frac{5\pi}{14}$. Then $7\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$, so $\cos 7\theta = 0$, while $\cos\theta \neq 0$. By the remark above, $D(\tan\theta) = 0$, so $y = \tan^2\theta$ satisfies

$$7y^3 - 35y^2 + 21y - 1 = 0.$$

As in part (i), the three angles lie in $(0, \frac{\pi}{2})$ and are different, so the three values of $\tan^2\theta$ are different and are exactly the roots of this cubic. By Vieta's formulae their sum is $\frac{35}{7}$:

$$\tan^2\tfrac{\pi}{14} + \tan^2\tfrac{3\pi}{14} + \tan^2\tfrac{5\pi}{14} = 5.$$

There's a quick alternative using part (i). Since $\frac{\pi}{14} = \frac{\pi}{2} - \frac{3\pi}{7}$, and similarly $\frac{3\pi}{14} = \frac{\pi}{2} - \frac{2\pi}{7}$ and $\frac{5\pi}{14} = \frac{\pi}{2} - \frac{\pi}{7}$, the required sum is $\sum 1/x_i$ over the roots x_i of the cubic in part (i). That is

$$\frac{x_1x_2 + x_2x_3 + x_3x_1}{x_1x_2x_3} = \frac{35}{7} = 5.$$

Equivalently, substituting $x = 1/y$ in the part (i) cubic and clearing denominators gives the part (ii) cubic.

Checks

- Put $t = 1$, that is $\theta = \frac{\pi}{4}$. The formula gives $\frac{1-21+35-7}{7-35+21-1} = \frac{8}{-8} = -1$, and indeed $\tan\frac{7\pi}{4} = -1$.
- For small θ, $\tan 7\theta \approx 7t$, and the formula gives $\frac{-7t}{-1} = 7t$ to first order.
- Numerically, $\tan\frac{\pi}{7}\tan\frac{2\pi}{7}\tan\frac{3\pi}{7} \approx 0.4816 \times 1.2540 \times 4.3813 \approx 2.6458 \approx \sqrt{7}$, and $\tan^2\frac{\pi}{14} + \tan^2\frac{3\pi}{14} + \tan^2\frac{5\pi}{14} \approx 0.0521 + 0.6360 + 4.3119 = 5.0000$.

Common slips

- Forgetting to justify that the three values are distinct, so that they really are *all* the roots of the cubic.
- Answering $\pm\sqrt{7}$ in part (i): the sign is fixed because all three tangents are positive.
- Sign errors from the powers of i: $\mathrm{i}^6 = -1$ and $\mathrm{i}^7 = -\mathrm{i}$.

Beyond STEP

The numbers $\tan^2\frac{k\pi}{7}$ are conjugate algebraic numbers: they are the roots of one irreducible cubic with integer coefficients, so any symmetric expression in them is rational. This is the starting point of Galois theory, which studies the symmetries between the roots of a polynomial. The same trick also gives identities such as $\sum_{k=1}^{n}\cot^2\frac{k\pi}{2n+1} = \frac{n(2n-1)}{3}$, used in one elementary proof that $\sum 1/k^2 = \pi^2/6$. For a geometric view of de Moivre's theorem and roots of unity, see Needham, *Visual Complex Analysis*; for more problems of this kind, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III 1996, Question 6

Pure • Groups of Matrices

STEP III 1996, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q6

This question has a full guided treatment in *Beyond STEP*, Section 6.8, *A Group of Singular Matrices*: https://beyondstep.keithmcnulty.org/ch06/singular-group.html

The key idea

Every matrix in S is a multiple of one fixed matrix $\mathbf{J}$, and $\mathbf{J}^2$ is again a multiple of $\mathbf{J}$, so multiplying elements of S comes down to multiplying real numbers. The surprise is that the identity of this group is *not* the identity matrix $\mathbf{I}$: the group never needs $\mathbf{I}$ at all. Part (ii) explains why that has to happen: inside a group, every element can be written as a product involving any other, and determinants multiply.

Solution

Part (i)

Write

$$\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \text{so that} \qquad \mathbf{J}^2 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} = 2\mathbf{J}.$$

Every element of S is $a\mathbf{J}$ with $a \neq 0$, and we write $\mathbf{M}_a = a\mathbf{J}$.

Closure. For non-zero a and b,

$$\mathbf{M}_a\mathbf{M}_b = ab\,\mathbf{J}^2 = 2ab\,\mathbf{J} = \mathbf{M}_{2ab}.$$

Since $a \neq 0$ and $b \neq 0$, we have $2ab \neq 0$, so the product lies in S. Hence S is closed under matrix multiplication.

Now we check the remaining group axioms.

Associativity. Matrix multiplication is associative for all 2×2 matrices, so in particular for those in S.

Identity. We want $\mathbf{M}_e$ with $\mathbf{M}_e\mathbf{M}_b = \mathbf{M}_b$ for every b, that is, $2eb = b$. So we take $e = \frac{1}{2}$:

$$\mathbf{M}_{1/2}\mathbf{M}_b = \mathbf{M}_{2\cdot\frac{1}{2}b} = \mathbf{M}_b,$$

and similarly $\mathbf{M}_b\mathbf{M}_{1/2} = \mathbf{M}_b$. So $\mathbf{E} = \frac{1}{2}\mathbf{J}$, which lies in S, is an identity element.

> **Ask yourself** The identity matrix $\mathbf{I}$ isn't in S. Doesn't a group of matrices need it?

No. A group needs *some* element that acts as an identity on the other elements of the set, and $\frac{1}{2}\mathbf{J}$ does that job for every element of S. It isn't an identity for all 2×2 matrices (for example $\frac{1}{2}\mathbf{JI} \neq \mathbf{I}$), but $\mathbf{I}$ isn't in S, so that doesn't matter.

Inverses. Given $\mathbf{M}_a$, we want $\mathbf{M}_c$ with $2ac = \frac{1}{2}$, so $c = \dfrac{1}{4a}$, which exists and is non-zero because $a \neq 0$. Then

$$\mathbf{M}_a\mathbf{M}_{1/(4a)} = \mathbf{M}_{1/(4a)}\mathbf{M}_a = \mathbf{M}_{2a/(4a)} = \mathbf{M}_{1/2} = \mathbf{E}.$$

All four axioms hold, so S is a group under matrix multiplication. (It is even abelian, since $\mathbf{M}_a\mathbf{M}_b = \mathbf{M}_{2ab} = \mathbf{M}_b\mathbf{M}_a$.) Note that every element of S has determinant $a^2 - a^2 = 0$. These matrices have no inverse in the ordinary sense, yet each has an inverse within the group.

Part (ii)

Suppose $\mathbf{A} \in G$ has $\det\mathbf{A} = 0$, and let $\mathbf{B}$ be any element of G. Because G is a group, $\mathbf{A}$ has an inverse $\mathbf{A}'$ in G with $\mathbf{AA}' = \mathbf{A}'\mathbf{A} = \mathbf{E}$. (We write $\mathbf{A}'$, not $\mathbf{A}^{-1}$: it is the inverse inside G, not the matrix inverse, which doesn't exist.)

Let $\mathbf{C} = \mathbf{A}'\mathbf{B}$. By closure, $\mathbf{C} \in G$, and

$$\mathbf{AC} = \mathbf{A}(\mathbf{A}'\mathbf{B}) = (\mathbf{AA}')\mathbf{B} = \mathbf{EB} = \mathbf{B},$$

using associativity and the fact that $\mathbf{E}$ is the identity of G. So the equation $\mathbf{AC} = \mathbf{B}$ has a solution $\mathbf{C}$ in G. Taking determinants, and using $\det(\mathbf{XY}) = \det\mathbf{X}\det\mathbf{Y}$,

$$\det\mathbf{B} = \det\mathbf{A}\ \det\mathbf{C} = 0 \cdot \det\mathbf{C} = 0.$$

Since $\mathbf{B}$ was arbitrary, every element of G is singular. In particular the identity $\mathbf{E}$ of such a group is singular, so it can't be $\mathbf{I}$, just as we saw in part (i).

> **Ask yourself** Where did we use that G is a group, rather than just a set closed under multiplication?

We used the inverse $\mathbf{A}'$ and the identity $\mathbf{E}$ to write $\mathbf{B}$ as $\mathbf{A}$ times something in G. Without them the result fails. For example, $\{\mathbf{I}, \mathbf{0}\}$ is closed under multiplication and contains a singular matrix, but $\mathbf{I}$ is not singular. (It isn't a group: $\mathbf{0}$ has no inverse.)

An example with $n = 3$. We copy part (i). Let

$$\mathbf{K} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad G = \{\, a\mathbf{K} : a \in \mathbb{R},\ a \neq 0 \,\}.$$

Each entry of $\mathbf{K}^2$ is $1+1+1=3$, so $\mathbf{K}^2 = 3\mathbf{K}$ and

$$(a\mathbf{K})(b\mathbf{K}) = 3ab\,\mathbf{K}.$$

- *Closure:* if $a, b \neq 0$ then $3ab \neq 0$, so the product is in G.
- *Associativity:* this is inherited from matrix multiplication.
- *Identity:* $\mathbf{E} = \frac{1}{3}\mathbf{K}$, since $(\frac{1}{3}\mathbf{K})(b\mathbf{K}) = (b\mathbf{K})(\frac{1}{3}\mathbf{K}) = 3 \cdot \frac{1}{3}b\,\mathbf{K} = b\mathbf{K}$.
- *Inverses:* the inverse of $a\mathbf{K}$ is $\dfrac{1}{9a}\mathbf{K}$, since $3 \cdot a \cdot \dfrac{1}{9a} = \frac{1}{3}$.

So G is a group under matrix multiplication. Every $a\mathbf{K}$ has three equal rows, so $\det(a\mathbf{K}) = 0$: this is a group of singular 3×3 matrices.

(Another example that works just as well is the set of matrices $\operatorname{diag}(a, b, 0)$ with $a, b \neq 0$, whose identity is $\operatorname{diag}(1, 1, 0)$.)

Checks

- Test a case: $\mathbf{M}_3\mathbf{M}_{1/12}$ has every entry $3 \cdot \frac{1}{12} + 3 \cdot \frac{1}{12} = \frac{1}{2}$, so it equals $\mathbf{E} = \mathbf{M}_{1/2}$, as the inverse formula $\frac{1}{4a}$ predicts.
- We checked with exact fractions, for hundreds of random non-zero a, b, c, that the closure, identity and inverse formulas hold for both the 2×2 and the 3×3 families, that all these matrices have determinant 0, and that $\mathbf{A}(\mathbf{A}'\mathbf{B}) = \mathbf{B}$.

Common slips

- Taking $\mathbf{I}$ as the identity, or $\mathbf{A}^{-1}$ as the ordinary matrix inverse. Neither exists in S. Find the identity and inverses from the group's own multiplication rule.
- Forgetting to check that $2ab \neq 0$. Closure needs the product to have the right form *and* a non-zero parameter.

Beyond STEP

A group is defined by its operation, not by where its elements happen to live. Here a group sits inside the $n \times n$ matrices without sharing their identity. In fact $\mathbf{E}$ always satisfies $\mathbf{E}^2 = \mathbf{E}$, so it is a projection, and the group acts as invertible maps on the image of $\mathbf{E}$. The map $a \mapsto \mathbf{M}_{a/2}$ is an isomorphism from the non-zero reals under multiplication to S, which you can check directly. For more, see Allenby, *Rings, Fields and Groups*, and for projections, Strang, *Introduction to Linear Algebra*.

WORKED SOLUTION

STEP III 1996, Question 7

Pure • Symmetric Functions of Roots

STEP III 1996, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q7

The key idea

Both parts turn on the same fact: any symmetric expression in three numbers can be written in terms of their sum, the sum of their products in pairs, and their product, and these three quantities are exactly the coefficients of the cubic whose roots are the numbers. In (i) this turns a nonlinear system into a cubic equation. In (ii) we never need the roots at all: Heron's formula is symmetric in a, b, c, so the cubic's coefficients are enough.

Solution

Part (i)

First, the sum of squares:

$$\alpha^2 = (x+y+z)^2 = x^2+y^2+z^2+2(xy+yz+zx), \qquad \text{so} \qquad x^2+y^2+z^2 = \alpha^2 - 2\beta.$$

Next we use the factorisation

$$x^3+y^3+z^3-3xyz = (x+y+z)\left(x^2+y^2+z^2-xy-yz-zx\right),$$

which you can verify by expanding the right-hand side: the nine cross terms such as $x \cdot xy$ and $y \cdot (-xy)$ cancel in pairs, and the three terms of the form $-xyz$ remain. Substituting,

$$\begin{aligned} x^3+y^3+z^3 &= 3\gamma + \alpha\left(\alpha^2 - 2\beta - \beta\right) \\ &= \alpha^3 - 3\alpha\beta + 3\gamma. \end{aligned}$$

So $A = 1$, $B = -3$ and $C = 3\gamma$. (Here C is three times the product, so it depends on γ; A and B are pure numbers.)

Now for the equations. We have $\alpha = 1$, and the sum of squares gives

$$3 = \alpha^2 - 2\beta = 1 - 2\beta \quad \Longrightarrow \quad \beta = -1.$$

The sum of cubes then gives

$$4 = \alpha^3 - 3\alpha\beta + 3\gamma = 1 + 3 + 3\gamma \quad \Longrightarrow \quad \gamma = 0.$$

Ask yourself We know α, β and γ. How do we get back to x, y, z?

The numbers x, y, z have sum α, pairwise-product sum β and product γ exactly when they are the three roots of

$$(t-x)(t-y)(t-z) = t^3 - \alpha t^2 + \beta t - \gamma = 0.$$

Both directions hold: if x, y, z solve the system then they are the roots of this cubic, and conversely the roots of the cubic, in any order, have the right α, β, γ and hence (by the identities above) satisfy all three equations. Here the cubic is

$$t^3 - t^2 - t = t\,(t^2 - t - 1) = 0, \qquad \text{so} \qquad t = 0, \quad t = \tfrac{1}{2}(1+\sqrt{5}), \quad t = \tfrac{1}{2}(1-\sqrt{5}).$$

Therefore the solutions are

$$\{x, y, z\} = \left\{ 0,\ \tfrac{1}{2}(1+\sqrt{5}),\ \tfrac{1}{2}(1-\sqrt{5}) \right\},$$

in any order: six ordered triples (x, y, z) in all, since the three values are different.

Part (ii)

Since a, b, c are the roots of the cubic,

$$x^3 - 16x^2 + 81x - 128 = (x-a)(x-b)(x-c)$$

for every x. Comparing the x^2 coefficients, $a + b + c = 16$, so $s = 8$. Now the trick: put $x = s = 8$ into this identity. The right-hand side becomes $(s-a)(s-b)(s-c)$, the very product in Heron's formula:

$$(s-a)(s-b)(s-c) = 8^3 - 16 \cdot 8^2 + 81 \cdot 8 - 128 = 512 - 1024 + 648 - 128 = 8.$$

Hence

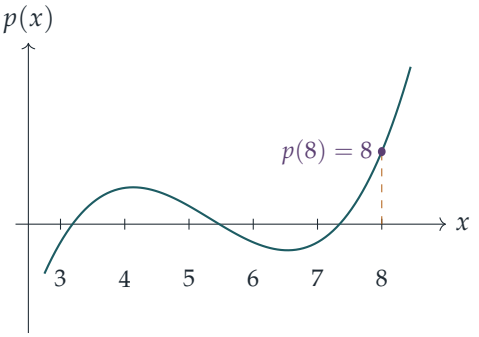
$$\text{area} = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{8 \cdot 8} = 8.$$

Ask yourself Is there really a triangle with these sides?

We should check, because Heron's formula only describes a triangle when a, b, c are positive and each is less than the sum of the other two. Since $a < b + c \iff 2a < a + b + c \iff a < s$, we need all three roots to be real, positive and less than 8. Write $p(x) = x^3 - 16x^2 + 81x - 128$. Then

$$p(3) = -2, \quad p(4) = 4, \quad p(5) = 2, \quad p(6) = -2, \quad p(7) = -2, \quad p(8) = 8,$$

so by the changes of sign there is a root in each of $(3,4)$, $(5,6)$ and $(7,8)$. A cubic has at most three roots, so these are all of them, and every root lies strictly between 0 and 8. The triangle exists, and its area is 8.



Checks

- In (i), try $x = 1$, $y = 2$, $z = 3$: $\alpha = 6$, $\beta = 11$, $\gamma = 6$ and $216 - 198 + 18 = 36 = 1 + 8 + 27$.
- With $x = 0$, $y = \frac{1}{2}(1 + \sqrt{5}) \approx 1.618$, $z = \frac{1}{2}(1 - \sqrt{5}) \approx -0.618$: the sum is 1, and since $y^2 = y + 1$ and $z^2 = z + 1$, the squares sum to $(y + z) + 2 = 3$ and the cubes to $y^2 + z^2 + (y + z) = 3 + 1 = 4$.
- Numerically the roots in (ii) are $a \approx 3.186$, $b \approx 5.471$, $c \approx 7.343$. Heron's formula with these values gives 8.000, and so does $\frac{1}{2}ab\sin C$ with C from the cosine rule.

Common slips

- Giving one triple instead of all six orderings, or forgetting that $t = 0$ is a root of $t^3 - t^2 - t = 0$.
- Getting a sign wrong in Vieta's formulas: for $t^3 - \alpha t^2 + \beta t - \gamma$ the signs alternate.
- Taking the area without checking that the roots really form a triangle.

Beyond STEP

The fundamental theorem of symmetric polynomials says that every polynomial in x, y, z that is unchanged by permuting them is a polynomial in the elementary symmetric functions α, β, γ. Newton's identities give the power sums $x^k + y^k + z^k$ recursively. These ideas lead into Galois theory, which explains why there are formulas for the roots of cubics and quartics but none for general quintics. Allenby's *Rings, Fields and Groups* is a good first step in that direction.

WORKED SOLUTION

STEP III 1996, Question 8

Pure • Möbius Transformations

STEP III 1996, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q8

The key idea

These are Möbius (fractional linear) transformations, and almost everything follows from two observations. Clearing the denominator in $y = T(x)$ gives an equation that is linear in x *and* linear in y, so it can be solved for either one. And writing $T(x)$ as a constant plus a multiple of $1/(cx-d)$ shows at once which value T never takes. For the composition, just substitute and multiply through; the result is easiest to organise by matching each T with the matrix $\begin{pmatrix} a & -b \\ c & -d \end{pmatrix}$.

Solution

Fixed points

For $x \neq d/c$, $T(x) = x$ means $ax - b = x(cx - d)$, that is,

$$cx^2 - (a+d)x + b = 0. \qquad (*)$$

First we check that $(*)$ cannot produce the forbidden value $x = d/c$ (when $c \neq 0$). Substituting gives $d^2/c - (a+d)d/c + b = (bc - ad)/c \neq 0$, so every root of $(*)$ is a genuine fixed point.

If $c \neq 0$, $(*)$ is a quadratic, and the fixed points are

$$x = \frac{(a+d) \pm \sqrt{(a+d)^2 - 4bc}}{2c}.$$

There are two of them if $(a+d)^2 > 4bc$, one if $(a+d)^2 = 4bc$, and none if $(a+d)^2 < 4bc$.

If $c = 0$, the condition $ad \neq bc$ becomes $ad \neq 0$, so $d \neq 0$ and $T(x) = (b - ax)/d$ is defined everywhere. Now $(*)$ reads $(a+d)x = b$. If $a + d \neq 0$ there is exactly one fixed point, $x = b/(a+d)$. If $a + d = 0$ and $b \neq 0$ there are none. If $a + d = 0$ and $b = 0$, then $T(x) = -ax/d = x$, so T is the identity and every x is fixed.

The inverse

From $y(cx-d) = ax-b$ we get $cxy - dy = ax - b$, so $x(cy-a) = dy - b$, and therefore

$$x = T^{-1}(y) = \frac{dy-b}{cy-a}.$$

This has the same form as T, with

$$a' = d, \qquad b' = b, \qquad c' = c, \qquad d' = a,$$

and the condition still holds because $a'd' - b'c' = da - bc \neq 0$. (Dividing by $cy - a$ is legitimate, because $y = a/c$ is never a value of T; see the next part.)

The sets S_r

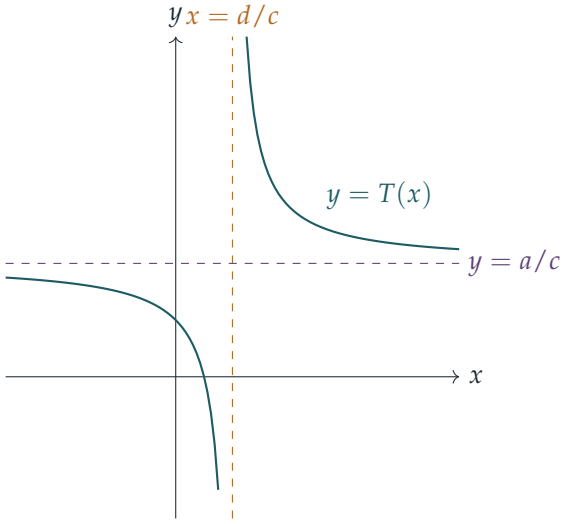
Suppose $c \neq 0$. $T(x)$ is defined exactly when $cx \neq d$, so we take $r = d/c$. Then T is defined for every $x \in S_{d/c}$. To find the image, we split off the constant part:

$$ax - b = \frac{a}{c}(cx-d) + \frac{ad-bc}{c}, \qquad \text{so} \qquad T(x) = \frac{a}{c} + \frac{ad-bc}{c\,(cx-d)}.$$

Since $ad - bc \neq 0$, the second term is never zero, so $T(x) \neq a/c$. Conversely, every $y \neq a/c$ is attained. The inverse formula gives $x = (dy-b)/(cy-a)$, and this x is not d/c, because d/c is not a value of T^{-1} (apply the same argument to T^{-1}, whose "a'/c'" is d/c). Hence

$$T(S_{d/c}) = S_{a/c},$$

and T is a bijection from $S_{d/c}$ to $S_{a/c}$. The graph is a rectangular hyperbola with asymptotes $x = d/c$ and $y = a/c$:



(The sketch uses $T(x) = (2x-1)/(x-1) = 2 + 1/(x-1)$.)

If $c = 0$, then $d \neq 0$ and $a \neq 0$ (because $ad \neq 0$), and $T(x) = (b - ax)/d$ is a non-constant linear function. So T is defined for *all* real x, with no value to exclude, and it maps $\mathbb{R}$ onto $\mathbb{R}$ bijectively. If we still remove a point r, then $T(S_r) = S_{T(r)} = S_{(b-ar)/d}$.

Composition

Substitute $T_1(x) = \dfrac{a_1x - b_1}{c_1x - d_1}$ into T_2, and multiply the top and bottom by $c_1x - d_1$:

$$\begin{aligned} T_2(T_1(x)) &= \frac{a_2(a_1x - b_1) - b_2(c_1x - d_1)}{c_2(a_1x - b_1) - d_2(c_1x - d_1)} \\ &= \frac{(a_1a_2 - b_2c_1)\,x - (a_2b_1 - b_2d_1)}{(a_1c_2 - c_1d_2)\,x - (b_1c_2 - d_1d_2)}. \end{aligned}$$

So T_3 has the same form, with

$$a_3 = a_1a_2 - b_2c_1, \quad b_3 = a_2b_1 - b_2d_1, \quad c_3 = a_1c_2 - c_1d_2, \quad d_3 = b_1c_2 - d_1d_2.$$

? **Ask yourself** Is "the same form" really proved? We also need $a_3d_3 \neq b_3c_3$.

Yes, we do need it. These coefficients are exactly the entries of the matrix product

$$\begin{pmatrix} a_2 & -b_2 \\ c_2 & -d_2 \end{pmatrix} \begin{pmatrix} a_1 & -b_1 \\ c_1 & -d_1 \end{pmatrix} = \begin{pmatrix} a_3 & -b_3 \\ c_3 & -d_3 \end{pmatrix},$$

and each matrix has determinant $bc - ad$. Since the determinant of a product is the product of the determinants,

$$b_3c_3 - a_3d_3 = (b_1c_1 - a_1d_1)(b_2c_2 - a_2d_2) \neq 0.$$

(You can also expand directly; it is a few lines of algebra.) So T_3 is a transformation of the same type.

When is T^2 the identity?

Take $T_1 = T_2 = T$ in the composition formula. Then $T^2(x) = \dfrac{Ax - B}{Cx - D}$ with

$$A = a^2 - bc, \qquad B = b(a - d), \qquad C = c(a - d), \qquad D = bc - d^2.$$

T^2 is the identity when $Ax - B = x(Cx - D)$ for all x in its domain, which is all but at most a few points. A polynomial identity holding at infinitely many points holds identically, so we compare coefficients:

$$C = 0, \qquad A + D = 0, \qquad B = 0,$$

that is,

$$c(a - d) = 0, \qquad (a - d)(a + d) = 0, \qquad b(a - d) = 0.$$

- If $a = d$, all three hold. Indeed $T^2(x) = \dfrac{(a^2 - bc)x}{-(bc - a^2)} = x$, where $a^2 - bc = ad - bc \neq 0$.
- If $a \neq d$, we need $b = c = 0$ and $a = -d$. Then $T(x) = ax/(-d) = x$, so T is the identity itself.

So T^2 is the identity **if and only if** $a = d$**, or** $b = c = 0$ **and** $a + d = 0$ (that is, T is the identity). The second case is trivial, so the interesting involutions are exactly the maps $x \mapsto \dfrac{ax - b}{cx - a}$ with $a^2 \neq bc$.

An example

By the condition just found, we need two maps with $a = d$ whose composition has $a \neq d$. Take

$$T_1(x) = -x \quad (a_1, b_1, c_1, d_1) = (1, 0, 0, 1), \qquad T_2(x) = 1 - x \quad (a_2, b_2, c_2, d_2) = (1, 1, 0, 1).$$

Each has $a = d$, so T_1^2 and T_2^2 are the identity. You can see this directly too: $-(-x) = x$ and $1 - (1 - x) = x$. Their composition is

$$T_3(x) = T_2(T_1(x)) = 1 - (-x) = x + 1,$$

which in the standard form has $(a_3, b_3, c_3, d_3) = (1, -1, 0, -1)$, so $a_3 \neq d_3$. Indeed $T_3^2(x) = x + 2 \neq x$.

Geometrically, two reflections of the line (in 0 and in $\frac{1}{2}$) compose to a translation, and a translation is not its own inverse. For an example with $c \neq 0$, try $T_1(x) = -1/x$ with the same T_2. Then $T_3(x) = (x + 1)/x$, and $T_3^2(x) = (2x + 1)/(x + 1)$.

Checks

- Use the sketched example $T(x) = (2x - 1)/(x - 1)$ ($a = 2$, $b = 1$, $c = d = 1$). Its fixed points are the roots of $x^2 - 3x + 1 = 0$, namely $x = \frac{1}{2}(3 \pm \sqrt{5})$, and you can check that $T(2.618\ldots) = 2.618\ldots$. Its inverse is $(y - 1)/(y - 2)$, which is undefined at $y = 2 = a/c$, as predicted.
- The composition formula with $T_1 = T$, $T_2 = T^{-1}$, i.e. $(a_2, b_2, c_2, d_2) = (d, b, c, a)$, gives $a_3 = ad - bc$, $b_3 = 0$, $c_3 = 0$, $d_3 = bc - ad$. This is the identity, as it should be.
- A computer verified the inverse, image and composition formulae, using exact fractions on thousands of random (a, b, c, d). It also tested the condition for T^2 to be the identity against direct evaluation for every integer choice of $a, b, c, d \in [-3, 3]$ with $ad \neq bc$.

Common slips

- Forgetting the trivial case: $a = d$ is not *necessary*, because the identity ($b = c = 0$, $a = -d$) also squares to the identity.

- Showing the composition is a ratio of linear functions but not checking $a_3d_3 \neq b_3c_3$.
- In the $c = 0$ case, looking for an excluded point. A non-constant linear map is defined everywhere and hits every real number.

Beyond STEP

Allowing $x = \infty$, with $T(d/c) = \infty$ and $T(\infty) = a/c$, makes every such T a bijection of the extended line. With complex coefficients you get the Möbius transformations of the Riemann sphere. The matrix correspondence used above says that these maps form a group, isomorphic to invertible 2×2 matrices modulo scalars (the projective linear group). Composition becomes matrix multiplication, and a map other than the identity is an involution exactly when its matrix has trace zero; here that is $a - d = 0$. Needham's *Visual Complex Analysis* has a beautiful chapter on Möbius transformations.

WORKED SOLUTION

STEP III 1996, Question 9

Mechanics • Circular Motion

STEP III 1996, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q9

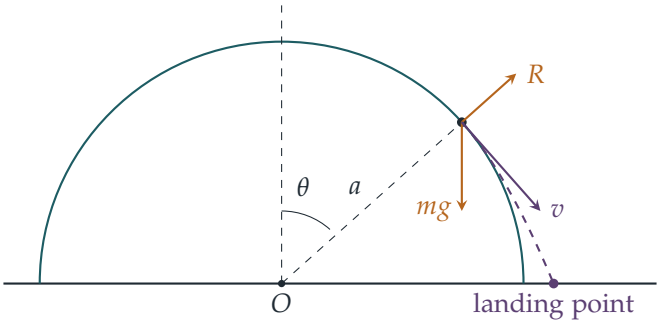
The key idea

The motion has two stages. While the particle stays on the sphere it moves in a circle, so energy conservation gives its speed and the radial equation of motion gives the normal reaction; it leaves the sphere at the moment that reaction drops to zero. After that it is an ordinary projectile, launched along the tangent from a known point with a known speed, and we just need to find where it crosses the level of the centre.

Solution

Model

Treat the particle as a point mass of mass m on a smooth fixed sphere with centre O, with no air resistance. The only forces are its weight mg and the normal reaction R, which acts along the radius away from O. Let θ be the angle between OP and the upward vertical, where P is the particle, and let v be its speed. Since the displacement is small, we take the initial speed to be zero (in the limit).



Stage 1: on the sphere

The reaction is perpendicular to the motion, so it does no work, and energy is conserved. The particle has fallen $a - a\cos\theta$ from the top, so

$$\tfrac{1}{2}mv^2 = mga(1-\cos\theta), \qquad \text{that is} \qquad v^2 = 2ga(1-\cos\theta).$$

While it stays on the sphere it moves in a circle of radius a, so resolving towards O gives

$$mg\cos\theta - R = \frac{mv^2}{a} \quad \Longrightarrow \quad R = mg\cos\theta - 2mg(1-\cos\theta) = mg(3\cos\theta - 2).$$

Ask yourself Why does the particle leave the sphere exactly when $R = 0$?

The sphere can only push, so $R \geq 0$ while there is contact. As θ increases from 0, R decreases steadily from mg, and the formula above would give $R < 0$ beyond $\cos\theta = \frac{2}{3}$. That would need the sphere to pull the particle inwards, which it can't do. So contact is lost when

$$\cos\theta = \tfrac{2}{3}, \qquad \sin\theta = \tfrac{\sqrt{5}}{3}, \qquad v^2 = 2ga\left(1-\tfrac{2}{3}\right) = \tfrac{2}{3}ga.$$

Stage 2: the projectile

Take axes with origin O, x horizontal (in the direction of motion) and y vertically up. At the moment of leaving, the particle is at

$$x_0 = a\sin\theta = \tfrac{\sqrt{5}}{3}a, \qquad y_0 = a\cos\theta = \tfrac{2}{3}a,$$

and its velocity is along the tangent, at angle θ below the horizontal:

$$u_x = v\cos\theta = \tfrac{2}{3}v, \qquad u_y = -v\sin\theta = -\tfrac{\sqrt{5}}{3}v, \qquad \text{where } v = \sqrt{\tfrac{2}{3}ga}.$$

Now $y = y_0 + u_y t - \frac{1}{2}gt^2$, so the particle reaches $y = 0$ when

$$\tfrac{1}{2}gt^2 + \tfrac{\sqrt{5}}{3}v\,t - \tfrac{2}{3}a = 0.$$

Taking the positive root,

$$t = \frac{-\frac{\sqrt{5}}{3}v + \sqrt{\frac{5}{9}v^2 + \frac{4}{3}ga}}{g}.$$

With $v^2 = \frac{2}{3}ga$, the expression under the root is $\frac{10}{27}ga + \frac{36}{27}ga = \frac{46}{27}ga$, and $\frac{\sqrt{5}}{3}v = \sqrt{\frac{10}{27}ga}$. Hence

$$t = \sqrt{\frac{a}{27g}}\,(\sqrt{46} - \sqrt{10}).$$

The horizontal distance covered in the flight is

$$\begin{aligned} u_x t &= \tfrac{2}{3}\sqrt{\tfrac{2}{3}ga}\cdot\sqrt{\frac{a}{27g}}\,(\sqrt{46}-\sqrt{10}) = \frac{2\sqrt{2}\,a}{27}(\sqrt{46}-\sqrt{10}) \\ &= \frac{2a}{27}(\sqrt{92}-\sqrt{20}) = \frac{2a}{27}(2\sqrt{23}-2\sqrt{5}) = \frac{a}{27}(4\sqrt{23}-4\sqrt{5}). \end{aligned}$$

Adding the starting distance $x_0 = \frac{\sqrt{5}}{3}a = \frac{9\sqrt{5}}{27}a$, the particle meets the horizontal plane through O at distance

$$x_0 + u_x t = \frac{a}{27}\left(9\sqrt{5} + 4\sqrt{23} - 4\sqrt{5}\right) = \frac{a\left(5\sqrt{5} + 4\sqrt{23}\right)}{27}$$

from the centre, as required.

The flight doesn't touch the sphere

To be thorough, we should check the projectile really is clear of the sphere. Let $f(t) = x^2 + y^2 - a^2$ along the flight path. Expanding in powers of t:

- the constant term is $x_0^2 + y_0^2 - a^2 = 0$;
- the coefficient of t is $2(x_0 u_x + y_0 u_y) = 0$, because the velocity is perpendicular to the radius;
- the coefficient of t^2 is $u_x^2 + u_y^2 - g y_0 = v^2 - ga\cos\theta = 0$, which is exactly the condition $R = 0$;
- what's left is $-g u_y t^3 + \frac{1}{4}g^2 t^4$, and $u_y < 0$.

So $f(t) = g|u_y|\, t^3 + \frac{1}{4}g^2 t^4 > 0$ for $t > 0$. The particle stays outside the sphere, and the landing distance above is the one it actually reaches.

Checks

- The answer is about $1.125a$. This is more than a, as it must be, because the particle has to clear the sphere, which reaches out to distance a at the level of O.
- The leaving height $\frac{2}{3}a$ above the centre is a well-known result, and it doesn't depend on g or m. The landing distance doesn't depend on them either, which dimensional analysis requires: a is the only length in the problem.
- We stepped the motion numerically: a Runge–Kutta integration of $\ddot{\theta} = (g/a)\sin\theta$ from a tiny displacement, stopped when $R = 0$ and continued as a projectile, lands at $1.12458a$. The exact formula gives $1.124580\ldots a$.

Common slips

- Using $v^2 = ga\cos\theta$ (from $R = 0$) but then forgetting to combine it with energy, or using the wrong speed: at leaving, $v^2 = \frac{2}{3}ga$, not $\frac{1}{3}ga$.
- Launching the projectile horizontally. The velocity is along the tangent, at angle θ *below* the horizontal, so it has a downward component from the start.
- Measuring the landing distance from the point of leaving rather than from the centre, which forgets x_0.

Beyond STEP

The condition “contact ends when the reaction vanishes” is an example of a *unilateral constraint*. These appear throughout mechanics, from a bead on a wire to a car going over a hump-backed bridge. The coincidence in the last step of the solution, where the projectile’s path and the sphere agree to second order at the point of leaving, says that at that moment the parabola and the circle have the same curvature. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats circular motion, constraint forces and energy methods in this spirit.

WORKED SOLUTION

STEP III 1996, Question 10

Mechanics • Equilibrium with Friction

STEP III 1996, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q10

The key idea

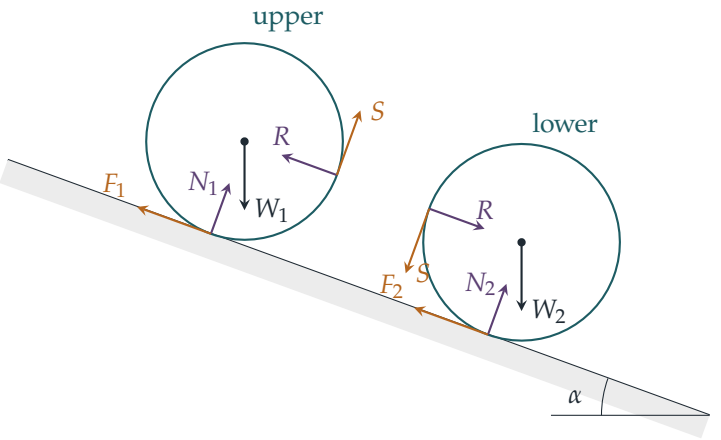
Write down the six equilibrium equations (two resolutions and one moment equation for each cylinder) and you will find that they determine every contact force uniquely. The moment equations are what make it work: each cylinder has just two frictional forces with a moment about its centre, so these must be equal in size. After that, each of the four inequalities is simply "friction $\leq$ coefficient $\times$ normal reaction" at one of the three lines of contact, plus the condition that the push between the cylinders is not negative.

Solution

Model. Each cylinder is a uniform rigid body of radius a, so its weight acts at its centre (on the axis). Work in a vertical cross-section through the centres. Since the radii are equal and both cylinders rest on the plane, the line of centres is parallel to the plane, and the contact between the cylinders is at the midpoint of that line. Take the x-direction *down* the slope and the y-direction perpendicular to the plane, pointing out of it.

Define the forces (all components may turn out to be of either sign):

- N_1, N_2: normal reactions of the plane on the upper and lower cylinders;
- F_1, F_2: frictional forces of the plane on them, measured *up* the slope;
- R: the normal force between the cylinders, along the line of centres, pushing the lower cylinder down the slope and the upper one up the slope;
- S: the frictional force between the cylinders, perpendicular to the plane, acting *away from* the plane on the upper cylinder and so *towards* the plane on the lower one (by Newton's third law).



(For clarity the cylinders are drawn apart; R and S act at the single line of contact. Each weight acts vertically downwards through the centre.)

Moments. Take moments about the centre of each cylinder. The normal forces N_1, N_2, R and the weights all pass through the centres, so only the frictional forces have moments. Each frictional force acts at distance a from the centre, tangentially. For the upper cylinder, F_1 (up the slope, at the bottom of the cylinder) and S (away from the plane, at its downhill side) have opposite moments, giving

$$aF_1 = aS, \qquad \text{so} \qquad F_1 = S.$$

In the same way, for the lower cylinder F_2 (up the slope, at its bottom) and S (towards the plane, at its uphill side) have opposite moments, so $F_2 = S$. Hence

$$F_1 = F_2 = S.$$

Ask yourself Why must the friction from the plane on each cylinder equal the friction between the cylinders?

Because a cylinder that is free to roll is only held in equilibrium if the couples about its axis cancel, and friction forces are the only forces with a moment about the axis. You can check the senses: on the upper cylinder, the plane's friction up the slope at the bottom would roll it downhill, i.e. clockwise in the diagram, while an upward frictional force on its downhill side turns it anticlockwise.

Resolving. Parallel to the plane (positive down the slope):

$$\begin{aligned} \text{upper:} \quad & W_1 \sin\alpha - F_1 - R = 0, \\ \text{lower:} \quad & W_2 \sin\alpha - F_2 + R = 0. \end{aligned}$$

Adding, and using $F_1 = F_2 = S$,

$$F_1 = F_2 = S = \tfrac{1}{2}(W_1 + W_2)\sin\alpha,$$

which is half the total down-slope component of weight, as you would expect. Subtracting then gives

$$R = \tfrac{1}{2}(W_1 - W_2)\sin\alpha.$$

Perpendicular to the plane (positive outwards):

$$\begin{aligned} \text{upper:}\quad & N_1 - W_1\cos\alpha + S = 0 \quad \Longrightarrow \quad N_1 = W_1\cos\alpha - \tfrac{1}{2}(W_1 + W_2)\sin\alpha, \\ \text{lower:}\quad & N_2 - W_2\cos\alpha - S = 0 \quad \Longrightarrow \quad N_2 = W_2\cos\alpha + \tfrac{1}{2}(W_1 + W_2)\sin\alpha. \end{aligned}$$

So the six equations fix all six forces. Equilibrium is possible exactly when these forces are physically allowed: every normal reaction is non-negative and every frictional force is at most the coefficient of friction times the corresponding normal reaction.

Part (i)

The cylinders can only push on each other, so $R \geq 0$. Since $\sin\alpha > 0$, this gives $W_1 \geq W_2$.

Part (ii)

At the contact between the cylinders we need $S \leq \mu R$:

$$\tfrac{1}{2}(W_1 + W_2)\sin\alpha \leq \mu \cdot \tfrac{1}{2}(W_1 - W_2)\sin\alpha.$$

The left-hand side is strictly positive, so this forces $R > 0$, i.e. $W_1 > W_2$ (equal weights are in fact impossible). Dividing by the positive quantity $\tfrac{1}{2}(W_1 - W_2)\sin\alpha$,

$$\mu \geq \frac{W_1 + W_2}{W_1 - W_2},$$

as required.

Part (iii)

At the upper cylinder's contact with the plane we need $N_1 \geq 0$ and $F_1 \leq \mu_1 N_1$. Since $F_1 > 0$, the second condition already forces $N_1 > 0$. Dividing $F_1 \leq \mu_1 N_1$ by $\tfrac{1}{2}(W_1 + W_2)\sin\alpha > 0$:

$$1 \leq \mu_1 \left(\frac{2W_1\cos\alpha}{(W_1 + W_2)\sin\alpha} - 1 \right) = \mu_1 \left(\frac{2W_1\cot\alpha}{W_1 + W_2} - 1 \right).$$

The bracket is $N_1 / \left(\tfrac{1}{2}(W_1 + W_2)\sin\alpha\right)$, which is positive, so we may divide by it:

$$\mu_1 \geq \left(\frac{2W_1\cot\alpha}{W_1 + W_2} - 1 \right)^{-1}.$$

(Equilibrium also needs the bracket to be positive, i.e. $\tan\alpha < 2W_1/(W_1 + W_2)$; otherwise the upper cylinder would be lifted off the plane.)

The inequality for μ_2

Here $N_2 = W_2\cos\alpha + \frac{1}{2}(W_1+W_2)\sin\alpha$ is automatically positive, and $F_2 \le \mu_2 N_2$ becomes, after dividing by $\frac{1}{2}(W_1+W_2)\sin\alpha$,

$$1 \le \mu_2\left(\frac{2W_2\cot\alpha}{W_1+W_2}+1\right), \qquad \text{i.e.} \qquad \mu_2 \ge \left(\frac{2W_2\cot\alpha}{W_1+W_2}+1\right)^{-1}.$$

Ask yourself Why is the sign $+1$ for the lower cylinder but -1 for the upper one?

The friction S between the cylinders pulls the upper cylinder away from the plane and pushes the lower one into it. So the plane presses less on the upper cylinder and more on the lower one, and the upper cylinder needs a larger ratio of friction to normal reaction.

Checks

- Adding the two cylinders' equations gives the whole system: total plane friction $(W_1+W_2)\sin\alpha$ up the slope and total normal reaction $(W_1+W_2)\cos\alpha$, exactly as for one rigid block.
- As $\alpha \to 0$ all frictional forces and R tend to 0, and both bounds on μ_1, μ_2 tend to 0, as they should on level ground.
- Numerically, with $W_1 = 3$, $W_2 = 1$, $\alpha = 20°$: solving the six equations directly gives $R \approx 0.342$, $S = F_1 = F_2 \approx 0.684$, $N_1 \approx 2.135$, $N_2 \approx 1.624$, so $\mu \ge 2$, $\mu_1 \ge 0.320$ and $\mu_2 \ge 0.421$, agreeing with the formulae. The same agreement was checked for many random choices of W_1, W_2 and α.

Common slips

- Forgetting the moment equations. Without them there are six unknowns and only four equations, and the problem looks indeterminate.
- Getting the direction of the friction between the cylinders wrong, which swaps the ± 1 in the last two answers.
- Overlooking that the lighter cylinder must be the lower one: if the upper cylinder were lighter, R would have to be negative.

Beyond STEP

This is a statically determinate system: the equilibrium equations fix every force, so the friction laws become checkable conditions. When there are more contacts than equations (a ladder against a rough wall on rough ground, say) the system is statically

indeterminate, and rigid-body statics alone cannot say how the forces are shared. Engineers then bring in the elasticity of the bodies. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers rigid-body equilibrium and torque carefully.

WORKED SOLUTION

STEP III 1996, Question 11

Mechanics • Small Oscillations

STEP III 1996, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q11

The key idea

Two pieces of circle geometry do the work. AC is a diameter, so the angle ABC is a right angle. That gives both string lengths, and so both tensions, in terms of θ. Also, the angle at the centre is twice the angle at the circumference, so B has turned through 2θ about the centre. Its arc length from C is therefore $2a\theta$, and Newton's second law along the tangent gives the equation of motion. For the oscillation, linearise about the equilibrium.

A warning before we start: the printed equation has $mg\sin\theta$ where it should have $mg\sin 2\theta$. The derivation below shows why, and only the corrected equation gives the stated $5\lambda = 24mg$ and the stated period.

Solution

The equation of motion

Model. The ring B is a particle of mass m on a smooth wire, so the wire exerts only a normal reaction R along the radius. The strings are light and elastic and obey Hooke's law: a string of natural length a stretched to length ℓ has tension $\lambda(\ell - a)/a$.

Geometry. Since AC is a diameter (length $2a$), the angle in a semicircle gives $\angle ABC = 90°$. In the right-angled triangle ABC,

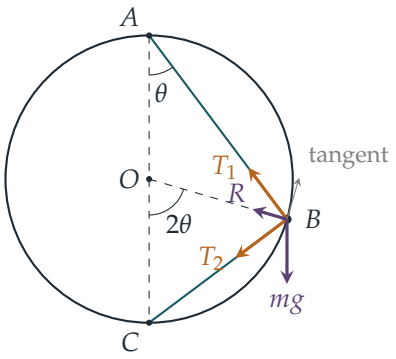
$$AB = 2a\cos\theta, \qquad CB = 2a\sin\theta.$$

The inscribed angle $\angle CAB = \theta$ stands on the arc CB, so the angle at the centre is $\angle COB = 2\theta$. The arc length from C to B is $s = 2a\theta$, so the acceleration of B along the wire, in the direction of increasing θ (towards A), is $\ddot{s} = 2a\ddot{\theta}$.

Tensions. The tensions in BA and BC are

$$T_1 = \lambda\,\frac{2a\cos\theta - a}{a} = \lambda(2\cos\theta - 1), \qquad T_2 = \lambda(2\sin\theta - 1).$$

Both strings are taut exactly when $\cos\theta > \frac{1}{2}$ and $\sin\theta > \frac{1}{2}$, that is, when $\frac{\pi}{6} < \theta < \frac{\pi}{3}$.



Resolving along the tangent at B. The tangent is perpendicular to OB, and R has no component along it.

- Triangle OAB is isosceles ($OA = OB = a$), so $\angle OBA = \angle OAB = \theta$. Hence BA makes an angle $90° - \theta$ with the tangent, and T_1 contributes $T_1 \sin\theta$ towards A.
- Triangle OCB is isosceles, and $\angle OCB = \angle ACB = 90° - \theta$, so $\angle OBC = 90° - \theta$. Hence BC makes an angle θ with the tangent, and T_2 contributes $T_2 \cos\theta$ away from A.
- OB makes an angle 2θ with the downward vertical, so the tangent makes an angle 2θ with the horizontal. Gravity contributes $mg \sin 2\theta$ away from A.

Newton's second law along the tangent gives

$$\begin{aligned} 2ma\ddot{\theta} &= \lambda(2\cos\theta - 1)\sin\theta - \lambda(2\sin\theta - 1)\cos\theta - mg\sin 2\theta \\ &= \lambda(\cos\theta - \sin\theta) - mg\sin 2\theta, \end{aligned}$$

since the $2\lambda \sin\theta\cos\theta$ terms cancel. This is the required equation, except that the gravity term is $mg\sin 2\theta$, not the printed $mg\sin\theta$.

? **Ask yourself** Is the printed $mg\sin\theta$ perhaps right after all?

No. At $\theta = \frac{\pi}{4}$ the ring is level with O, the tangent is vertical, and gravity's full weight mg acts along the wire. That gives $mg\sin 2\theta = mg$, but $mg\sin\theta = mg/\sqrt{2}$. The next part settles it too: the printed equation would give $\lambda = 3mg$ at equilibrium, not $\frac{24}{5}mg$.

The equilibrium

With $\sin\theta = \frac{3}{5}$ we have $\cos\theta = \frac{4}{5}$ (taking θ acute), and θ lies in the taut range because $\frac{3}{5} > \frac{1}{2}$ and $\frac{4}{5} > \frac{1}{2}$. Setting $\ddot{\theta} = 0$:

$$\lambda\left(\tfrac{4}{5} - \tfrac{3}{5}\right) = mg \cdot 2 \cdot \tfrac{3}{5} \cdot \tfrac{4}{5} \quad\Longrightarrow\quad \frac{\lambda}{5} = \frac{24mg}{25} \quad\Longrightarrow\quad 5\lambda = 24mg,$$

as required.

Small oscillations

Write $f(\theta) = \lambda(\cos\theta - \sin\theta) - mg\sin 2\theta$, so that $2ma\ddot{\theta} = f(\theta)$ and $f(\theta_0) = 0$ at the equilibrium θ_0. Put $\theta = \theta_0 + \varepsilon$ with ε small. To first order, $f(\theta_0 + \varepsilon) \approx f'(\theta_0)\,\varepsilon$, where

$$\begin{aligned} f'(\theta_0) &= -\lambda(\sin\theta_0 + \cos\theta_0) - 2mg\cos 2\theta_0 \\ &= -\tfrac{24}{5}mg\cdot\tfrac{7}{5} - 2mg\left(\tfrac{16}{25} - \tfrac{9}{25}\right) = -\tfrac{168}{25}mg - \tfrac{14}{25}mg = -\tfrac{182}{25}mg. \end{aligned}$$

So

$$2ma\ddot{\varepsilon} = -\tfrac{182}{25}mg\,\varepsilon \quad\Longrightarrow\quad \ddot{\varepsilon} = -\frac{91g}{25a}\,\varepsilon.$$

This is simple harmonic motion with $\omega^2 = \dfrac{91g}{25a}$. Because $f'(\theta_0) < 0$, the equilibrium is stable. The displacement is slight, so the strings stay taut (θ stays near θ_0, well inside $(\tfrac{\pi}{6}, \tfrac{\pi}{3})$). The period is approximately

$$\frac{2\pi}{\omega} = 2\pi\cdot 5\sqrt{\frac{a}{91g}} = 10\pi\sqrt{\frac{a}{91g}}.$$

Checks

- At $\theta = \frac{\pi}{4}$ the strings are equally stretched, so their tangential effects cancel ($\cos\theta - \sin\theta = 0$), and the whole weight acts along the vertical tangent. Our equation gives $2ma\ddot{\theta} = -mg$, which is right.
- With $a = g = m = 1$ and $\lambda = 4.8$, we computed the tangential force from position vectors. It vanishes at $\sin\theta = 0.6$. Stepping the motion numerically from a small displacement gave a period of 3.29328, matching $10\pi/\sqrt{91} \approx 3.29328$.

Common slips

- Taking the arc length as $a\theta$. The ring turns through 2θ about the centre, not θ, so its tangential acceleration is $2a\ddot{\theta}$. This is where the 2 on the left comes from.
- Resolving along the chord directions instead of the tangent. The reaction is unknown, so resolve along the tangent to eliminate it.

Beyond STEP

Linearising about an equilibrium, $\ddot{x} = f(x) \approx f'(x_0)(x - x_0)$, is the first step in the stability analysis of nonlinear systems. It gives stability when $f'(x_0) < 0$ and simple harmonic motion with $\omega^2 = -f'(x_0)$. At university this extends to several degrees of freedom through normal modes and the eigenvalues of a matrix. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to read more.

WORKED SOLUTION

STEP III 1996, Question 12

Probability and Statistics • Markov Chains

STEP III 1996, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q12

This question has a full guided treatment in *Beyond STEP*, Section 12.16, *Markov Chains Forwards and Backwards*: https://beyondstep.keithmcnulty.org/ch12/markov.html

The key idea

This is a Markov chain: the type of the next theorem depends only on the type of the current one. Part (i) is the law of total probability, conditioning on the type of the current theorem. Part (ii) asks for a distribution that the chain leaves unchanged (a *stationary* distribution), which is a small linear system. Part (iii) runs the chain backwards, and reversing a conditional probability is exactly what Bayes' theorem does.

Solution

Number the theorems in order. Let X be the type of a given theorem and Y the type of the theorem after it, so that

$$\mathrm{P}(X = i) = a_i, \qquad \mathrm{P}(Y = j \mid X = i) = p_{ij}, \qquad \mathrm{P}(Y = j) = b_j.$$

Part (i)

The events $\{X = 1\}$, $\{X = 2\}$, $\{X = 3\}$ are mutually exclusive, and one of them must happen, because every theorem has exactly one of the three types. So they partition the sample space, and the event $\{Y = j\}$ splits into three disjoint pieces:

$$\{Y = j\} = \{X = 1,\, Y = j\} \cup \{X = 2,\, Y = j\} \cup \{X = 3,\, Y = j\}.$$

Adding the probabilities of the pieces, and writing each one as $\mathrm{P}(X = i)\,\mathrm{P}(Y = j \mid X = i)$,

$$b_j = \sum_{i=1}^{3} \mathrm{P}(X = i)\,\mathrm{P}(Y = j \mid X = i) = a_1 p_{1j} + a_2 p_{2j} + a_3 p_{3j},$$

which is the law of total probability. The given observation is used here: $\mathrm{P}(Y = j \mid X = i)$ is p_{ij} whatever happened before the current theorem.

Part (ii)

We need $b_j = a_j$ for each j, together with $a_1 + a_2 + a_3 = 1$, since the a_i form a probability distribution. Reading down the columns of the array, the three equations are

$$\begin{aligned} a_1 &= 0.3a_1 + 0.2a_2 + 0.1a_3, \\ a_2 &= 0.3a_1 + 0.4a_2 + 0.3a_3, \\ a_3 &= 0.4a_1 + 0.4a_2 + 0.6a_3. \end{aligned}$$

> **Ask yourself** Three equations in three unknowns, plus $a_1 + a_2 + a_3 = 1$: isn't that one equation too many?

It isn't, because the three equations are not independent. Adding them gives $a_1 + a_2 + a_3 = a_1 + a_2 + a_3$, since each row of the array sums to 1. So any two of them imply the third, and the normalising condition is what pins down the solution.

The middle equation is the easy one, because a_1 and a_3 have the same coefficient there. Using $a_1 + a_3 = 1 - a_2$,

$$a_2 = 0.3(1 - a_2) + 0.4a_2 \quad \Longrightarrow \quad 0.9a_2 = 0.3 \quad \Longrightarrow \quad a_2 = \tfrac{1}{3}.$$

Then $a_1 + a_3 = \frac{2}{3}$. Multiplying the first equation by 10 gives $7a_1 = 2a_2 + a_3 = \frac{2}{3} + a_3$, and substituting $a_3 = \frac{2}{3} - a_1$,

$$7a_1 = \tfrac{4}{3} - a_1 \quad \Longrightarrow \quad a_1 = \tfrac{1}{6}, \qquad a_3 = \tfrac{2}{3} - \tfrac{1}{6} = \tfrac{1}{2}.$$

Check the third equation: $0.4 \cdot \frac{1}{6} + 0.4 \cdot \frac{1}{3} + 0.6 \cdot \frac{1}{2} = \frac{1}{15} + \frac{2}{15} + \frac{3}{10} = \frac{1}{2} = a_3$. So

$$a_1 = \tfrac{1}{6}, \qquad a_2 = \tfrac{1}{3}, \qquad a_3 = \tfrac{1}{2}.$$

In the long run, then, half of Professor Ecks's theorems are false.

Part (iii)

Now take X to be the type of the preceding theorem and Y the type of the particular theorem, with X distributed as in part (ii). We want $q_{ij} = \mathrm{P}(X = i \mid Y = j)$. By Bayes' theorem,

$$q_{ij} = \frac{\mathrm{P}(X = i)\,\mathrm{P}(Y = j \mid X = i)}{\mathrm{P}(Y = j)} = \frac{a_i p_{ij}}{b_j} = \frac{a_i p_{ij}}{a_j},$$

where the last step uses $b_j = a_j$ from part (ii). This is why the question fixes these values of the a_i: then the particular theorem has the same distribution as the one before it.

Column by column:

- $j = 1$ ($a_1 = \frac{1}{6}$): $q_{11} = \dfrac{\frac{1}{6}\cdot 0.3}{\frac{1}{6}} = \frac{3}{10}$, $\quad q_{21} = \dfrac{\frac{1}{3}\cdot 0.2}{\frac{1}{6}} = \frac{2}{5}$, $\quad q_{31} = \dfrac{\frac{1}{2}\cdot 0.1}{\frac{1}{6}} = \frac{3}{10}$.
- $j = 2$ ($a_2 = \frac{1}{3}$): $q_{12} = \dfrac{\frac{1}{6}\cdot 0.3}{\frac{1}{3}} = \frac{3}{20}$, $\quad q_{22} = 0.4 = \frac{2}{5}$, $\quad q_{32} = \dfrac{\frac{1}{2}\cdot 0.3}{\frac{1}{3}} = \frac{9}{20}$.
- $j = 3$ ($a_3 = \frac{1}{2}$): $q_{13} = \dfrac{\frac{1}{6}\cdot 0.4}{\frac{1}{2}} = \frac{2}{15}$, $\quad q_{23} = \dfrac{\frac{1}{3}\cdot 0.4}{\frac{1}{2}} = \frac{4}{15}$, $\quad q_{33} = 0.6 = \frac{3}{5}$.

Arranged like the original array (row i, column j):

$$(q_{ij}) = \begin{pmatrix} \frac{3}{10} & \frac{3}{20} & \frac{2}{15} \\ \frac{2}{5} & \frac{2}{5} & \frac{4}{15} \\ \frac{3}{10} & \frac{9}{20} & \frac{3}{5} \end{pmatrix} = \begin{pmatrix} 0.3 & 0.15 & 0.1\dot{3} \\ 0.4 & 0.4 & 0.2\dot{6} \\ 0.3 & 0.45 & 0.6 \end{pmatrix}.$$

Here it is the *columns* that sum to 1: for a fixed type j of the particular theorem, the preceding theorem must have been of some type.

Checks

- Each column of (q_{ij}) sums to 1; for example $\frac{2}{15} + \frac{4}{15} + \frac{3}{5} = 1$.
- The diagonal entries satisfy $q_{ii} = p_{ii}$, as the formula $q_{ii} = a_i p_{ii} / a_i$ predicts.
- A simulation of 10^6 steps of the chain gave long-run frequencies 0.167, 0.333, 0.501 and backward proportions within 0.001 of every q_{ij} above.

Common slips

- Multiplying the array by a column vector of the a_i (using rows instead of columns). The equation is $b_j = \sum_i a_i p_{ij}$, so the sum runs down column j.
- Forgetting $a_1 + a_2 + a_3 = 1$, and then being puzzled that the three balance equations do not determine the a_i.
- Answering part (iii) with p_{ji}. Going backwards in time is not the same as reading the array the other way round; this chain is not reversible, since $q_{12} = 0.15$ but $p_{12} = 0.3$.

Beyond STEP

The vector (a_1, a_2, a_3) is the stationary distribution of the chain: a left eigenvector of the transition matrix with eigenvalue 1. Because every entry of the matrix is positive, the distribution of the type of the nth theorem converges to it whatever type the first theorem had. The array (q_{ij}) describes the *time-reversed* chain, and chains whose reversal looks the same as the original (with $a_i p_{ij} = a_j p_{ji}$, the detailed balance condition) are central to random walks and to Monte Carlo methods. Grimmett and Welsh, *Probability: An Introduction*, has an accessible chapter on Markov chains.

WORKED SOLUTION

STEP III 1996, Question 13

Probability and Statistics • Chebyshev's Inequality

STEP III 1996, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q13

The key idea

The variance is a sum of non-negative terms $p_i(x_i - \mu)^2$, one for each value. Split that sum into the values that are far from the mean (at least λ away) and those that are near. For an *upper* bound on the far probability, throw the near terms away and bound each far term below. For a *lower* bound, do the opposite: bound every term above, the near ones by λ^2 and the far ones by k^2.

Solution

Definitions and $aX + b$

Write $p_i = \mathrm{P}(X = x_i)$, so that $p_i \geq 0$ and $\sum_{i=1}^{n} p_i = 1$. Then

$$\mathrm{E}(X) = \sum_{i=1}^{n} x_i p_i, \qquad \mathrm{var}(X) = \mathrm{E}\big((X-\mu)^2\big) = \sum_{i=1}^{n} (x_i - \mu)^2 p_i,$$

where $\mu = \mathrm{E}(X)$. (Equivalently $\mathrm{var}(X) = \mathrm{E}(X^2) - \mu^2$, but the first form is the one we shall use.)

The random variable $aX + b$ takes the value $ax_i + b$ with probability p_i. (If $a = 0$ these values coincide, but adding the probabilities of equal values does not change the sums below.) So

$$\mathrm{E}(aX+b) = \sum_i (ax_i + b)p_i = a\sum_i x_i p_i + b\sum_i p_i = a\mathrm{E}(X) + b,$$

using $\sum_i p_i = 1$. Hence $aX + b - \mathrm{E}(aX+b) = a(X - \mu)$, and

$$\mathrm{var}(aX+b) = \sum_i \big(a(x_i - \mu)\big)^2 p_i = a^2 \sum_i (x_i - \mu)^2 p_i = a^2\,\mathrm{var}(X).$$

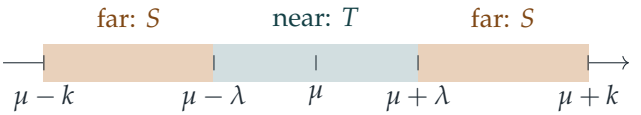
Adding b shifts every value and the mean together, so it has no effect on the spread.

The upper bound

Let $\mu = \mathrm{E}(X)$, let S be the set of indices i with $|x_i - \mu| \geq \lambda$ (the "far" values), and let T be the rest, where $|x_i - \mu| < \lambda$. Then

$$\mathrm{P}\big(|X - \mu| \geq \lambda\big) = \sum_{i \in S} p_i.$$

Call this probability P.



Every term of the variance is non-negative, so dropping the terms in T can only make the sum smaller; and for $i \in S$ we have $(x_i - \mu)^2 \geq \lambda^2$. Therefore

$$\begin{aligned}
\mathrm{var}(X) &= \sum_{i \in S} (x_i - \mu)^2 p_i + \sum_{i \in T} (x_i - \mu)^2 p_i \\
&\geq \sum_{i \in S} (x_i - \mu)^2 p_i \\
&\geq \lambda^2 \sum_{i \in S} p_i = \lambda^2 P.
\end{aligned}$$

Since $\lambda > 0$ we may divide by λ^2, giving

$$\mathrm{P}\big(|X - \mu| \geq \lambda\big) \leq \frac{\mathrm{var}(X)}{\lambda^2},$$

as required. (If S is empty both sums over S are zero and the inequality is trivially true.)

The lower bound

Now suppose also that $|x_i - \mu| \leq k$ for every i. This time we bound every term of the variance from *above*:

- for $i \in S$, $(x_i - \mu)^2 \leq k^2$ by the new hypothesis;
- for $i \in T$, $(x_i - \mu)^2 < \lambda^2$ by the definition of T.

Since $\sum_{i \in T} p_i = 1 - P$, this gives

$$\mathrm{var}(X) \leq k^2 \sum_{i \in S} p_i + \lambda^2 \sum_{i \in T} p_i = k^2 P + \lambda^2 (1 - P) = \lambda^2 + (k^2 - \lambda^2) P.$$

Rearranging,

$$(k^2 - \lambda^2)\, P \geq \mathrm{var}(X) - \lambda^2.$$

Ask yourself Can we divide by $k^2 - \lambda^2$?

We are told $k \geq \lambda > 0$, so $k^2 - \lambda^2 \geq 0$. If $k > \lambda$ it is strictly positive, and dividing keeps the direction of the inequality:

$$\mathrm{P}\big(|X - \mu| \geq \lambda\big) \geq \frac{\mathrm{var}(X) - \lambda^2}{k^2 - \lambda^2},$$

which is the required result. In the boundary case $k = \lambda$ the right-hand side has a zero denominator and the statement has no meaning; the inequality just before it still holds and says $\mathrm{var}(X) \leq \lambda^2$, which is clear since every $(x_i - \mu)^2 \leq k^2 = \lambda^2$.

When $\mathrm{var}(X) \leq \lambda^2$ the lower bound is zero or negative and so says nothing new; it has content only when the variance is large compared with λ^2, forcing some probability out into the tails.

Checks

- Take $X = \pm 1$ with probability 0.4 each and ± 4 with probability 0.1 each, so $\mu = 0$ and $\mathrm{var}(X) = 0.8 \times 1 + 0.2 \times 16 = 4$. With $\lambda = 1$, $k = 4$: $P = 1$, the upper bound is 4 and the lower bound is $\frac{3}{15} = \frac{1}{5}$, and indeed $\frac{1}{5} \leq 1 \leq 4$.
- Take $X = 0$ or $\pm c$, with $\mathrm{P}(X = \pm c) = q/2$ each. Then $\mathrm{var}(X) = qc^2$ and with $\lambda = c$ the upper bound is exactly $q = P$: Chebyshev's inequality cannot be improved in general.
- We tested both bounds, and the formulae for $\mathrm{E}(aX + b)$ and $\mathrm{var}(aX + b)$, in exact rational arithmetic on 20 000 random distributions with up to seven values; no case failed.

Common slips

- Writing $\mathrm{var}(aX + b) = a\,\mathrm{var}(X)$ or $a^2\mathrm{var}(X) + b$: the variance is quadratic in deviations, and a shift changes no deviation.
- In the lower bound, bounding the near terms by 0 instead of λ^2. That gives $\mathrm{var}(X) \leq k^2 P$, which is true but not what is asked.
- Dividing by $k^2 - \lambda^2$ without remarking that it is positive (and that $k = \lambda$ must be excluded).

Beyond STEP

The first inequality is Chebyshev's inequality, and the trick of discarding terms is the same one behind Markov's inequality $\mathrm{P}(Y \geq c) \leq \mathrm{E}(Y)/c$ for $Y \geq 0$ (apply it to $Y = (X - \mu)^2$). In a first-year probability course Chebyshev's inequality gives a two-line proof of the weak law of large numbers: the average of n independent copies of X has variance $\mathrm{var}(X)/n$, so it is unlikely to stray far from μ. The second inequality is a "reverse" version that needs boundedness, in the same spirit as the Paley–Zygmund inequality. Grimmett and Welsh, *Probability: An Introduction*, covers all of this.

WORKED SOLUTION

STEP III 1996, Question 14

Probability and Statistics • Two-State Markov Process

STEP III 1996, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q14

The key idea

Chop time into tiny intervals of length δt and ask what can happen in one of them. Expected counts then add up by linearity of expectation, and the probability of being mid-repair obeys a simple "in minus out" balance that turns into a first-order linear differential equation as $\delta t \to 0$. Once we have $p(t)$, the expected repair time is just its integral.

Solution

Expected number of punctures

Only time spent actually cycling can produce a puncture, and while I am cycling all is well. Split the total cycling time T into n small pieces of length $\delta t = T/n$. Let X_k be the number of punctures in the kth piece; to first order in δt, X_k is 1 with probability $\alpha\,\delta t$ and 0 otherwise, so $\mathrm{E}(X_k) = \alpha\,\delta t$. By linearity of expectation (no independence needed),

$$\mathrm{E}(\text{number of punctures}) = \sum_{k=1}^{n} \mathrm{E}(X_k) = n\alpha\,\delta t = \alpha T.$$

Time spent repairing doesn't enter, because the clock here is cycling time only.

The equation for $p(t)$

Now let t be real (clock) time, including repairs. At time t I am either repairing (probability $p(t)$) or cycling (probability $1 - p(t)$). I am repairing at time $t + \delta t$ if either

- I was repairing at t and the repair was *not* completed in $(t, t+\delta t)$: probability $p(t)(1 - \beta\,\delta t)$; or
- I was cycling at t and got a puncture in $(t, t+\delta t)$: probability $(1 - p(t))\,\alpha\,\delta t$.

These are exclusive, so, ignoring terms of order $(\delta t)^2$ (two events in one tiny interval),

$$p(t+\delta t) = p(t)(1-\beta\,\delta t) + \big(1-p(t)\big)\alpha\,\delta t.$$

Rearranging,

$$\frac{p(t+\delta t)-p(t)}{\delta t} = \alpha - (\alpha+\beta)\,p(t),$$

and letting $\delta t \to 0$ gives

$$p'(t) = \alpha - (\alpha+\beta)\,p(t).$$

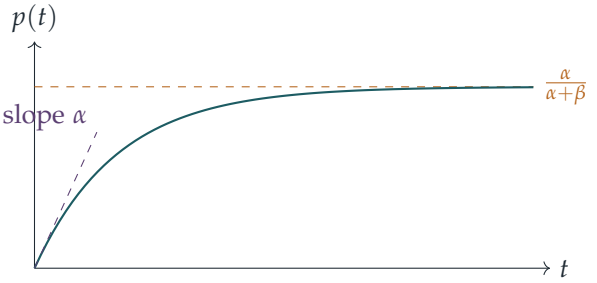
I start with the bike in good order, so the initial condition is $p(0) = 0$.

Checking the given solution

For $p(t) = \dfrac{\alpha}{\alpha+\beta}\big(1-\mathrm{e}^{-(\alpha+\beta)t}\big)$ we have $p(0) = \frac{\alpha}{\alpha+\beta}(1-1) = 0$, and

$$\begin{aligned} p'(t) &= \alpha\,\mathrm{e}^{-(\alpha+\beta)t}, \\ \alpha - (\alpha+\beta)p(t) &= \alpha - \alpha\big(1-\mathrm{e}^{-(\alpha+\beta)t}\big) = \alpha\,\mathrm{e}^{-(\alpha+\beta)t}. \end{aligned}$$

These agree, so p satisfies the differential equation and the initial condition, as required.



The graph shows p rising from 0 with initial slope α towards the long-run level $\frac{\alpha}{\alpha+\beta}$.

Expected repair time

Let R be the time spent mending during $[0, T]$, and let $I(t)$ be 1 if I am repairing at time t and 0 otherwise, so $R = \int_0^T I(t)\,\mathrm{d}t$. Since $\mathrm{E}(I(t)) = p(t)$, taking expectations (again just linearity, applied to the integral as a limit of sums over small intervals) gives

$$\begin{aligned} \mathrm{E}(R) = \int_0^T p(t)\,\mathrm{d}t &= \frac{\alpha}{\alpha+\beta}\left[t + \frac{\mathrm{e}^{-(\alpha+\beta)t}}{\alpha+\beta}\right]_0^T \\ &= \frac{\alpha}{\alpha+\beta}\left(T - \frac{1-\mathrm{e}^{-(\alpha+\beta)T}}{\alpha+\beta}\right). \end{aligned}$$

Ask yourself Why is the expected time mending equal to the integral of a probability?

In each small interval $(t, t+\delta t)$, the time spent mending is about δt with probability $p(t)$ and 0 otherwise, so its expectation is $p(t)\,\delta t$. Adding over all the intervals gives $\sum p(t)\,\delta t \to \int_0^T p(t)\,\mathrm{d}t$.

The two approximations

Write $x = (\alpha+\beta)T$. The expected fraction of the journey spent mending is

$$F = \frac{\mathrm{E}(R)}{T} = \frac{\alpha}{\alpha+\beta}\left(1 - \frac{1-\mathrm{e}^{-x}}{x}\right).$$

x small. Using $\mathrm{e}^{-x} = 1 - x + \frac{1}{2}x^2 - \cdots$, we get $\dfrac{1-\mathrm{e}^{-x}}{x} = 1 - \frac{1}{2}x + O(x^2)$, so

$$F \approx \frac{\alpha}{\alpha+\beta}\cdot\frac{x}{2} = \frac{\alpha}{\alpha+\beta}\cdot\frac{(\alpha+\beta)T}{2} = \frac{\alpha T}{2}.$$

This makes sense: over a short journey $p(t) \approx \alpha t$, which averages to $\alpha T/2$ over $[0, T]$.

x large. Then $0 < \dfrac{1-\mathrm{e}^{-x}}{x} < \dfrac{1}{x}$, which is small, so

$$F \approx \frac{\alpha}{\alpha+\beta}.$$

Over a long journey $p(t)$ has settled at its equilibrium value $\frac{\alpha}{\alpha+\beta}$ for almost all of the time.

Checks

- Equilibrium: setting $p' = 0$ gives $\alpha(1-p) = \beta p$, "rate in equals rate out", so $p = \frac{\alpha}{\alpha+\beta}$, matching the large-$T$ limit.
- If $\beta \to \infty$ (instant repairs) everything mending-related tends to 0, as it should.
- With $\alpha = 0.7$, $\beta = 2.3$, $T = 2.5$ the formula gives $\mathrm{E}(R) \approx 0.5056$. A simulation of 2×10^5 journeys (alternating exponential cycling and repair spells) gave 0.5058, and a simulated puncture count over cycling time 2.5 averaged $1.750 = \alpha T$.

Common slips

- Forgetting the factor $(1 - p(t))$ on the puncture term: you can only puncture while you are cycling.
- Using $p(0) = \frac{\alpha}{\alpha+\beta}$ or leaving the initial condition unstated; the bike starts in good order, so $p(0) = 0$.

- Dropping the second-order term in the small-x expansion, which leaves $F \approx 0$ instead of $\alpha T/2$.

Beyond STEP

This is a two-state continuous-time Markov chain: the bike flips between "working" and "being mended" at rates α and β, and the equation for p is the simplest case of the Kolmogorov forward equations. The same model describes machines that break down and get repaired, ion channels opening and closing, and much of queueing theory. The puncture count in cycling time is a Poisson process with rate α. Grimmett and Welsh, *Probability: An Introduction*, covers Poisson processes and Markov chains at the right level.

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